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CareerCompass: AI-Powered Career Guidance and Job Recommendation Platform

CareerCompass

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Table of Contents

List of Figures

List of Tables

Acknowledgment

The project team would like to express sincere appreciation to Dr. Amena Mahmoud for academic supervision, technical guidance, and continuous feedback during the preparation of CareerCompass. The team also thanks the Faculty of Computers and Information at Kafr El-Sheikh University for providing the academic setting in which this graduation project was designed, implemented, tested, and documented.

The work presented in this book reflects a collaborative software engineering effort. It combines web application development, database design, AI-assisted document analysis, explainable matching, containerized deployment, testing, and technical documentation. The two supervisor-provided graduation books were used only to understand expected report structure and visual formality; no content, wording, project-specific claims, diagrams, or references were copied from them.

Abstract

CareerCompass is a graduation/demo career guidance platform that helps students and early-career users understand their CV profile, explore imported job opportunities, and compare their current skills against job requirements. The system consists of a React and Vite frontend, a Laravel API backend, a MySQL database, a FastAPI-based CV analyzer, a FastAPI/Scrapy-based job miner, MinIO-compatible private file storage, Nginx routing, and Prometheus/Grafana monitoring. The platform supports registration, login, CV upload, AI-assisted CV parsing, normalized profile and skills storage, job recommendation, gap analysis, an application tracker, and administrator dashboards for job and source diagnostics.

The AI CV Analyzer is documented as a hybrid implementation rather than a single opaque model. It combines PDF/image text extraction, OCR fallback, section segmentation, a BERT-family token-classification path for named-entity recognition, rule-based contact/date/experience extraction, skill canonicalization, domain and seniority classification, sentence embeddings, and TF-IDF-style matching. The runtime code can load an exported local NER artifact when that ignored deployment folder is present, and a Colab-oriented training notebook documents how the artifact is produced. A user-provided exported Colab PDF now records the overall NER training metrics, while the cleaned dataset content and model weights remain outside Git; therefore, the report separates recorded training-run evidence from repository-alone reproducibility and production benchmark claims.

The implementation is intentionally described as a graduation/demo system rather than a production product. The AI outputs are estimates, the job data depends on imported and demo sources, and the security posture is appropriate for demonstration but requires further production hardening. Validation was performed through Docker Compose configuration checks, backend tests, frontend lint/build, Python service tests or syntax checks, HTTP probes, and manual browser screenshots. Backend tests passed with 39 tests and 297 assertions, the AI job miner tests passed with 75 tests, and the frontend build completed successfully. The AI CV analyzer container did not include pytest, so its pytest suite was marked as skipped while Python syntax compilation passed.

Abbreviations

Abbreviation	Meaning
API	Application Programming Interface
CV	Curriculum Vitae
DFD	Data Flow Diagram
ERD	Entity Relationship Diagram
HTTP	Hypertext Transfer Protocol
JSON	JavaScript Object Notation
JWT	JSON Web Token
ML	Machine Learning
NLP	Natural Language Processing
OCR	Optical Character Recognition
RBAC	Role-Based Access Control
REST	Representational State Transfer
S3	Simple Storage Service compatible object storage
TF-IDF	Term Frequency-Inverse Document Frequency
UI	User Interface

Chapter 1: Introduction

1.1 Introduction to the Project

CareerCompass is an AI-assisted career guidance and job recommendation platform developed as a Computer Science graduation project for Kafr El-Sheikh University. The system helps a student upload a CV, receive a structured profile, view estimated job matches, inspect skill gaps, and save opportunities in an application tracker. The project combines web engineering, backend service design, natural language processing support, web scraping support, data persistence, and containerized operations.

The platform uses a modern web stack. The frontend is implemented with React, a component-based UI library [4], and bundled with Vite [5]. The backend uses Laravel, which provides routing, controllers, validation, middleware, Eloquent ORM, queues, testing support, and file storage abstractions [1]. The AI services are implemented with FastAPI, a Python API framework that supports type-hinted request and response handling [6]. The deployment is orchestrated through Docker Compose [11].

1.2 Background and Motivation

Many students prepare CVs and search for jobs without a clear view of how their current skills relate to job descriptions. Career guidance is often fragmented across CV feedback, job boards, learning resources, and manual advice. CareerCompass addresses this academic problem by placing those steps into one integrated demonstration system. The objective is not to replace human career advising, but to provide a practical software prototype that can parse a CV, normalize profile data, import job records, estimate matches, and present explainable gap analysis.

1.3 Problem Statement

Students frequently face three connected problems. First, they may not know whether their CV communicates their skills clearly. Second, job listings often describe requirements in different formats, making comparison difficult. Third, students may save opportunities across separate tools without a coherent tracker. CareerCompass proposes a centralized graduation/demo system that links CV data, jobs, matching, gap analysis, and tracking.

1.4 Purpose of the Project

The purpose of CareerCompass is to demonstrate how a distributed web system can support career exploration using explainable AI-assisted workflows. The project demonstrates authentication, CV upload, private storage, parsing, skill extraction, job import, recommendation, gap analysis, admin diagnostics, and monitoring in a Dockerized environment.

1.5 Project Objectives

- Provide a student-facing web interface for account creation, login, CV upload, profile review, job recommendations, gap analysis, and application tracking.
- Provide an administrator interface for dashboard statistics, job review, source diagnostics, and target role management.
- Implement a Laravel API that protects authenticated workflows using token-based access and role checks.
- Implement Python services for CV analysis and job mining support.
- Store uploaded CV files privately and expose downloads through controlled URLs.
- Validate the system with backend tests, frontend lint/build, Python tests where available, Docker checks, HTTP probes, and browser screenshots.

1.6 Proposed Solution

The proposed solution is a multi-service application. React renders the browser interface. Nginx serves as the gateway. Laravel handles REST-style APIs over HTTP [16], authentication, validation, data models, business services, and admin routes. MySQL stores normalized data [8]. MinIO provides S3-compatible object storage [9]. The CV analyzer parses PDFs and images using Python text extraction and OCR-related libraries. The job miner imports jobs from demo, API, and scraping-style sources using FastAPI, Scrapy, and HTML parsing concepts [16], [17], [18].

1.7 Project Scope

The scope covers a graduation/demo environment: local Docker deployment, student workflows, admin workflows, testing, screenshots, and documentation. It does not claim production readiness, job placement certainty, full job market coverage, legal compliance for production privacy, or complete AI accuracy.

1.8 Graduation Demo Positioning

CareerCompass should be presented as a working academic prototype with realistic engineering boundaries. The screenshots and tests show that the core workflows can run locally. However, large-scale evaluation, production identity management, cloud infrastructure, observability hardening, legal privacy review, and complete market integrations remain future work.

1.9 Report Organization

Chapter 2 analyzes requirements and users. Chapter 3 presents architecture, diagrams, database design, and deployment. Chapter 4 lists software and tools with references. Chapter 5 documents implementation modules from the repository. Chapter 6 presents the AI CV Analyzer deep technical analysis as a standalone academic contribution. Chapter 7 presents testing and evaluation results. Chapter 8 discusses security and privacy. Chapter 9 concludes with achievements, limitations, and future work.

Chapter 2: System Analysis

2.1 Introduction

System analysis defines what CareerCompass should do, who uses it, and what constraints influence implementation. The analysis is based on the reviewed repository, including the root documentation, Docker configuration, Laravel API, React frontend, Python services, database migrations, seeders, tests, and finalization notes.

2.2 Development Methodology

The project followed an iterative engineering approach. Each feature was implemented, checked through local commands or tests, integrated with Docker services, then documented. This approach is appropriate for a graduation project because it supports incremental progress while keeping the system demonstrable. The architecture uses separate services, which follows a common microservice-style pattern where independently deployable services communicate over network APIs [29].

2.3 Existing Career Guidance Problems

The system targets common problems in student career preparation:

- CV content is not structured for direct software comparison.
- Job descriptions vary in wording and completeness.
- Students need explainable feedback rather than unexplained match numbers.
- Admin users need visibility into imported jobs and scraping sources.
- Demo systems need reliable startup, testing, and evidence for academic evaluation.

2.4 Target Users and Stakeholders

Stakeholder	Description	Main Interest
Student user	A university student or early-career user.	Upload CV, view profile, discover jobs, analyze gaps, save opportunities.
Administrator	A project operator or supervisor/demo administrator.	Review jobs, source diagnostics, users, target roles, and system health.
Supervisor	Academic supervisor evaluating the graduation project.	Correctness, completeness, originality, and honest evaluation.
Project team	Developers responsible for design and implementation.	Maintainable code, demonstrable workflows, testing, and documentation.

Table 1. Stakeholder summary.

2.5 User Roles

CareerCompass implements two practical roles. The student role can register, login, upload a CV, view recommendations, run gap analysis, and track applications. The admin role can access protected admin routes for dashboard statistics, job administration, scraping sources, target roles, and user review.

2.6 Functional Requirements

ID	Requirement	Implementation Evidence
FR-01	Register and login users.	Laravel AuthController, RegisterRequest, LoginRequest, React Login/Register pages.
FR-02	Upload a CV file.	CvUploadRequest validates PDF/JPEG/PNG and max size; Dashboard appends file field cv.
FR-03	Store CV files privately.	CvStorageService and MinIO/S3-compatible disk configuration.
FR-04	Parse CV and extract profile/skills.	CvProcessingService and AI CV Analyzer /api/parse-cv.
FR-05	Display normalized profile and skills.	UserResource, Profile page, Dashboard AI insight components.
FR-06	Import and display jobs.	JobPosting model, ScrapedJobController, JobController, AI Job Miner.
FR-07	Estimate job recommendations.	JobController and GapAnalysisService use CV/profile data and matching service.
FR-08	Analyze skill gaps.	GapAnalysisController and GapAnalysis page.
FR-09	Track applications.	ApplicationController, ApplicationTrackerService, Applications page.
FR-10	Provide admin dashboards.	Admin Dashboard, Jobs, Sources, Targets pages and admin API routes.
FR-11	Provide health and metrics endpoints.	HealthController, MetricsController, Prometheus/Grafana compose services.

Table 2. Functional requirements summary.

2.7 Non-Functional Requirements

Category	Requirement	CareerCompass Approach
Usability	The UI should guide students through CV upload and recommendations.	React pages, dashboard cards, profile score, and action buttons.
Maintainability	Code should be modular.	Controllers, requests, services, resources, models, React pages, and Python services are separated.
Reliability	The system should degrade honestly if AI services fail.	CV processing returns timeout/error/no_text statuses and preserves prior data when appropriate.
Security	Authentication, validation, private files, and admin role checks are required.	Sanctum tokens, request validation, admin middleware, signed URLs, service tokens.
Observability	Health and metrics should be available.	/api/health, /api/ready, /api/metrics, Prometheus, Grafana.
Portability	The demo should run locally.	Docker Compose services and environment examples.

Table 3. Non-functional requirements summary.

2.8 Hardware Requirements

For local demonstration, a developer machine capable of running Docker Desktop and multiple containers is required. CV parsing and OCR-like processing can be CPU-intensive; therefore, enough memory should be available for the Laravel backend, MySQL, frontend, Python services, MinIO, Prometheus, and Grafana. GPU acceleration is not required for the demonstrated flow.

2.9 Software Requirements

Layer	Software
Frontend	React, Vite, Tailwind-style CSS classes, lucide-react icons, Recharts.
Backend	Laravel 12, PHP, Composer dependencies, Sanctum-style token authentication.
AI services	Python, FastAPI, text extraction, OCR-related dependencies, Scrapy-style job mining.
Data	MySQL 8.0, MinIO/S3-compatible storage.
Infrastructure	Docker, Docker Compose, Nginx, Prometheus, Grafana.
Testing	PHPUnit/Pest-style Laravel tests, pytest for Python, ESLint and Vite build.

Table 4. Hardware and software environment.

2.10 Input and Output Flow

Primary inputs include user account data, uploaded CV files, imported job records, target role settings, and administrator source configurations. Primary outputs include normalized user profiles, skill lists, CV analysis metadata, estimated job matches, gap reports, application records, admin statistics, health checks, and monitoring metrics.

2.11 Use Case Summary

The main use cases are shown in Figure 5. The system separates student and administrator responsibilities while sharing the same backend API and database.

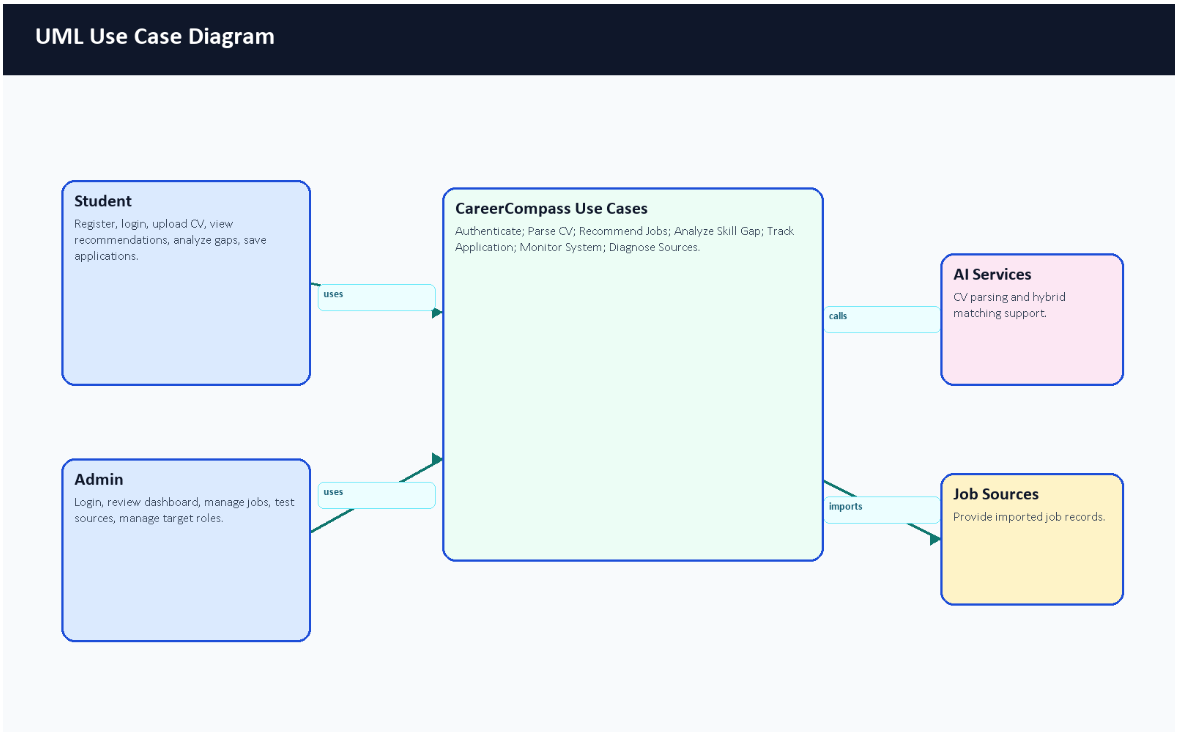


Figure 5. UML use case diagram.

Chapter 3: System Design and Architecture

3.1 Introduction

CareerCompass is designed as a Dockerized multi-service application. This design separates browser UI, API logic, AI services, data storage, object storage, reverse proxy routing, and monitoring. Docker containers help package runtime dependencies consistently [10], while Docker Compose coordinates the multi-container local deployment [11].

3.2 High-Level System Architecture

The high-level architecture is shown in Figure 1. Browser users interact with the React frontend through Nginx. The frontend calls the Laravel API. Laravel persists records in MySQL, stores CV files in MinIO-compatible storage, calls the AI CV Analyzer for parsing, calls matching logic for recommendations/gaps, and receives job imports from the job miner. This diagram was reviewed during the final evidence pass and already represents the important deployment boundaries: React, Nginx, Laravel, MySQL, MinIO, AI CV Analyzer, AI Job Miner, and monitoring.

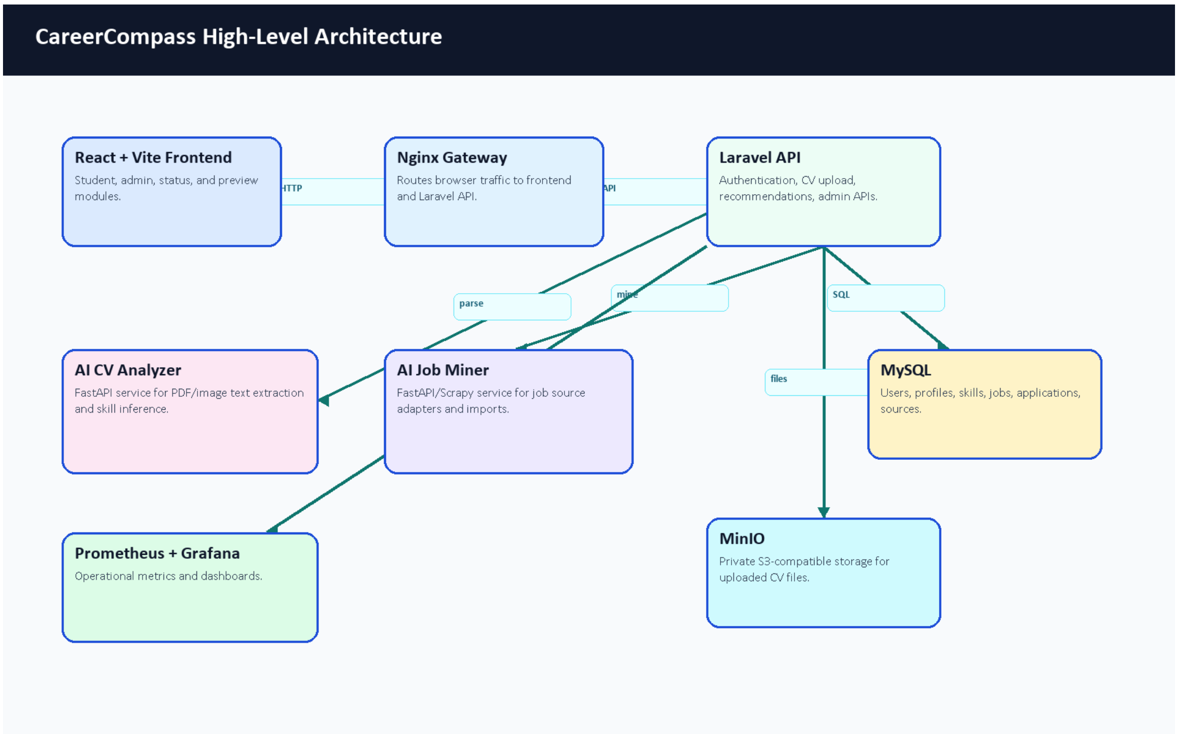


Figure 1. High-level architecture of CareerCompass.

3.3 Frontend Architecture

The frontend is implemented with React and organized around routes in frontend/src/App.jsx. Public pages include Home, Login, Register, About, Privacy, Terms, and System Status. Protected student routes include Dashboard, Jobs, Gap Analysis, Profile, Settings, Market Intelligence, Applications, CV Builder, Mock Interview, Learning, Career Planner, Mentorship, and Tools Hub. Protected admin routes include Admin Dashboard, Jobs, Users, Sources, and Target Roles.

The frontend API layer is located under frontend/src/api. Axios is configured in client.js, including base URL resolution, bearer token injection, request IDs, retry behavior for safe GET requests, and 401 handling. Authentication state is managed by AuthContext.jsx, which stores the user and token in local storage. Localization files exist under frontend/src/locales.

3.4 Backend API Architecture

The backend is a Laravel API. Routes are defined in backend-api/routes/api.php and are registered both at /api and /api/v1. The API includes public health/readiness/metrics endpoints, guest authentication routes, public job listing routes, internal scraper import routes protected by a service token, authenticated student routes, and admin routes protected by middleware.

Laravel provides structured controllers, form requests, resources, services, models, migrations, seeders, and tests. This aligns with Laravel's documented framework responsibilities, including routing, validation, database access, queues, and testing [1], [3].

3.5 AI CV Analyzer Architecture

The AI CV Analyzer is a FastAPI service. Laravel sends CV files to this service for parsing through /api/parse-cv. The analyzer routes PDFs and images differently, enforces timeout/error fallbacks, extracts readable text, runs structured extraction, and returns fields such as predicted role, seniority, domain, skills, strengths, gaps, red flags, confidence, document statistics, and parsing status. The backend handles statuses such as success, OCR fallback, timeout, error, empty file, and no text.

The analyzer is not a pure pretrained-model wrapper and not a model built entirely from scratch. It is a hybrid pipeline. The runtime prefers a local exported token-classification model under ai-cv-analyzer/models/ner_weights/career_compass_ner_final when that ignored deployment artifact exists; if the local artifact is unavailable, the NER engine has a fallback model path. Around that model-loading path, the team implemented practical CV-specific logic: spatial PDF parsing, OCR fallback, semantic sectioning, contact extraction, date/experience parsing, noisy-skill filtering, canonicalization, domain inference, seniority inference, and hybrid matching. PDF and OCR-related libraries are supported by external tools such as PyMuPDF, pdfplumber, and EasyOCR [22], [23], [24]. Transformer token classification and training concepts follow Hugging Face documentation [31], [32], [33].

Figure 9 summarizes the runtime path from the browser upload to Laravel persistence, FastAPI parsing, model/rule extraction, and dashboard output.

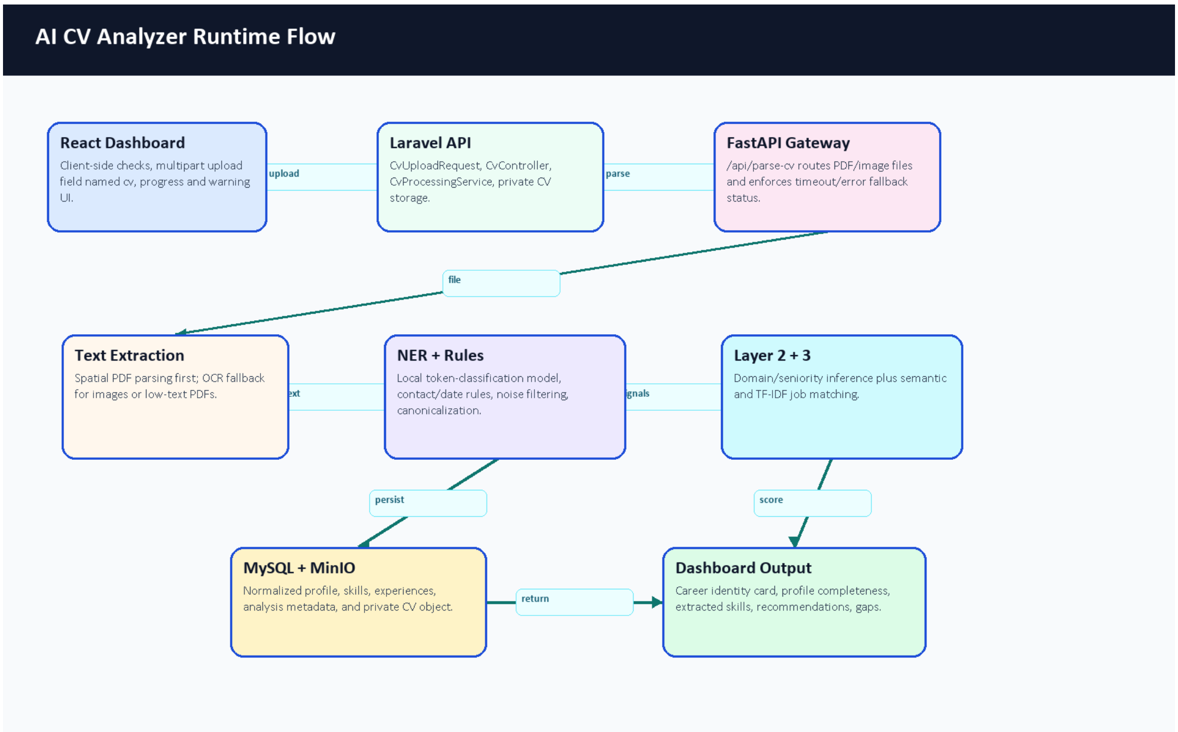


Figure 9. AI CV Analyzer runtime flow.

3.6 AI Job Miner Architecture

The AI Job Miner is a FastAPI service with scraping and import support. It includes source adapters for demo/local data, public APIs, and HTML/Scrapy-style extraction. Scrapy is a Python framework for extracting structured data from websites [17], and Beautiful Soup is commonly used to parse HTML documents [18]. CareerCompass uses a quality gate and honest source classifications so that demo/imported data is not overstated as complete market coverage.

3.7 Database Design

MySQL stores users, user profiles, skills, user-skill pivots, experience records, CV analyses, job postings, job-skill pivots, applications, scraping sources, target job roles, scraping jobs, failed URLs, and related metadata. MySQL is a relational database system documented by Oracle [8]. The Laravel migrations define schema constraints, indexes, foreign keys, and unique combinations such as job title/company uniqueness.

3.8 ERD

Figure 8 summarizes the main database tables and relationships. It is not a complete replacement for migrations, but it provides a readable graduation-book view of the data model. The final diagram was reviewed against migrations and updated to show the actual user_profiles, user_experiences, user_skills, and job_skills relationships.

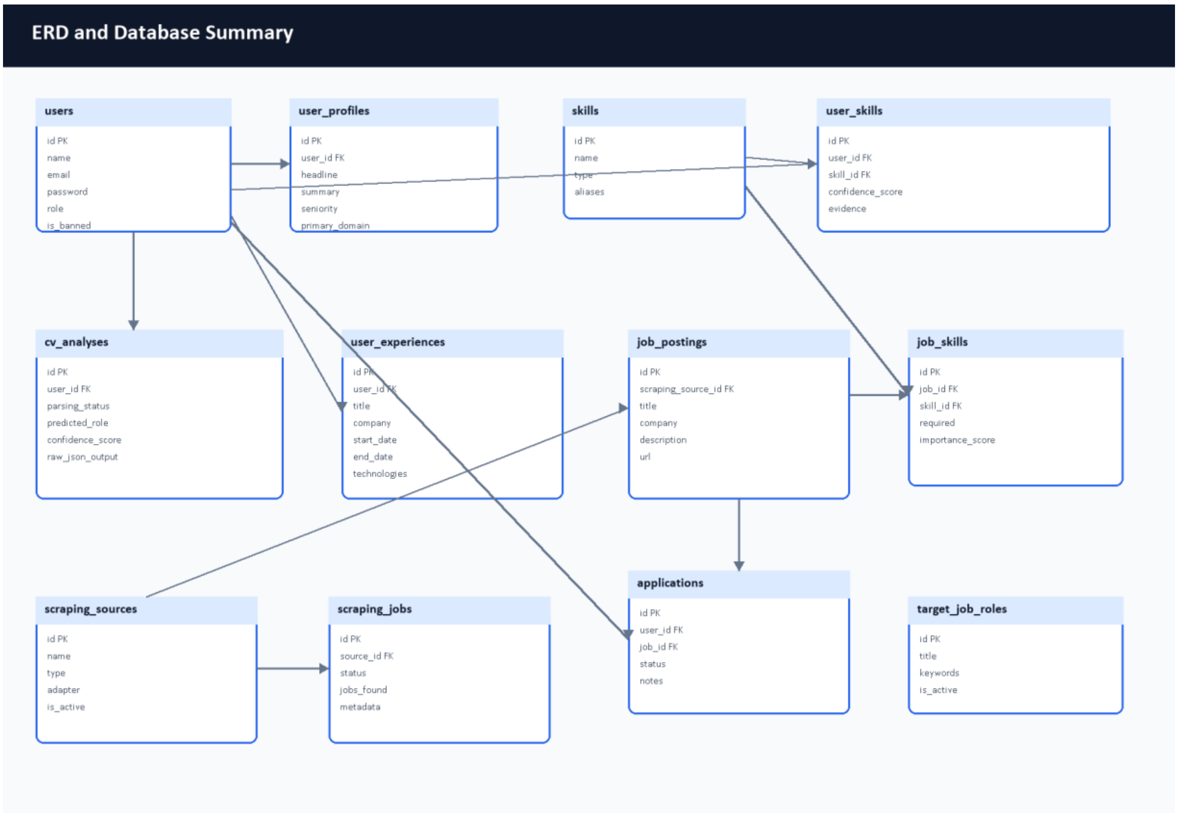


Figure 8. ERD and database summary diagram.

3.9 Data Flow Diagrams

The context-level data flow is shown in Figure 3, and the expanded process-level view is shown in Figure 4. Student and administrator workflows enter the same system boundary, while external job sources and AI services interact with controlled backend processes.

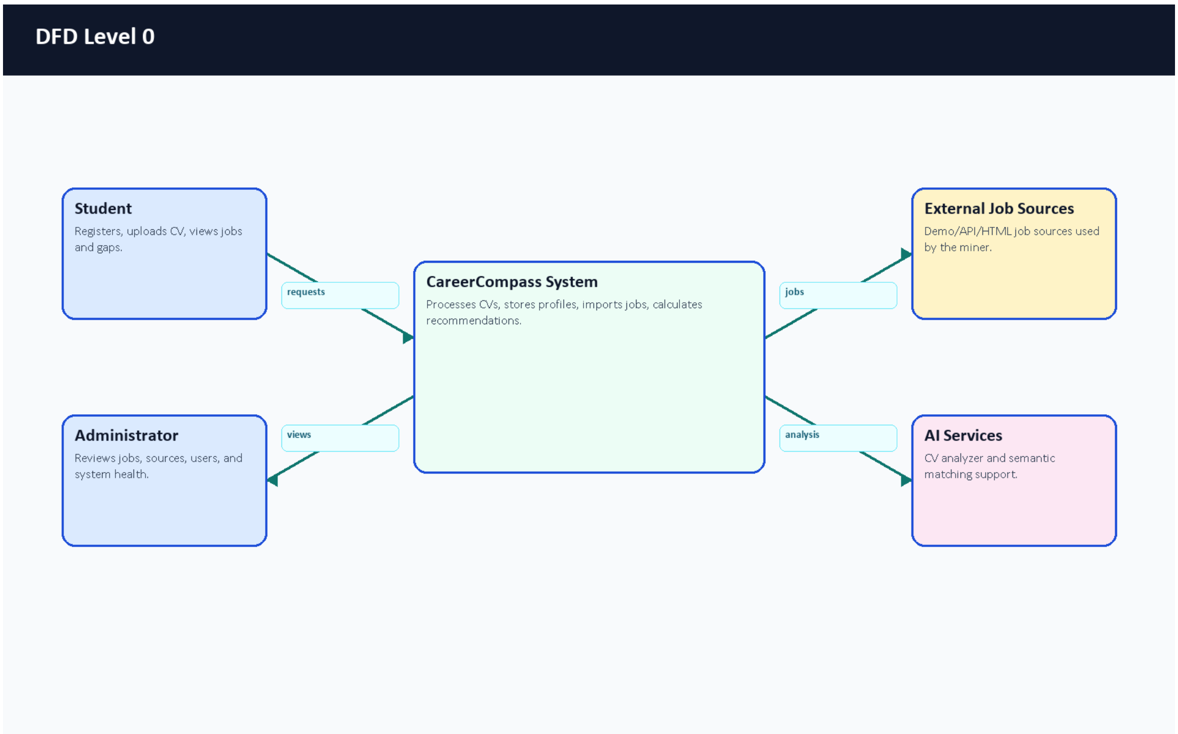


Figure 3. DFD Level 0 context diagram.

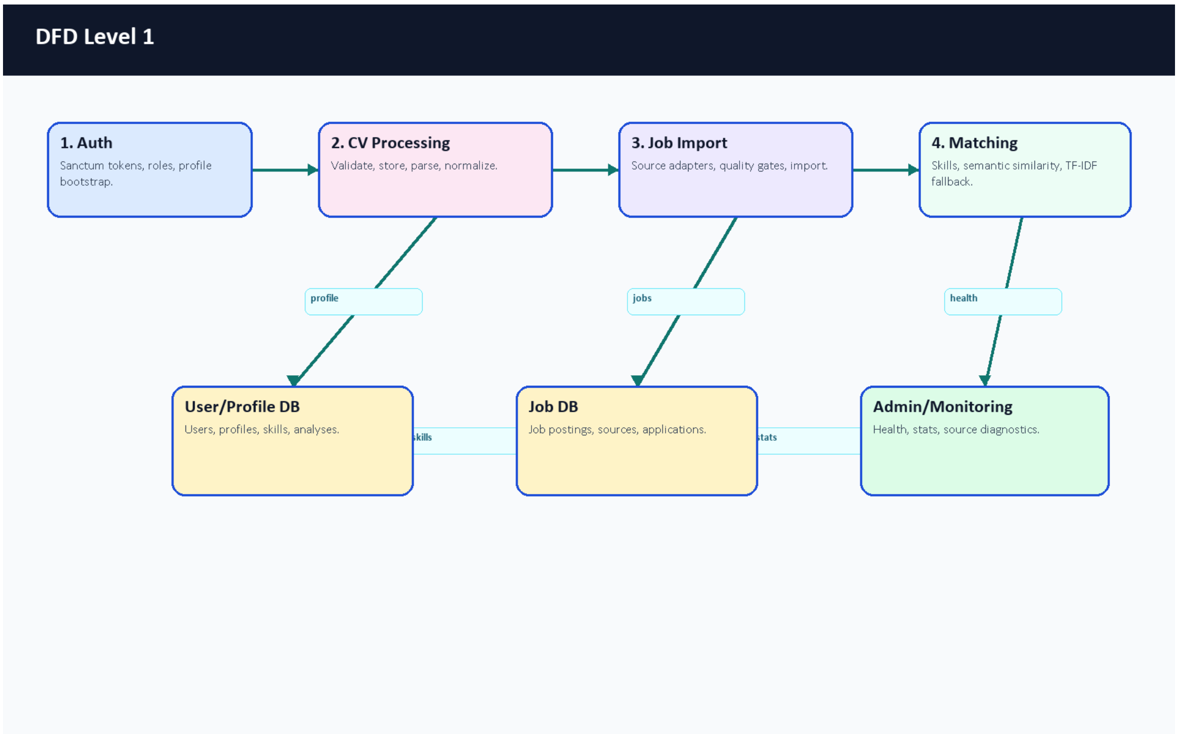


Figure 4. DFD Level 1 process diagram.

3.10 UML Use Case Diagram

The use case diagram separates student actions from administrator actions. Student workflows focus on career exploration. Admin workflows focus on operating and inspecting the imported job ecosystem.

3.11 UML Sequence Diagrams

Figure 6 shows the CV upload and analysis sequence. Figure 7 shows recommendation and gap analysis.

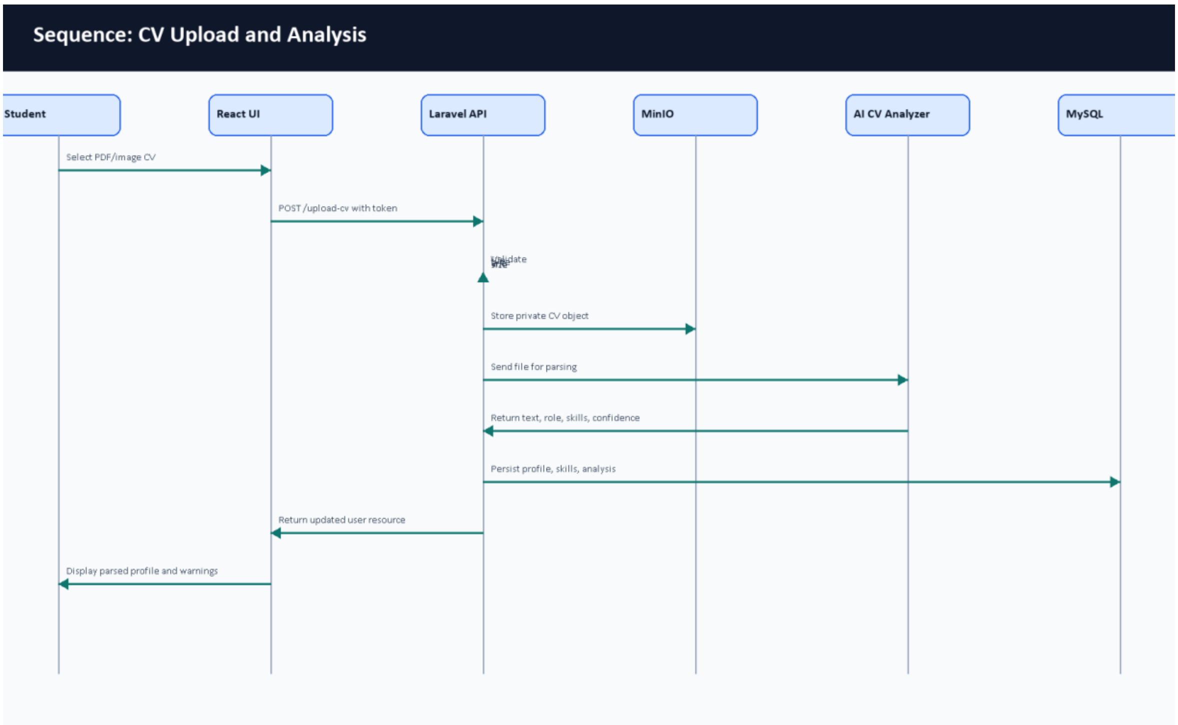


Figure 6. Sequence diagram for CV upload and analysis.

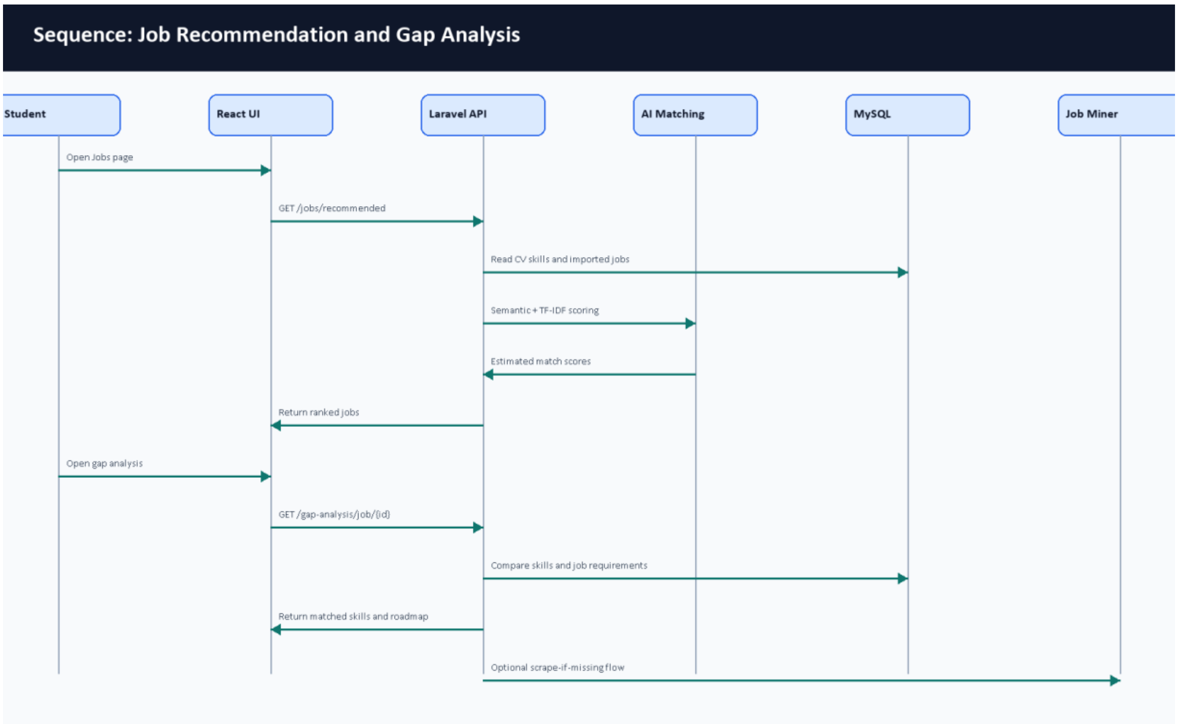


Figure 7. Sequence diagram for recommendation and gap analysis.

3.12 Deployment Architecture with Docker

The deployment is defined by Docker Compose files. Nginx exposes the application, the frontend serves built React assets, the Laravel API and workers handle backend work, MySQL stores structured data, MinIO stores private CV objects, Python services provide AI CV parsing and job mining, and Prometheus/Grafana provide monitoring. Figure 2 summarizes the container layout.

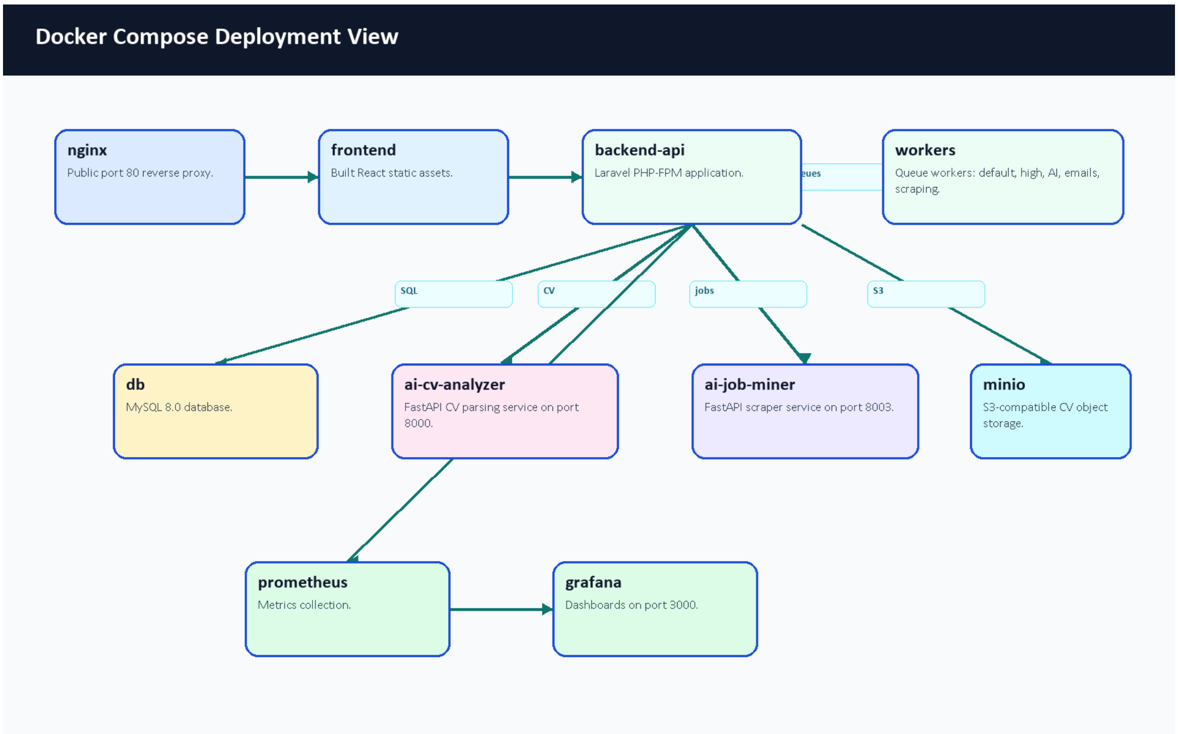


Figure 2. Docker deployment architecture.

3.13 Monitoring Architecture

CareerCompass includes live, readiness, and metrics endpoints. Prometheus is used for scraping and time-series metrics collection [13], while Grafana visualizes metrics and dashboard panels [14]. The admin dashboard also exposes application-level health information for the graduation demo.

3.14 Design Decisions and Justification

Decision	Justification
Use Laravel for the main API.	The project benefits from built-in routing, validation, Eloquent ORM, queues, resources, tests, and middleware [1].
Use React for UI.	Component-based pages support student/admin route separation and reusable cards [4].
Use FastAPI for AI services.	Python AI/NLP dependencies are easier to isolate behind typed HTTP services [6].
Use Docker Compose.	Multiple services can be started consistently for a graduation defense [11].
Use private object storage for CV files.	CV files are sensitive; private storage and signed downloads reduce accidental exposure [9], [27].
Keep AI wording honest.	Match scores and CV parsing are estimates; the demo should avoid claiming certain outcomes.

Table 5. Design decisions summary.

Chapter 4: Software and Tools Used

4.1 Laravel

Laravel is used for the backend API, controllers, middleware, validation requests, Eloquent models, migrations, resources, queues, tests, and storage integration [1]. In CareerCompass, Laravel centralizes authenticated business logic and data persistence.

4.2 React

React is used to build the browser-based interface for students and administrators [4]. The project uses React routes and components for dashboard cards, forms, pages, tables, modals, and visualization panels.

4.3 Vite

Vite is used as the frontend build tool [5]. The successful production build transformed 2904 modules during validation.

4.4 FastAPI

FastAPI is used for Python services because it supports efficient API development using Python type hints and modern web service patterns [6]. CareerCompass uses it for CV analysis and job mining service endpoints.

4.5 Python

Python is used for AI and scraping-related services [7]. It supports the ecosystem of text extraction, OCR, NLP, testing, and scraping libraries used by the AI services.

4.6 MySQL

MySQL stores relational project data including users, profiles, skills, jobs, applications, scraping sources, and CV analyses [8].

4.7 MinIO / S3-Compatible Storage

MinIO-compatible object storage is used to store CV files privately. This avoids placing uploaded CVs directly in public web directories and supports signed temporary download flows [9].

4.8 Docker and Docker Compose

Docker containers package each runtime, and Docker Compose coordinates multi-container execution [10], [11]. CareerCompass uses Compose for Nginx, frontend, backend, workers, MySQL, MinIO, AI services, Prometheus, and Grafana.

4.9 Nginx

Nginx acts as the reverse proxy and public gateway for the local deployment [12]. It routes browser and API traffic to the correct containers.

4.10 Prometheus and Grafana

Prometheus collects metrics [13], and Grafana visualizes metrics and dashboards [14]. CareerCompass uses these tools to support monitoring-oriented evaluation.

4.11 GitHub Actions

GitHub Actions is configured for repository automation and CI/CD-style validation [15]. The report treats workflow definitions as part of the development process, while final PR status should be reviewed after the draft PR is opened.

4.12 NLP / AI Libraries

The project uses concepts and libraries related to text extraction, OCR, transformer token classification, TF-IDF, cosine similarity, and sentence embeddings. TF-IDF and cosine similarity are documented by scikit-learn [19], [20]. Sentence Transformers provides sentence embedding models and utilities [21]. PyMuPDF, pdfplumber, and EasyOCR support PDF/image text extraction and OCR-style workflows [22], [23], [24]. The token-classification path builds on BERT-style contextual representations [37].

Hugging Face Transformers is used for model loading and token-classification style inference/training [31], [32]. The training notebook uses Hugging Face Trainer concepts, token-label alignment, the seqeval metric family, and a BERT token-classification configuration [33]. Optional dynamic quantization is supported in the local model-loading code as an inference optimization concept [38]. The deployed matching layer also uses sentence embeddings and a custom TF-IDF fallback so that a service outage or weak semantic signal does not become a silent failure.

4.13 Synthetic Data and Training Tools

The repository includes a model-training workflow designed around synthetic annotated CV snippets. The data-generation script uses the Gemini API through Google AI developer tooling to generate labeled examples, while the training notebook is designed for Google Colab GPU execution [34], [35], [36]. This report treats the external AI tooling as training-support tooling, not as a runtime dependency for private CV uploads.

4.14 Testing Tools

Backend tests use Laravel/PHP testing tools and PHPUnit concepts [25]. Python service tests use pytest where available [26]. Frontend validation uses ESLint and Vite build checks.

4.15 Development and Version Control Tools

Git and GitHub are used for version control and pull-request-based collaboration. Docker Desktop provides the local container runtime. Browser screenshots were captured through a Chrome DevTools Protocol helper to provide evidence images for this report.

Chapter 5: System Implementation

5.1 Introduction

This chapter documents the implemented CareerCompass modules based on repository files. The implementation is not a generic career platform; it is a Laravel/React/FastAPI/Docker system with specific routes, services, pages, models, seeders, and tests.

5.2 Authentication and User Management

Authentication is implemented in backend-api/app/Http/Controllers/Api/AuthController.php. Registration creates a user with role user, loads profile and related resources, and returns a token. Login validates credentials, checks the banned state, revokes old tokens, creates a new token, and returns a user resource. The frontend stores the token as auth_token and the user object in local storage through frontend/src/context/AuthContext.jsx.

The register request restricts emails to selected public email domains and validates password format. The admin role is protected by the IsAdmin middleware. Admin seed data creates a demo-only administrator account through AdminUserSeeder.

5.3 Student Dashboard

The student dashboard is implemented in frontend/src/pages/user/Dashboard.jsx. It presents the current profile state, CV upload/update controls, profile completeness, career identity, AI insights, and next actions. Before CV upload, it prompts the user to add a CV. After upload, it displays parsed CV availability, role inference, profile score, experience, and action buttons.

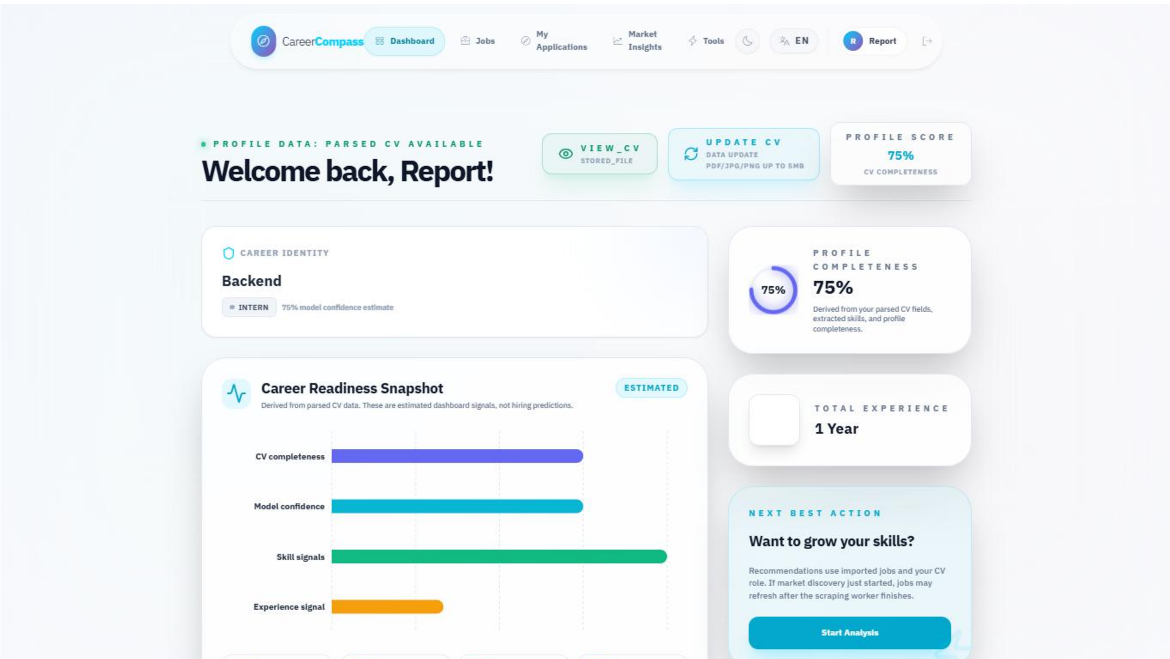


Figure 34. Student dashboard before CV upload.

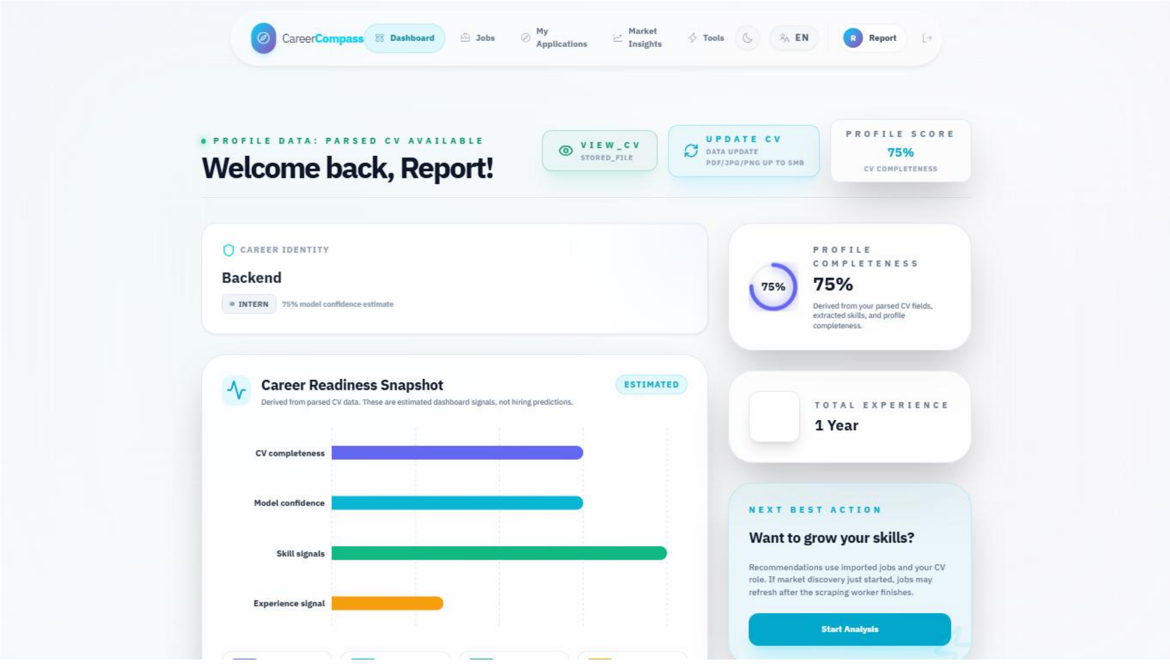


Figure 36. Dashboard after successful CV parsing.

5.4 CV Upload and Storage

CvUploadRequest requires a cv file and accepts PDF, JPEG, JPG, and PNG files up to 5 MB. The frontend appends the selected file as cv in a FormData object. CvController calls the CV processing service, persists the file path and metadata, and returns a unified user resource.

CV storage is handled as a private file workflow. The system supports signed download URLs, which is a better demo posture than public file exposure. OWASP recommends validating uploaded file type, extension, size, and storage handling carefully [27].

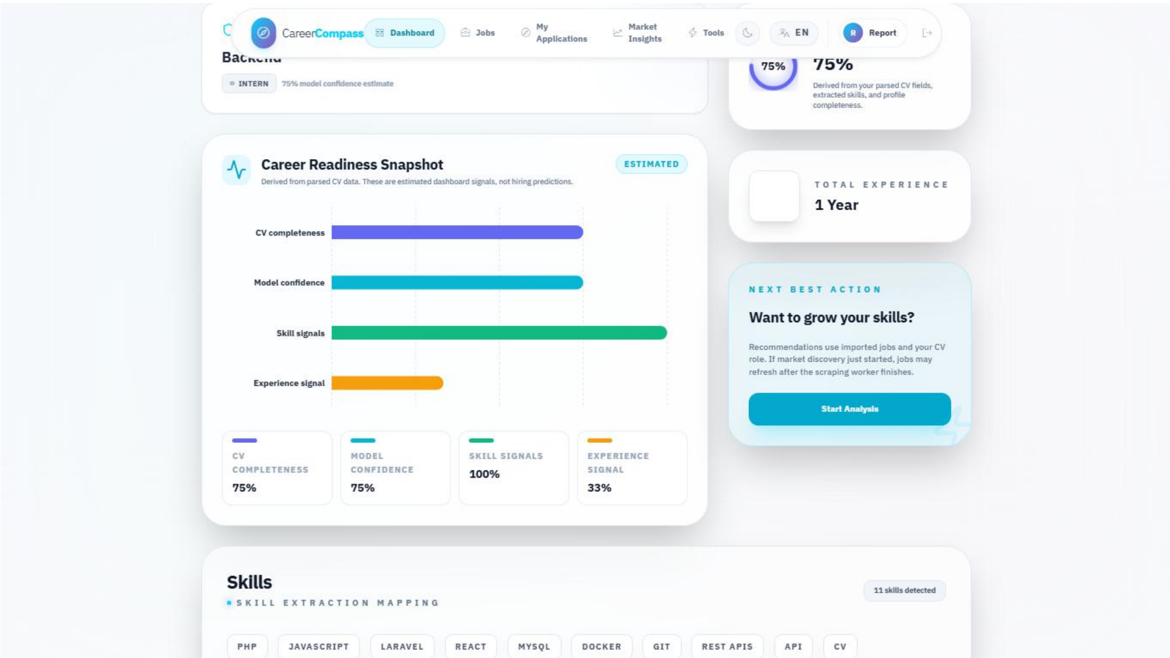


Figure 35. CV upload user interface.

5.5 CV Parsing and Skill Extraction

The CV processing flow sends the file to the AI CV Analyzer, receives parsed data, synchronizes skills, updates profile fields, and stores CV analysis metadata. The implementation handles multiple parsing statuses honestly. If analysis times out, fails, or finds no readable text, the backend returns warnings and preserves existing profile details rather than silently replacing data with low-quality output.

5.5.1 Analyzer Runtime Components

The analyzer is implemented as layered Python code rather than one monolithic function. main.py exposes FastAPI endpoints, CVOrchestrator coordinates extraction, AdvancedNEREngine runs transformer-based named-entity recognition, and supporting engines handle contacts, sections, experience blocks, canonicalization, domain classification, seniority classification, semantic embeddings, and hybrid job matching. Figure 11 summarizes these extraction components.

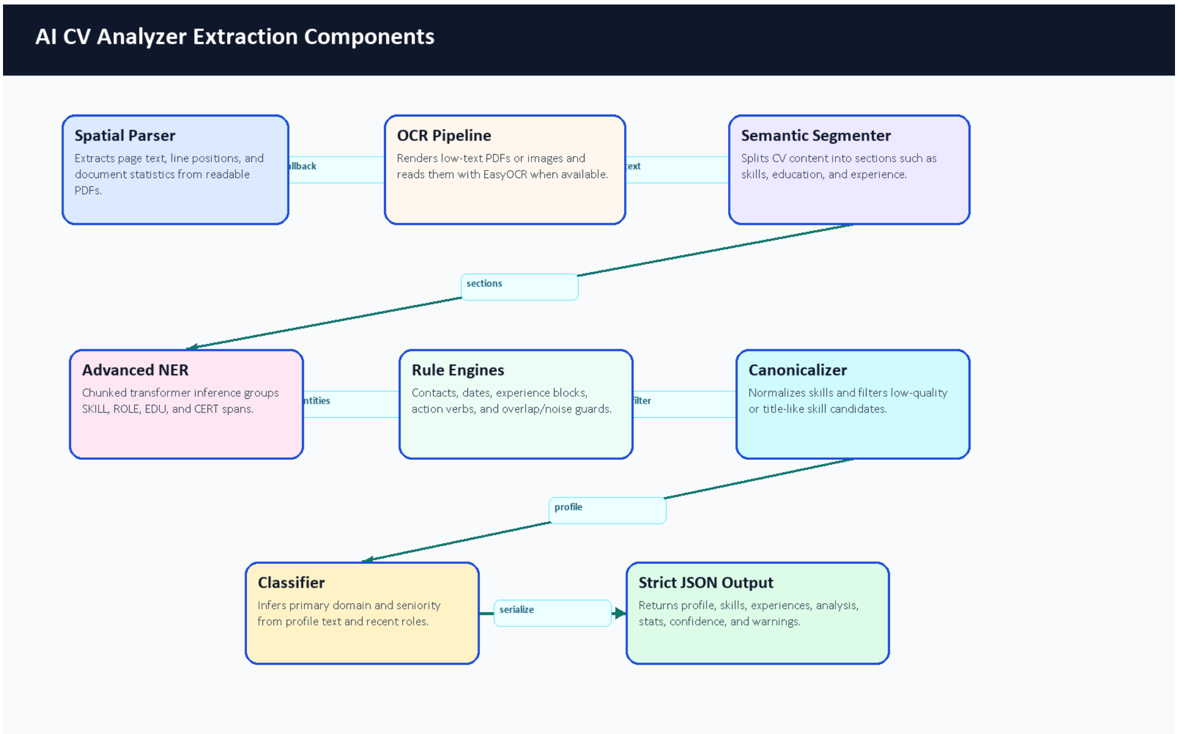


Figure 11. AI CV Analyzer extraction components.

Component	Repository Evidence	Responsibility	Output Used By
FastAPI gateway	ai-cv-analyzer/main.py	Receives /api/parse-cv and /api/hybrid-match requests; handles timeout/error fallbacks.	Laravel CvProcessingService and GapAnalysisService
Laravel CV service	backend-api/app/Services/CvProcessingService.php	Sends uploads to the analyzer, stores private CV objects, persists normalized analysis.	User profile, skills, experiences, recommendations
Spatial/OCR extraction	spatial_parser.py, ocr_pipeline.py	Reads PDF text first and falls back to image/OCR when needed.	Section segmenter and NER pipeline

Component	Repository Evidence	Responsibility	Output Used By
Advanced NER	advanced_ner.py and optional ignored local model folder	Loads the exported local token-classification model when deployed; chunks long CVs and groups entity spans.	Skills, roles, education, certifications
Rule engines	contact, experience, date, noise-filtering helpers	Extract contact details, experience blocks, dates, and remove title-like or noisy skill candidates.	Profile and experience persistence
Canonicalization/classification	Layer 1 and Layer 2 modules	Normalize skills and infer primary domain plus seniority.	Dashboard identity card and matching
Hybrid matching	Layer 3 matching modules	Combines semantic scores, skill text similarity, domain alignment, constraints, and TF-IDF fallback.	Recommendations and gap analysis
Frontend display	Dashboard.jsx, AiInsights.jsx	Shows upload status, confidence-style signals, role/seniority, and extracted skills.	Student-facing CV feedback

Table 6. AI CV Analyzer components.

5.5.2 Model Type and Customization

The NER part is a fine-tuned transformer token-classification design, not a model trained from randomly initialized architecture. The notebook uses bert-base-cased as the base checkpoint and defines a CV-specific BIO label set. The runtime checks for the exported local model at ai-cv-analyzer/models/ner_weights/career_compass_ner_final and uses it when available. That folder is ignored by Git, so committed repository evidence should be described as code and training workflow evidence. Local ignored metadata inspected during this documentation update identified a BERT token-classification configuration, 512-token maximum position setting, 28,996-token vocabulary, 12 transformer layers, 768 hidden size, and a cased tokenizer; the binary model weights were not copied into the report.

The project customization is mainly in the data, labels, orchestration, and post-processing. The team defined CV-specific labels, generated synthetic annotated examples, cleaned entity spans, aligned character spans to tokens, exported a local model, and connected it to a CV-specific extraction pipeline. The pipeline then adds deterministic rules for contacts, dates, experience blocks, noisy skill rejection, canonical skill names, domain/seniority inference, and safer fallback statuses.

Label	BIO Forms	Meaning in Training Data	Runtime Note
O	O	Token outside a labeled entity.	Used by the model to ignore ordinary text.
SKILL	B-SKILL, I-SKILL	Technical skill such as Laravel, React, Docker, SQL, or PyTorch.	Returned as skills after filtering and canonicalization.
ROLE	B-ROLE, I-ROLE	Job title or role such as Backend Developer or Data Analyst.	Used for predicted role and role evidence.

Label	BIO Forms	Meaning in Training Data	Runtime Note
EDU	B-EDU, I-EDU	Degree, major, faculty, university, or education phrase.	Used as education/profile evidence.
CERT	B-CERT, I-CERT	Certification such as AWS Cloud Practitioner.	Used as certification evidence.
SOFT	B-SOFT, I-SOFT	Soft skill phrase such as leadership or communication.	Present in training configuration; the runtime NER grouping mainly returns SKILL, ROLE, EDU, and CERT.

Table 7. NER entity label schema.

5.5.3 Synthetic Dataset Generation

The helper material under D:/Graduation/model-analys-helper was reviewed as documentation support. The top-level helper folders found were docs, layer1, layer2, and layer3. No raw datasets, screenshots, or secrets were copied into the graduation-book folder. The important training workflow is already represented in the repository by ai-cv-analyzer/training/generate_tech_dataset.py, clean_dataset.py, and train_ner.ipynb.

The dataset generator is designed to call the Gemini API through API keys stored outside source control. It asks for synthetic technical CV snippets across backend, frontend, DevOps, mobile, AI/data, cybersecurity, cloud, QA, and networking contexts. It intentionally includes positive labeled examples and negative decoy examples so that the model learns both what to tag and what to ignore. Because this uses external generation and keys, it was inspected rather than executed for this documentation update.

Step	Repository Evidence	Description	Documentation Decision
API-key loading	generate_tech_dataset.py	Reads GEMINI_API_KEYS from .env and rotates keys/models during generation.	Do not commit keys; no secrets were found in helper docs.
Synthetic sample generation	Generator system prompt	Requests batches with technical domains, noise, varied CV formats, and labeled entities.	Documented as synthetic data, not real student CV data.
Negative decoys	Generator distribution comments	Includes examples with no entities to reduce false positives.	Preserved in the description because it affects model behavior.
Cleaning	clean_dataset.py	Normalizes whitespace, deduplicates exact text, validates entity text, filters long skill spans.	Documented as a data-quality gate before training.
Output	Training docs and scripts	Expected cleaned JSON/JSONL input for notebook training.	Dataset files were not present in the repository, so metrics cannot be reproduced from repo files alone.

Table 8. Synthetic dataset generation workflow.

5.5.4 Training Notebook Workflow

The training notebook is structured for Google Colab rather than local execution. It installs model-training dependencies, loads cleaned JSON data, defines labels, tokenizes examples, aligns entity spans to token labels, initializes AutoModelForTokenClassification from bert-base-cased, trains with Hugging Face Trainer, evaluates with sequence-labeling metrics, then exports career_compass_ner_final for deployment [31], [32], [33], [36].

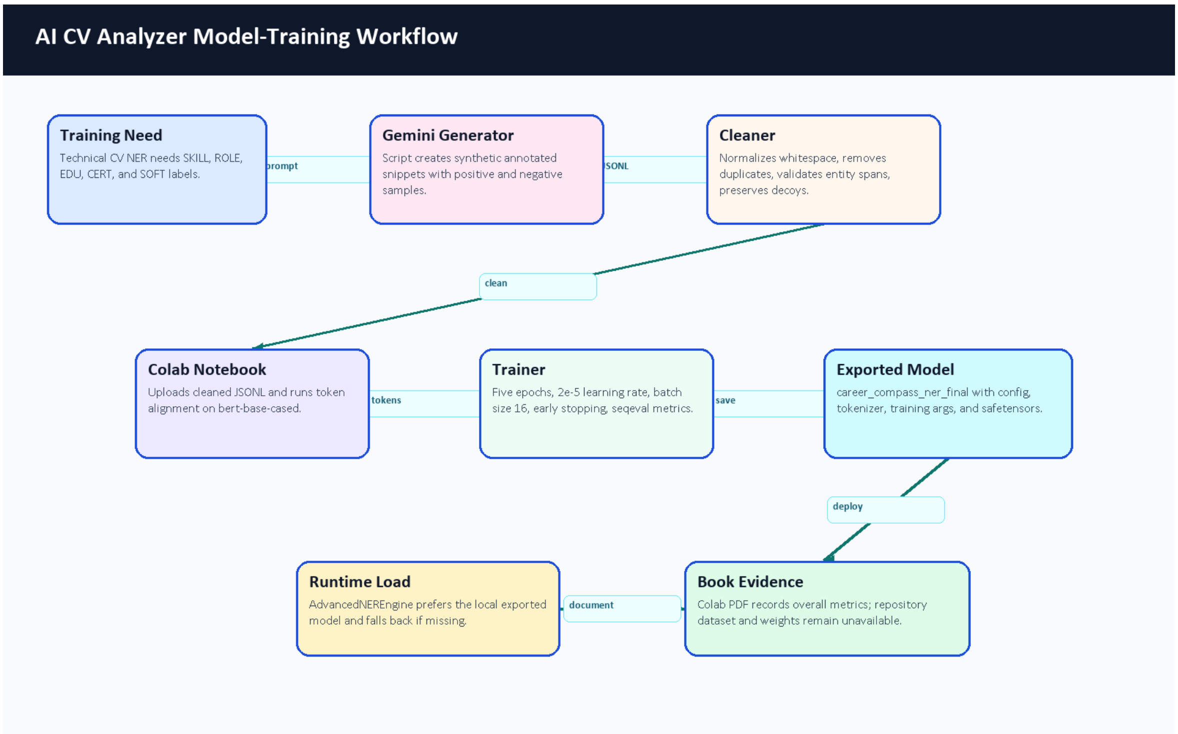


Figure 10. AI CV Analyzer model-training workflow.

Setting	Value Found in Notebook or Docs	Purpose	Evidence Limitation
Base checkpoint	bert-base-cased	Provides pretrained language representations for token classification.	Confirmed by training notebook and Colab PDF.
Labels	O plus B/I for SKILL, ROLE, EDU, CERT, SOFT	Encodes CV entity spans using BIO tagging.	Label map is reproducible from notebook/config.
Split	90 percent train, 10 percent test, seed 42	Creates a repeatable train/evaluation split.	Colab PDF records 41,319 train rows and 4,592 test rows; dataset content is not committed.
Max length	512 tokens	Fits BERT token-classification input limits.	Long runtime CVs are handled through chunking separately.
Epochs and learning rate	5 epochs, 2e-5	Standard fine-tuning style schedule for a small NER task.	No final epoch table was available.
Batch size	16 train/eval	Balances GPU memory and throughput on Colab/T4-style runtime.	Visible in the exported Colab PDF.
Metrics	precision, recall, F1, accuracy via sequence labeling	Evaluates entity extraction quality when labels are available.	Colab PDF records overall metrics; per-label report is not visible.
Export	career_compass_ner_final zip/model folder	Produces the deployable local model artifact.	Export success is visible in the PDF, but model weights are ignored by Git.

Table 9. Model training configuration.

5.5.5 Layer 1: CV Understanding Pipeline

Layer 1 is responsible for turning a noisy CV file into a structured candidate profile. The runtime begins at main.py, which accepts the uploaded file, chooses PDF or image handling, applies timeout/error wrappers, and delegates the actual extraction to CVOrchestrator. The orchestrator first tries ordered PDF text extraction. The spatial parser reads words from PDF pages, groups words into rows using an adaptive tolerance, splits row segments when large x-axis gaps imply columns, removes (cid:...) artifacts, and falls back to plain PDF extraction when the spatial output loses too much text.

If the file has little or no readable text, the OCR path renders PDF pages to images and uses EasyOCR after grayscale/blur preprocessing. After text recovery, the semantic segmenter finds CV sections, contact extraction parses email/phone/location fields, the NER engine extracts entity candidates, experience logic estimates date ranges and career signals, and the canonicalizer normalizes skills before the result is validated through strict Pydantic schema classes.

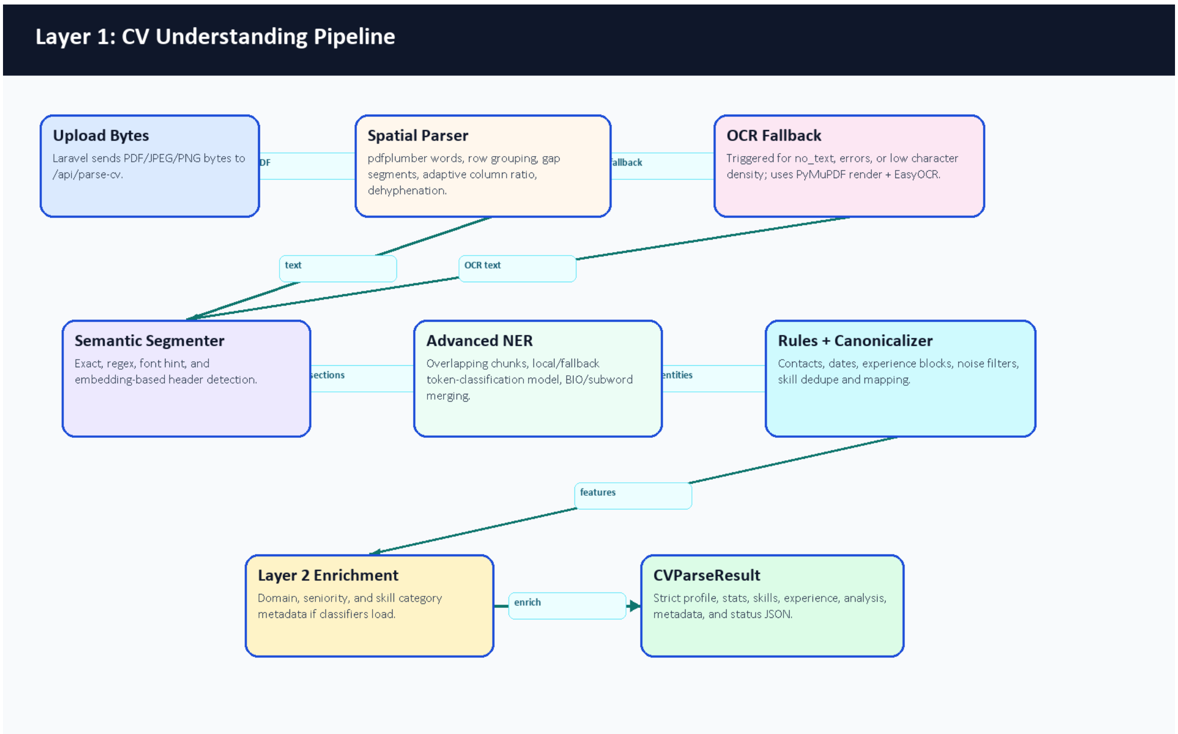


Figure 12. Layer 1 CV understanding pipeline.

Layer 1 Component	Main Files	Important Behavior	Risk or Fallback
API gateway	main.py	/api/parse-cv, /api/hybrid-match, timeout handling, health and metrics endpoints.	Timeout results are returned as explicit status dictionaries.
Spatial parser	core/layer1_understanding/spatial_parser.py	Word extraction, row grouping, column ordering, dehyphenation, plain-text fallback.	Falls back when spatial output is too weak.
OCR fallback	core/layer1_understanding/ocr_pipeline.py, orchestrator OCR helpers	Renders image-like PDFs, preprocesses pages, and extracts text when normal PDF parsing fails.	Triggered for short/no-text inputs.

Layer 1 Component	Main Files	Important Behavior	Risk or Fallback
Section segmenter	core/layer1_understanding/section_segmenter.py	Header detection from patterns and optional semantic header matching.	Missing headers fall back to profile-style grouping.
Contact and experience engines	contact_extractor.py, experience_engine.py	Extract emails/phones/location, date ranges, total years, skill durations, gaps, overlaps, and action-verb strength.	Ambiguous dates are treated conservatively.
NER and canonicalization	advanced_ner.py, canonicalizer.py	Extracts skills/roles/education/certifications, filters noise, deduplicates, and maps skills to canonical names.	Fallback model path exists when local deployment artifact is missing.
Output schema	schema.py	Strict typed response for profile, skills, experience, confidence, stats, and parsing status.	Invalid shapes are prevented before backend persistence.

Table 10. Layer 1 component details.

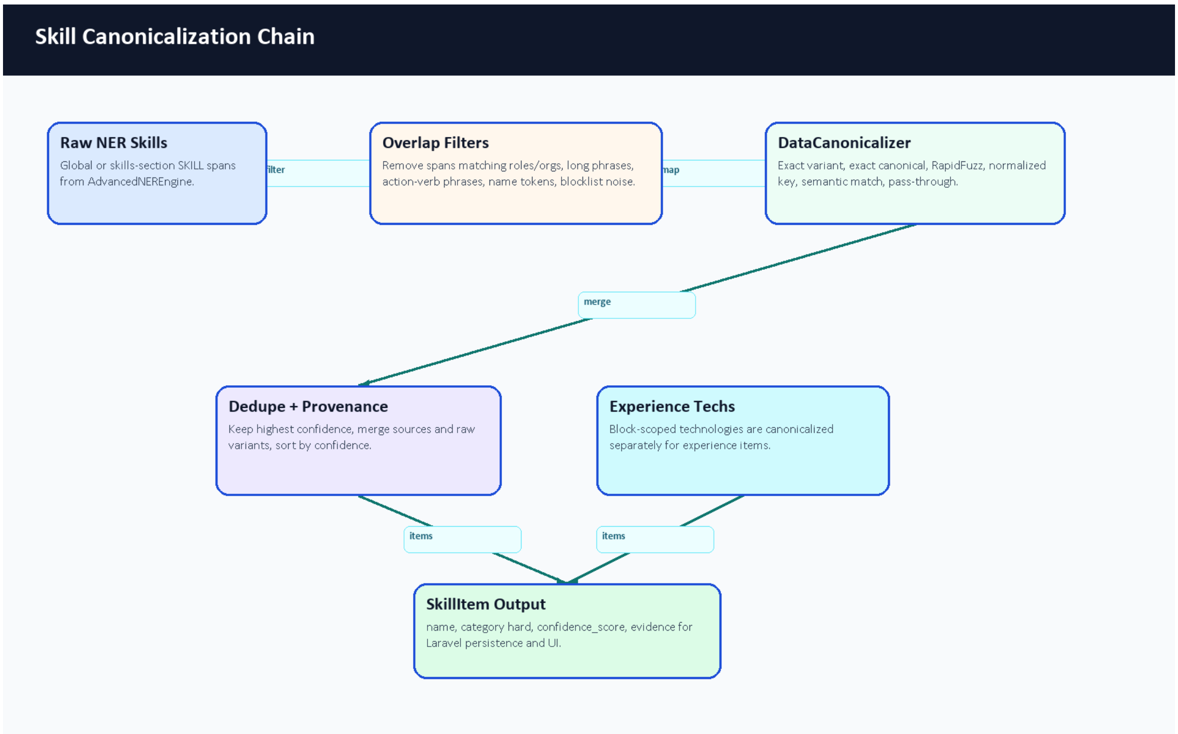


Figure 17. Skill canonicalization chain.

5.5.6 Layer 2: Classification Engine

Layer 2 enriches the extracted CV with a domain and seniority interpretation. The classification orchestrator reads the Layer 1 result, then combines title, experience, summary, skill categories, and taxonomy descriptions. DomainEngine compares CV context against taxonomy descriptions using semantic embeddings when available. SkillEngine separates hard, soft, and management-oriented skills using taxonomy rules. SeniorityEngine combines years of experience, title keywords, semantic title/summary hints, and action-verb strength.

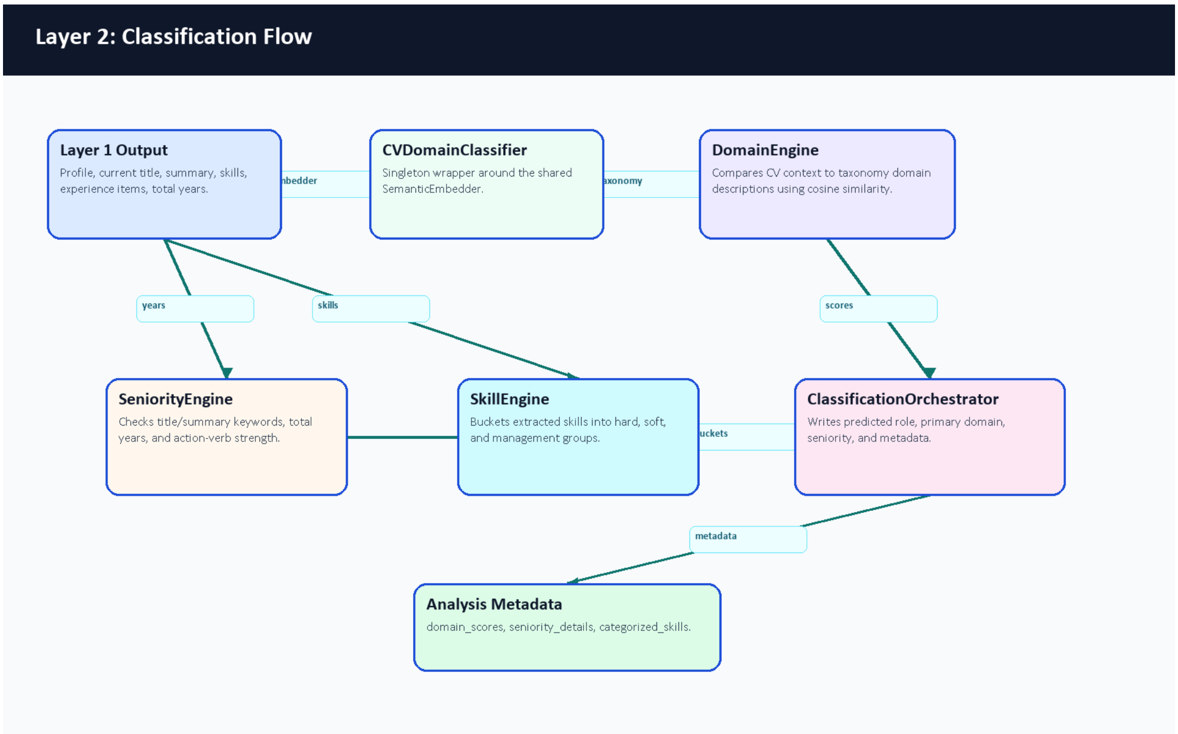


Figure 13. Layer 2 classification flow.

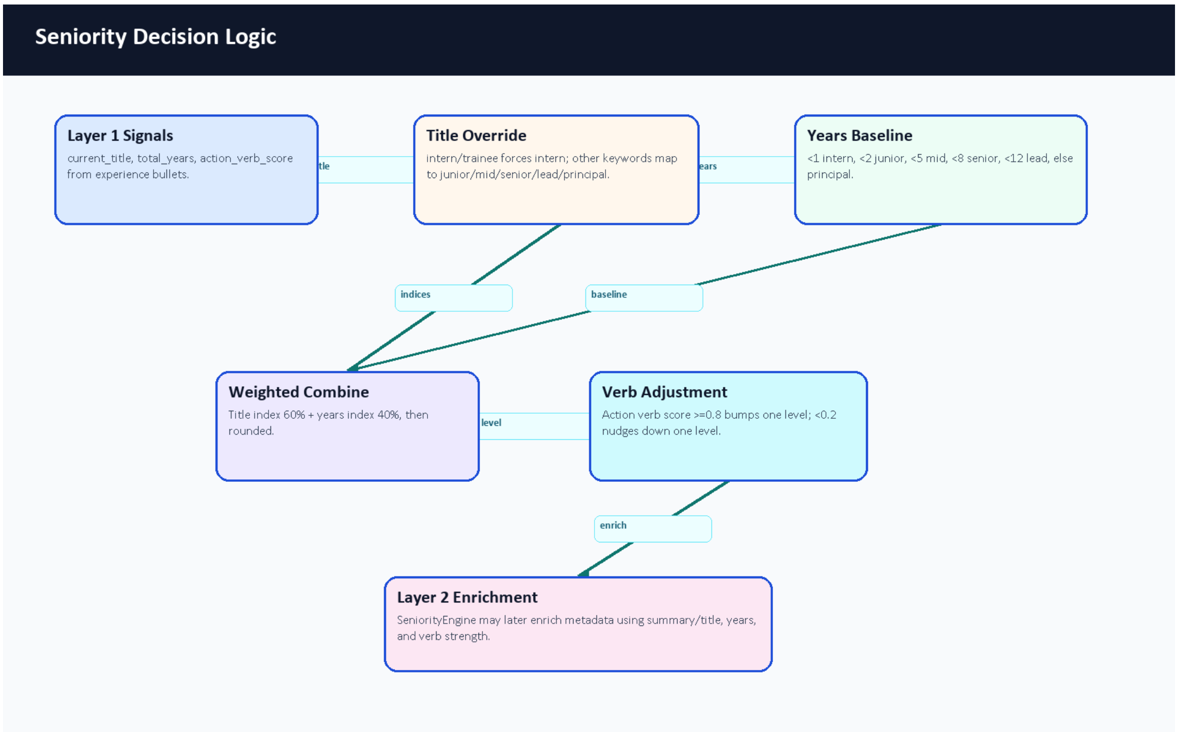


Figure 16. Seniority decision logic.

Layer 2 Component	Main Files	Input	Output
Classification orchestrator	core/layer2_classification/orchestrator.py	Parsed CV profile, skills, experience, and summary.	Adds primary domain, seniority level, skill categories, and confidence-style signals.
Domain engine	core/layer2_classification/domain_engine.py, data/taxonomy.json	Title, summary, and first experience titles.	Primary technical domain selected from taxonomy descriptions.

Layer 2 Component	Main Files	Input	Output
Seniority engine	core/layer2_classification/seniority_engine.py	Experience years, title, summary, and action verbs.	Intern, Junior, Mid-Level, Senior, or Lead / Manager estimate.
Skill engine	core/layer2_classification/skill_engine.py	Canonical skill names and taxonomy terms.	Hard, soft, and management skill buckets.
Taxonomy loader	core/layer2_classification/utils.py	JSON taxonomy file.	Shared configuration for domain and skill classification.

Table 11. Layer 2 classification engine details.

5.5.7 Layer 3: Matching Engine

Layer 3 compares a candidate profile with a job description. JobDescriptionEngine parses job text into seniority, required years, mandatory skills, bonus skills, domain, and summary. IntelligentMatcher calculates semantic similarity, skill-text similarity, and domain alignment using adaptive weights that change by seniority level. ConstraintValidator subtracts penalties for missing mandatory skills, experience shortfalls, and seniority mismatch. FitAnalysisGenerator turns the numeric result into strengths, gaps, red flags, and a verdict.

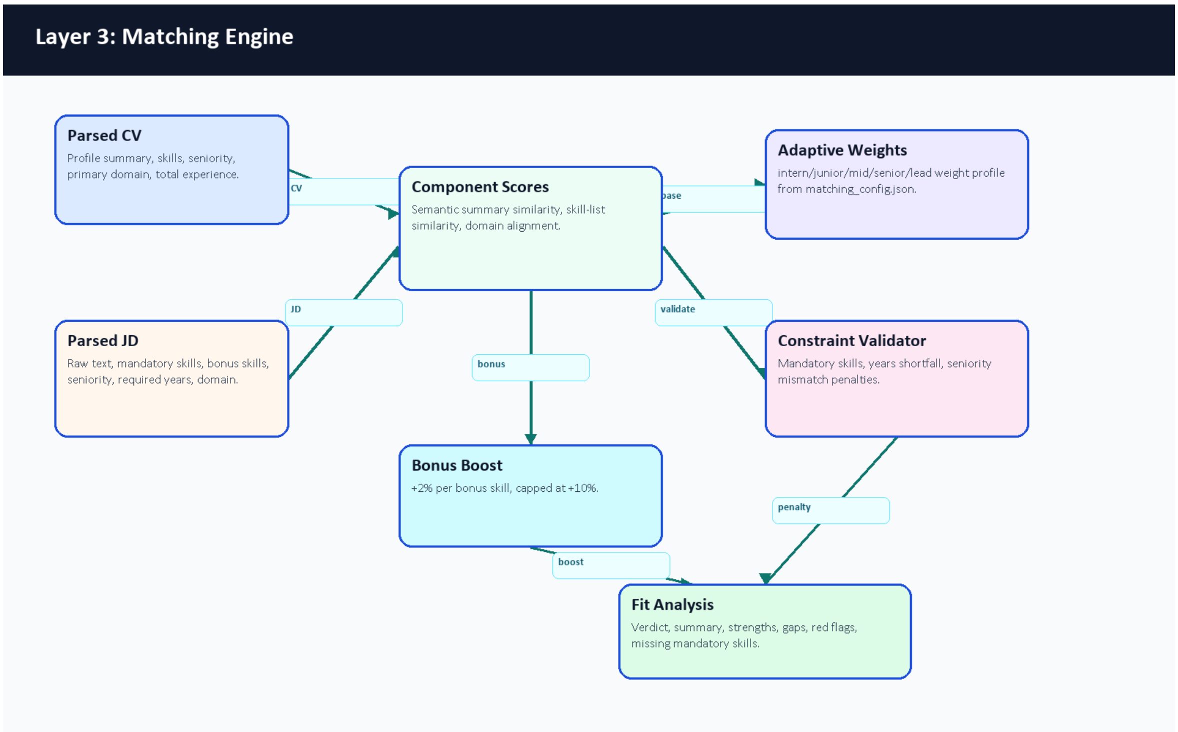


Figure 14. Layer 3 matching engine.

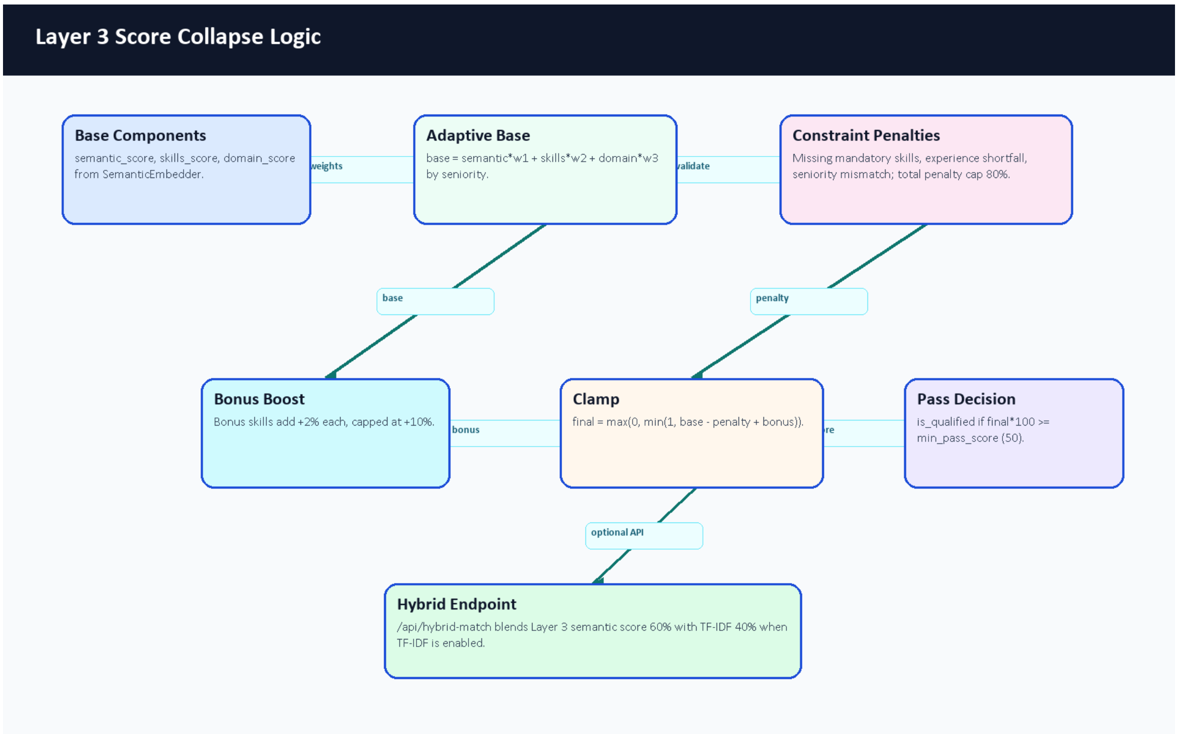


Figure 18. Layer 3 score collapse logic.

Layer 3 Component	Main Files	Scoring Contribution	Explanation Contribution
JD parser	job_description_engine.py	Extracts requirements that become matching inputs.	Explains what the system understood from the job post.
Semantic embedder	embedder.py	Summary similarity and domain-similarity fallback.	Captures meaning beyond exact keyword overlap when dependencies are available.
Intelligent matcher	similarity.py, matching_config.json	Combines semantic, skill, and domain scores using seniority-aware weights.	Produces score breakdown and qualification flag.
Constraint validator	constraint_validator.py	Applies capped penalties for mandatory gaps, experience gaps, and seniority mismatch.	Lists missing mandatory skills and mismatch reasons.
Fit analysis generator	fit_analysis_generator.py	Converts score ranges into verdict categories.	Generates strengths, gaps, and red flags for the UI.
Ranking orchestrator	ranking_orchestrator.py	Applies matcher repeatedly across candidates/jobs.	Sorts candidates or opportunities by explainable fit.

Table 12. Layer 3 matching engine details.

5.5.8 Semantic Embedding, Caching, and TF-IDF Fallback

The semantic embedder uses the configured sentence-transformer model name, with all-MiniLM-L6-v2 as the default. The class is implemented as a singleton so that model loading is not repeated for every request. It keeps a bounded embedding cache of 2,000 entries and can optionally apply dynamic quantization through EMBEDDER_QUANTIZE. If the embedding stack is unavailable, it returns zero vectors and the calling code falls back to transparent lower-confidence behavior instead of pretending semantic comparison succeeded.

The FastAPI /api/hybrid-match endpoint also combines semantic-style matching with the pure Python TF-IDF matcher when the TF-IDF module is importable. In that endpoint, semantic/adaptive scoring contributes 60 percent and TF-IDF contributes 40 percent. This gives the demo a useful exact-keyword safety check for skills such as Laravel, Docker, MySQL, React, AWS, or Kubernetes.

Method	Repository Evidence	Strength	Limitation
Sentence embeddings	core/layer3_matching/embedder.py	Captures meaning even when words differ.	Requires sentence-transformer dependencies and model loading.
Domain similarity	core/layer3_matching/similarity.py	Allows related domains to receive partial credit.	Low domain similarity is cut off below the configured threshold.
Skill text similarity	core/layer3_matching/similarity.py	Compares extracted CV skills against mandatory and bonus job skills.	Quality depends on upstream extraction and canonicalization.
TF-IDF fallback	core/layer3_matching/tfidf.py, /api/hybrid-match	Lightweight deterministic keyword overlap.	Does not understand synonyms or phrase meaning.
Constraint penalties	constraint_validator.py	Prevents high scores when hard requirements are missing.	Penalty weights need larger evaluation data for calibration.

Table 13. Semantic embedding and TF-IDF fallback comparison.

5.5.9 NER Token Processing and BIO Tagging

The training notebook uses character-span annotations and converts them into token labels. Each text sample is tokenized with offsets; special tokens are assigned -100 so they are ignored by the loss; tokens whose offsets fall inside an entity span are assigned B- or I- labels. The cased BERT tokenizer is appropriate for CV text because names, certificates, role titles, and technology names often rely on capitalization. At runtime, long CV text is chunked with overlap, model predictions are merged, subword prefixes are cleaned, and duplicate/noisy entities are filtered before canonicalization.

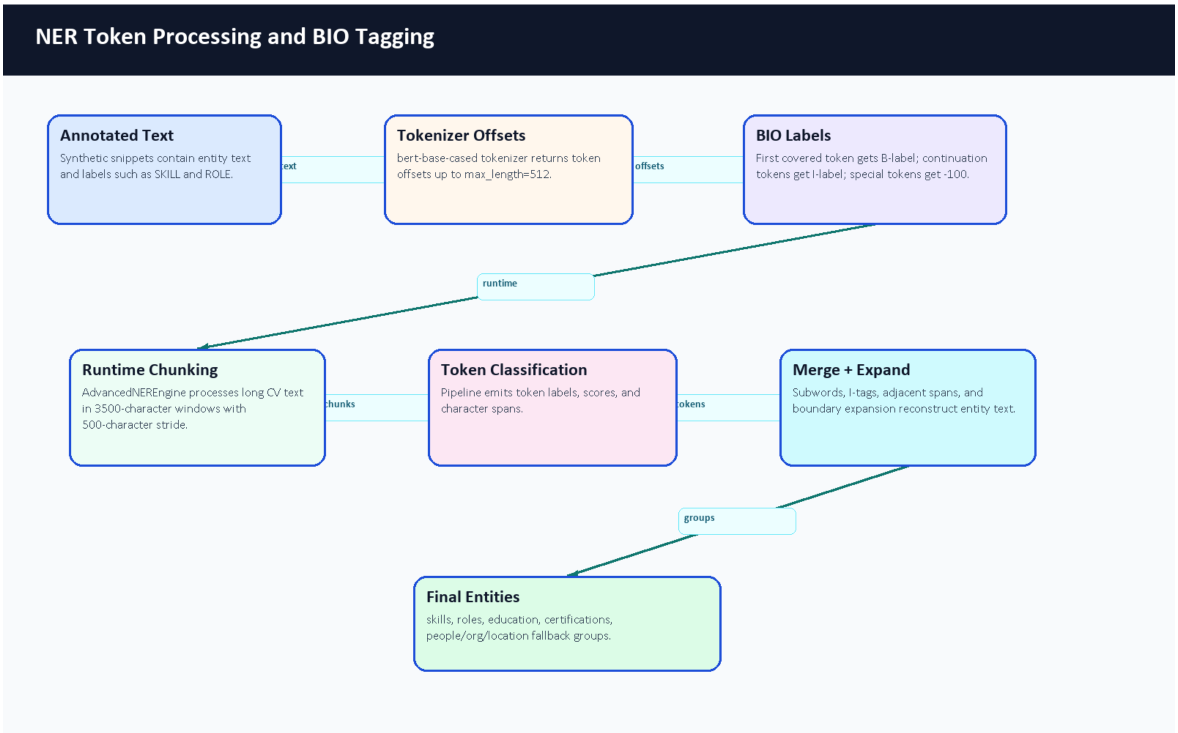


Figure 15. NER token processing and BIO tagging.

Simplified Text Token	BIO Label	Why It Matters
Experienced	O	Ordinary descriptive word, not extracted as an entity.
Backend	B-ROLE	Start of a role phrase.
Developer	I-ROLE	Continuation of the role phrase.
with	O	Connector word.
Laravel	B-SKILL	Skill entity.
Docker	B-SKILL	Skill entity.
MySQL	B-SKILL	Skill entity.

Table 14. Simplified BIO tagging example.

5.5.10 AI CV Analyzer Source Code Inventory

The AI CV Analyzer was audited as source code, not only as a running service. The inventory below summarizes the relevant repository areas. Full support notes are stored under docs/graduation-book/model-analysis/.

Area	Representative Files	Main Responsibility
FastAPI service	main.py, Dockerfile, requirements.txt, .env.example	Service startup, endpoints, container runtime, and documented configuration variables.
Layer 1 understanding	core/layer1_understanding/*.py, core/layer1_understanding/data/config.json	PDF/image text extraction, OCR fallback, sectioning, NER, contact extraction, experience analysis, and canonicalization.
Layer 2 classification	core/layer2_classification/*.py, core/layer2_classification/data/taxonomy.json	Domain, seniority, and skill-category enrichment.
Layer 3 matching	core/layer3_matching/*.py, core/layer3_matching/matching_config.json	Job parsing, semantic/TF-IDF matching, constraints, fit explanation, and ranking.
Training workflow	training/generate_tech_dataset.py, clean_dataset.py, train_ner.ipynb	Synthetic labeled dataset generation, cleaning, token alignment, Trainer setup, metrics code, and export.
Diagnostics and tests	scripts/verify_phase*.py, tests/test_service_api.py, tests/trace_cv.py, manual tests	Phase checks, service API tests with fakes, tracing, and manual validation helpers.
Documentation	Layer README and EXPLAIN files	Developer explanations for analyzer layers and matching logic.
Ignored deployment assets	.env, models/ner_weights/...	Local secrets and model weights are intentionally ignored; only safe metadata was inspected locally.

Table 15. AI CV Analyzer source inventory summary.

Algorithm or Workflow	Primary File(s)	Notes
Parse-CV request routing	main.py	File validation, image/PDF branching, timeout and error wrappers.

Algorithm or Workflow	Primary File(s)	Notes
Ordered PDF extraction	core/layer1_understanding/spatial_parser.py	Adaptive row/column ordering and dehyphenation.
OCR fallback	core/layer1_understanding/ocr_pipeline.py, CVOrchestrator	Used when PDF text is absent or too short.
NER extraction	core/layer1_understanding/advanced_ner.py	Singleton model loading, chunked inference, entity merging, and cleanup.
Skill normalization	core/layer1_understanding/canonicalizer.py	Exact, fuzzy, and semantic mapping toward canonical names.
Experience calculation	core/layer1_understanding/experience_engine.py	Date ranges, total years, skill durations, gaps, overlaps, and action verbs.
Seniority inference	core/layer1_understanding/orchestrator.py, core/layer2_classification/seniority_engine.py	Combines title keywords, years, semantic hints, and action verbs.
Domain classification	core/layer2_classification/domain_engine.py	Embedding comparison against taxonomy descriptions.
Job parsing	core/layer3_matching/job_description_engine.py	Seniority, years, mandatory/bonus skills, domain, and summary extraction.
Hybrid match scoring	core/layer3_matching/similarity.py, core/layer3_matching/constraint_validator.py	Weighted semantic/skill/domain score minus constraint penalties plus bonus boost.
TF-IDF fallback	core/layer3_matching/tfidf.py, main.py	Pure Python sparse cosine score used in hybrid endpoint.
Fit explanation	core/layer3_matching/fit_analysis_generator.py	Verdict, strengths, gaps, and red flags.
NER training	training/train_ner.ipynb	BERT base checkpoint, BIO labels, token alignment, Trainer, seqeval metrics.
Synthetic data generation	training/generate_tech_dataset.py	Gemini-based generation with key rotation and negative decoys.

Table 16. Algorithm-to-file mapping.

5.6 Profile and Skills Management

The profile page reads normalized user data, profile fields, experiences, skills, and CV analysis. The system distinguishes user fields, profile fields, extracted skills, predicted role, seniority, and completeness score. Skill synchronization is handled through backend services rather than only frontend state.

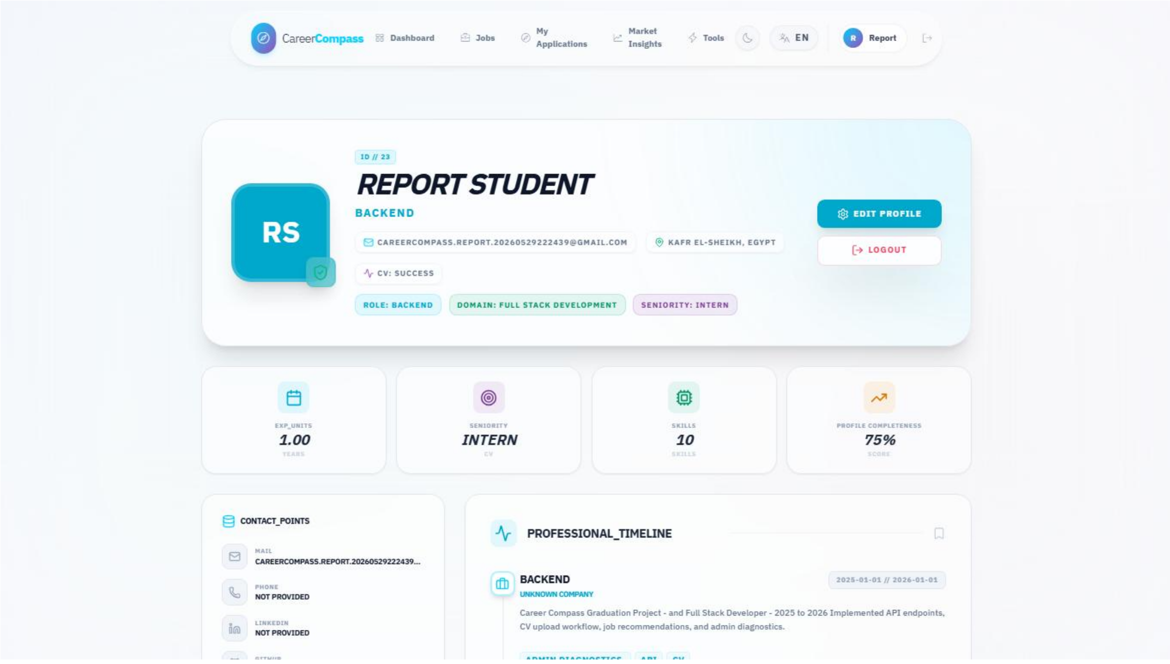


Figure 37. Extracted profile and skills page.

5.7 Job Data Model

Jobs are represented in the backend through job posting models and migrations. Fields include title, company, description/requirements, URL, source, and metadata. The seeders and import controllers enforce quality gates and uniqueness rules, including a title/company uniqueness constraint that prevented duplicate seed insertion during validation.

5.8 AI Job Miner and Scraping Sources

The job miner exposes a FastAPI service and imports jobs using configured sources. The backend protects scraper import routes with an internal service token. Admin pages expose source diagnostics, source status, testing, and target role management. The project differentiates demo/local sources, API sources, and HTML/scraping sources instead of claiming complete market coverage.

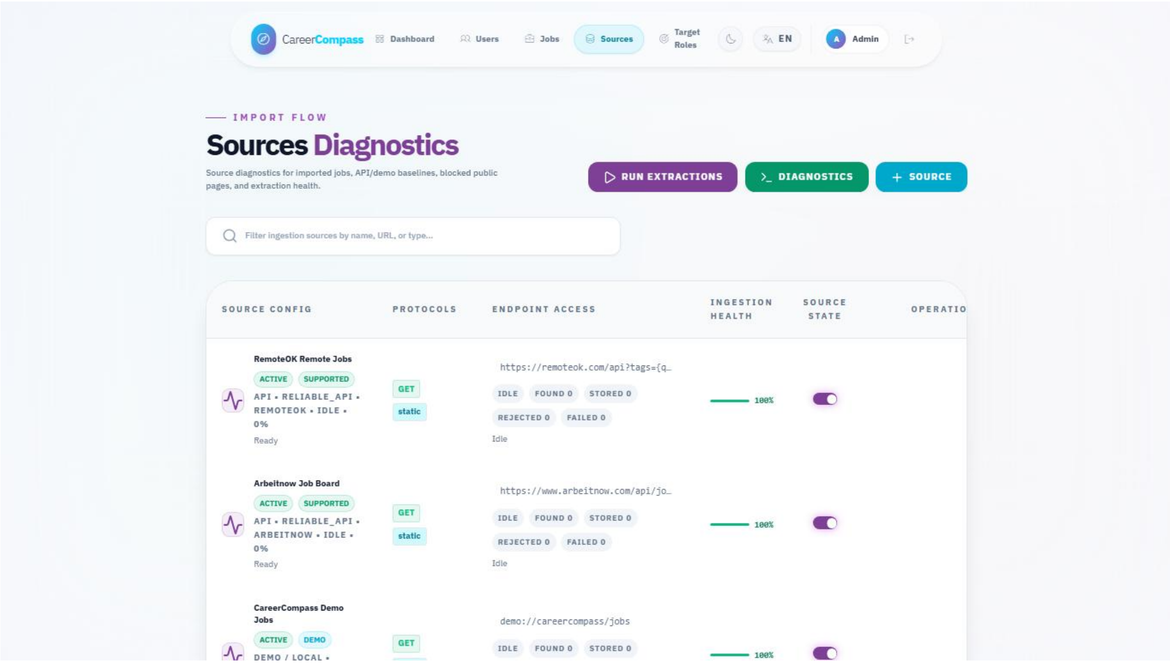


Figure 46. Admin sources diagnostics page.

5.9 Job Recommendations

The jobs page requests recommended jobs when no manual search query is active. Recommendations are based on CV/profile context when available. Matching combines normalized database data with semantic and TF-IDF-style comparison where available. TF-IDF represents text using term frequency and inverse document frequency weighting [19], while cosine similarity compares vector orientation [20].

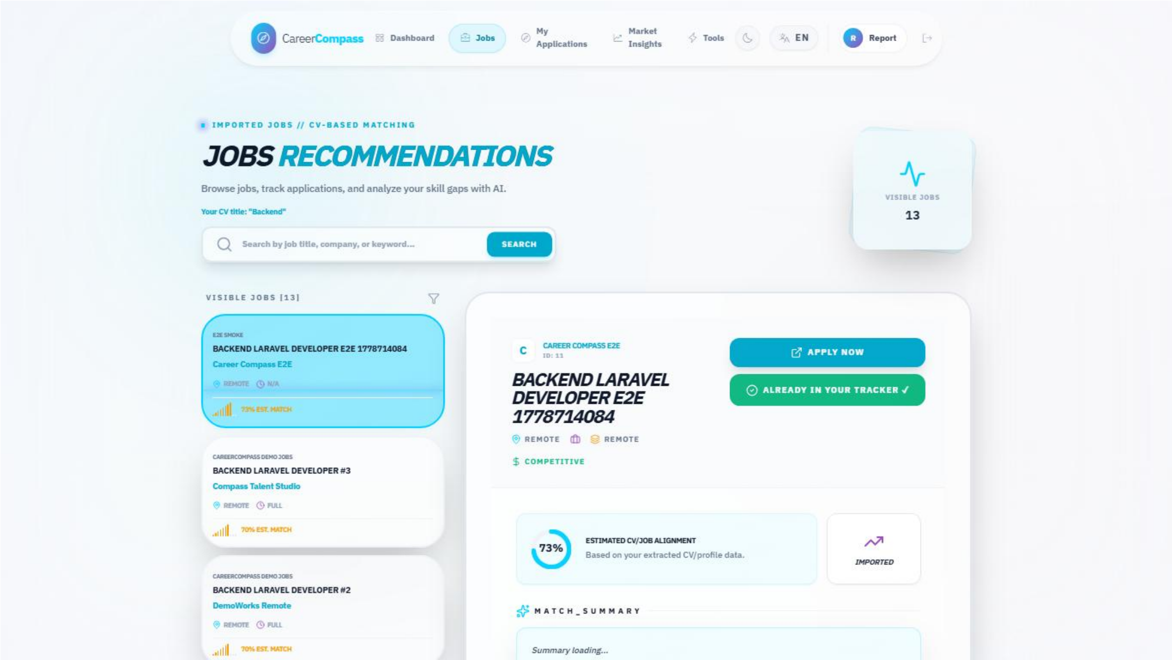


Figure 38. Jobs recommendations page.

5.10 Gap Analysis

Gap analysis compares a selected job or target role against the user's profile and extracted skills. It returns matched skills, critical/missing skills, recommendations, match percentage, and roadmap-like guidance. The frontend displays these outputs in an explainable layout rather than a single opaque score.

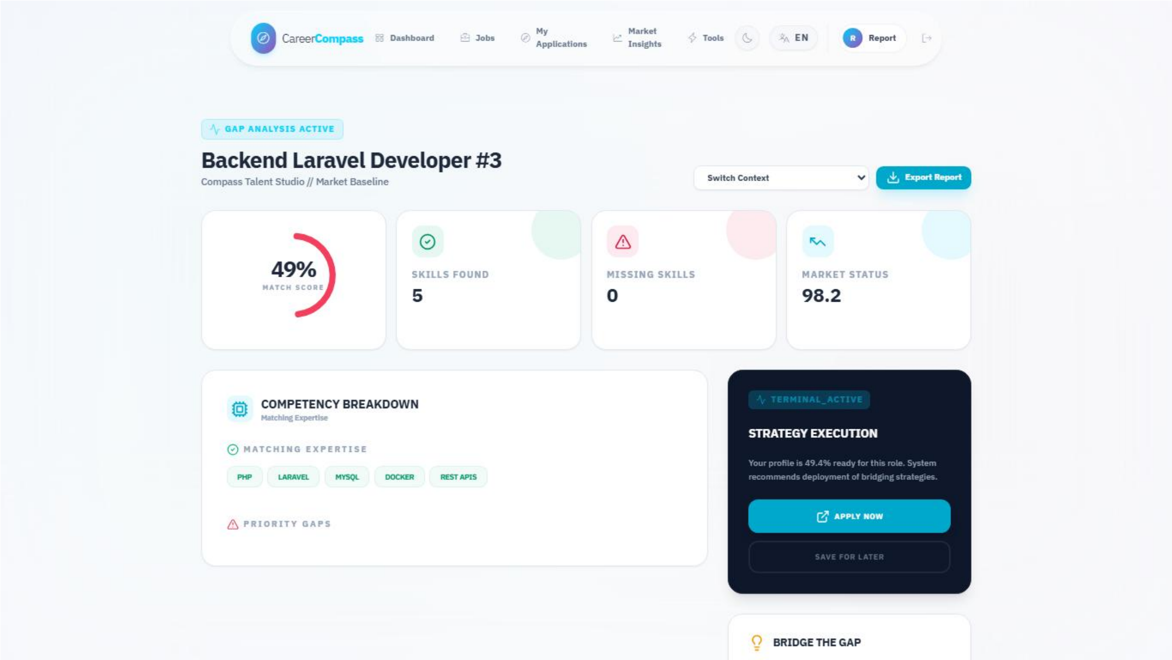


Figure 40. Gap analysis page.

5.11 Application Tracker

The application tracker is implemented through ApplicationController, ApplicationTrackerService, and frontend/src/pages/user/Applications.jsx. Students can save a job, update status, view counts, and delete tracked items. The backend validates job existence and allowed statuses.

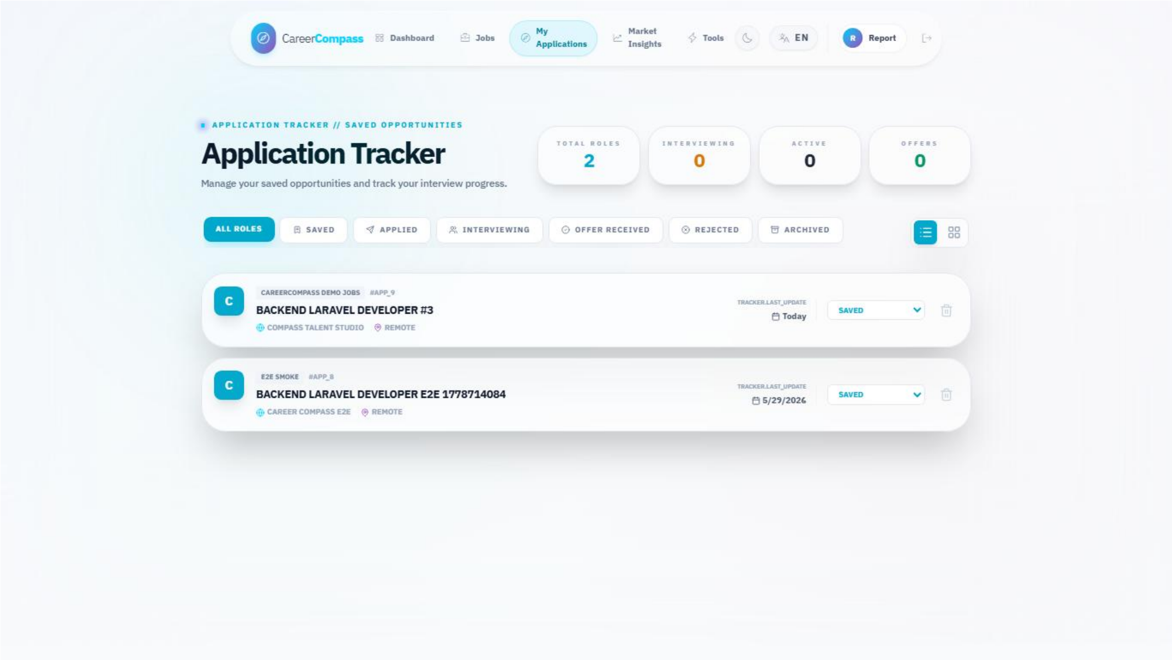


Figure 41. Applications tracker page.

5.12 Admin Dashboard

The admin dashboard summarizes users, imported jobs, active sources, target roles, health status, and scraping batch progress. It is protected by admin middleware and uses admin API routes.

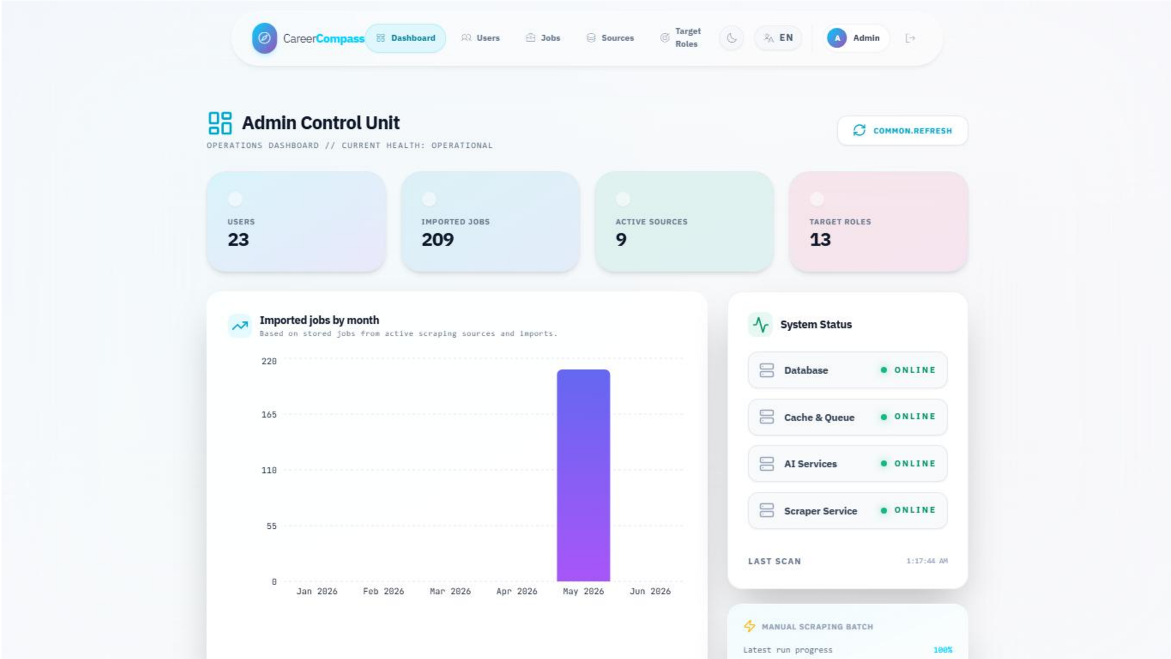


Figure 44. Admin dashboard.

5.13 Admin Source Diagnostics

The source diagnostics page lists configured scraping sources, supports source testing, and displays quality and scraping status information. The target roles page manages role names used by scraping and market discovery.

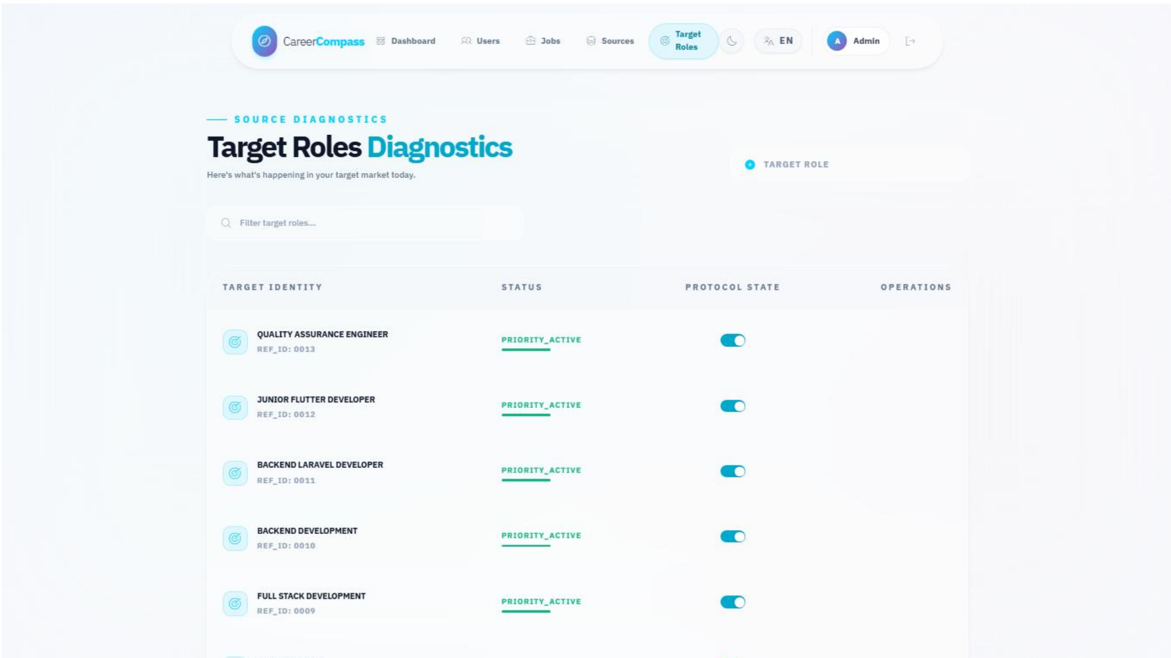


Figure 47. Admin target roles page.

5.14 System Health and Monitoring

Health endpoints include live and readiness checks. The system status page presents service state to users, while admin health data supports operational monitoring. Metrics are available for Prometheus and dashboards are available through Grafana.

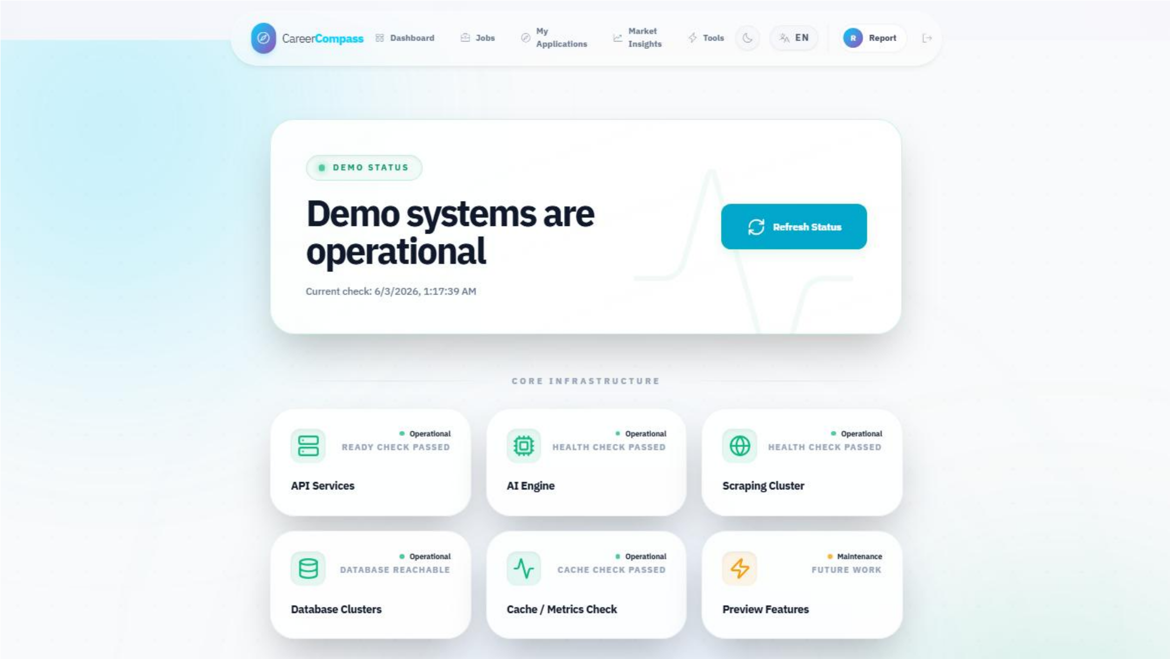


Figure 43. System status page.

5.15 Error Handling and Fallbacks

The code includes explicit handling for CV processing failures, AI gateway connection failures, validation errors, missing user data, empty job data, and unavailable services. The job recommendation and gap analysis code includes fallback behavior when AI services are not available.

5.16 Internationalization and UI Preview Modules

The frontend contains English and Arabic locale files. Several menu items intentionally appear as preview modules so the interface can show the intended product direction without presenting those areas as completed core deliverables. The completed graduation/demo core remains CV upload, parsed profile and skills, recommendations, gap analysis, application tracking, admin diagnostics, and system status.

Module	Current Status	Reason Included	Future Work
CV Builder	Preview placeholder	Shows where guided resume authoring would fit beside CV upload.	Add editable sections, export templates, and validation tests.
Mock Interview	Preview placeholder	Demonstrates a possible practice workflow after gap analysis.	Add question banks, scoring rubrics, and consent-aware recording rules.
Learning Paths	Preview placeholder	Connects missing skills to future study plans.	Integrate curated resources and progress tracking.
Career Planner	Preview placeholder	Shows longer-term roadmap direction.	Add milestone planning and advisor review.
Mentorship	Preview placeholder	Shows possible human-support extension.	Add mentor profiles, matching rules, and moderation.
Tools Hub	Preview screen	Groups preview tools in one visible place.	Replace placeholders with independently tested modules.
Market Intelligence	Supporting/preview view	Helps explain job-market context from imported jobs.	Add stronger statistics, freshness indicators, and source quality labels.

Preview module clarification table.

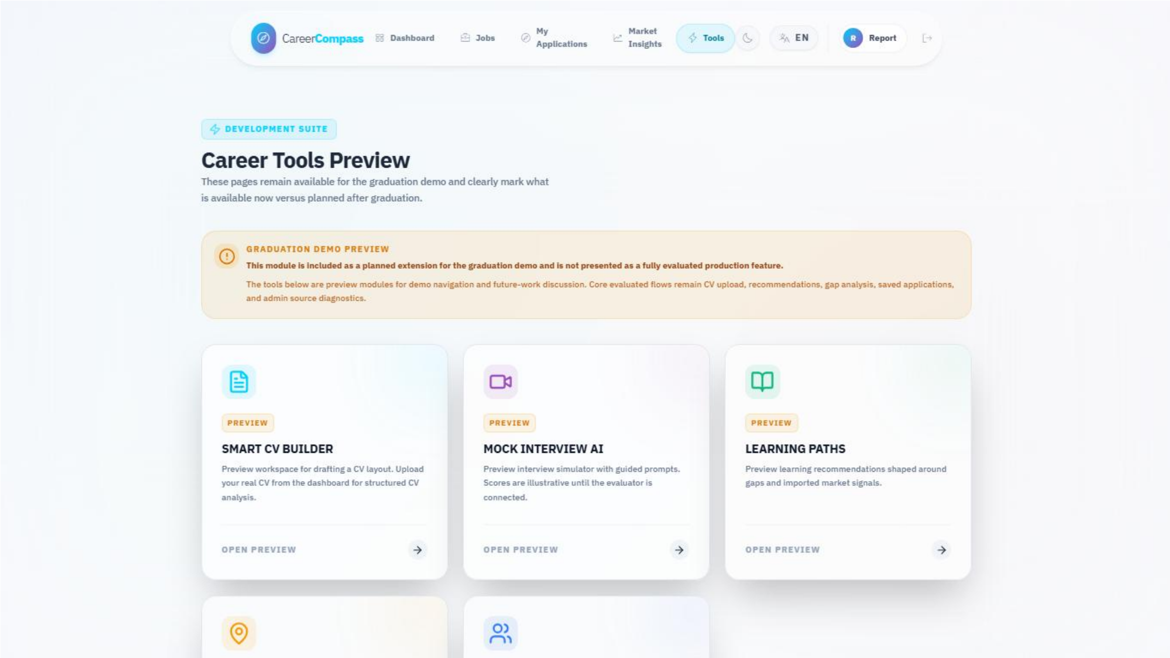
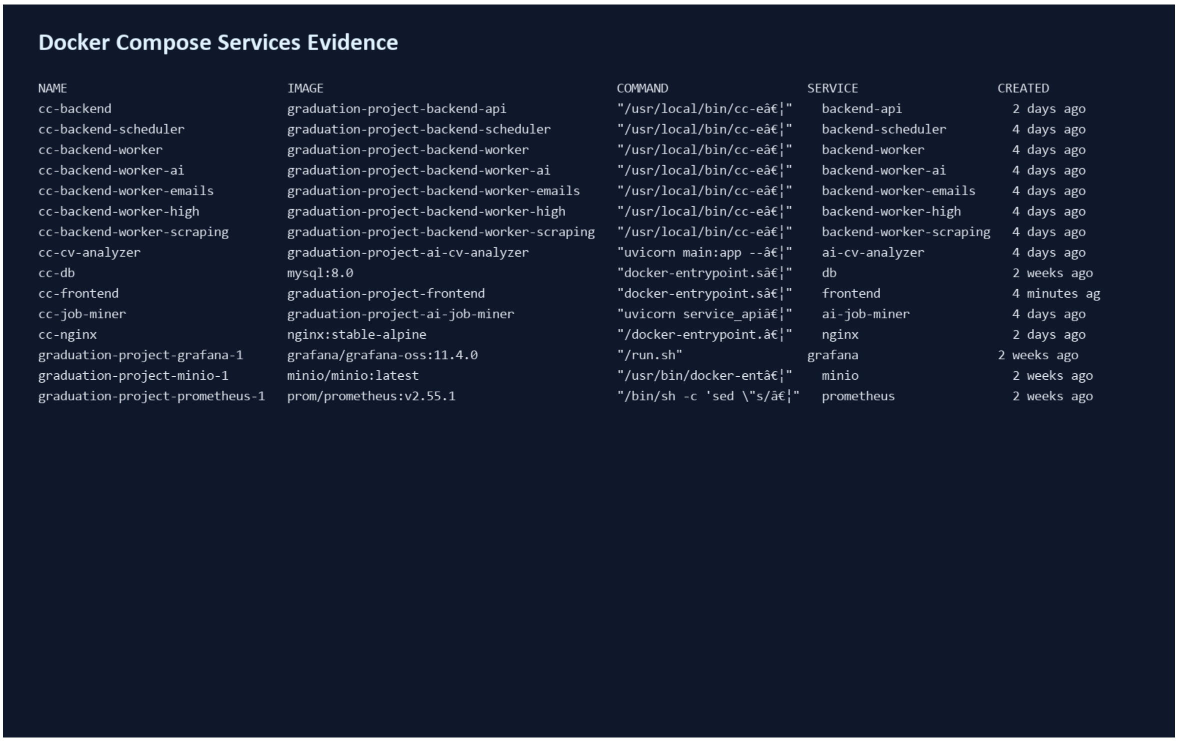


Figure 42. Tools Hub preview page.

5.17 Dockerized Runtime Flow

The runtime starts through Docker Compose. Nginx exposes the app, frontend and backend containers serve UI/API flows, backend workers process queues, Python services support AI workflows, MySQL and MinIO persist state, and monitoring services observe the stack.



Docker Compose Services Evidence

NAME	IMAGE	COMMAND	SERVICE	CREATED
cc-backend	graduation-project-backend-api	"/usr/local/bin/cc-eâ€¦"	backend-api	2 days ago
cc-backend-scheduler	graduation-project-backend-scheduler	"/usr/local/bin/cc-eâ€¦"	backend-scheduler	4 days ago
cc-backend-worker	graduation-project-backend-worker	"/usr/local/bin/cc-eâ€¦"	backend-worker	4 days ago
cc-backend-worker-ai	graduation-project-backend-worker-ai	"/usr/local/bin/cc-eâ€¦"	backend-worker-ai	4 days ago
cc-backend-worker-emails	graduation-project-backend-worker-emails	"/usr/local/bin/cc-eâ€¦"	backend-worker-emails	4 days ago
cc-backend-worker-high	graduation-project-backend-worker-high	"/usr/local/bin/cc-eâ€¦"	backend-worker-high	4 days ago
cc-backend-worker-scraping	graduation-project-backend-worker-scraping	"/usr/local/bin/cc-eâ€¦"	backend-worker-scraping	4 days ago
cc-cv-analyzer	graduation-project-ai-cv-analyzer	"uvicorn main:app --â€¦"	ai-cv-analyzer	4 days ago
cc-db	mysql:8.0	"docker-entrypoint.sâ€¦"	db	2 weeks ago
cc-frontend	graduation-project-frontend	"docker-entrypoint.sâ€¦"	frontend	4 minutes ag
cc-job-miner	graduation-project-ai-job-miner	"uvicorn service_apiâ€¦"	ai-job-miner	4 days ago
cc-nginx	nginx:stable-alpine	"/docker-entrypoint.â€¦"	nginx	2 days ago
graduation-project-grafana-1	grafana/grafana-oss:11.4.0	"/run.sh"	grafana	2 weeks ago
graduation-project-minio-1	minio/minio:latest	"/usr/bin/docker-entâ€¦"	minio	2 weeks ago
graduation-project-prometheus-1	prom/prometheus:v2.55.1	"/bin/sh -c 'sed \"s/â€¦"	prometheus	2 weeks ago

Figure 48. Docker services evidence.

Chapter 6: AI CV Analyzer Deep Technical Analysis

6.1 Introduction

The AI CV Analyzer is one of the main technical contributions of CareerCompass. It should not be understood as a thin wrapper around one pretrained model. The implemented analyzer is a layered hybrid pipeline that combines document-processing logic, NER, deterministic extraction rules, semantic enrichment, score composition, and explanation generation. This chapter separates that AI contribution from the general implementation chapter so that supervisors and examiners can evaluate the design as an academic system component.

6.2 AI Design Philosophy

CareerCompass does not use a pure NER model because CVs are noisy, multi-format documents. They can contain multiple columns, icons, section headers, table-like blocks, scanned pages, mixed date formats, and skill aliases. NER can extract entity candidates, but NER alone does not naturally explain seniority, primary technical domain, job-fit constraints, or recommendation reasons.

The system also does not use a pure rule-based parser. Rules are deterministic and useful for validation, but fixed rules are brittle when skill names, job titles, section headings, and CV layouts vary. A rule set can recognize known patterns, but it struggles with semantic similarity, synonyms, and role/domain interpretation.

The implemented design is therefore hybrid. NER extracts structured candidates, deterministic rules improve consistency and safety, canonicalization reduces noisy variants, Layer 2 adds domain and seniority interpretation, Layer 3 compares candidate and job evidence, and the explanation layer turns scores into strengths, gaps, red flags, and verdicts. TF-IDF fallback keeps the matching endpoint useful when heavier semantic components are unavailable.

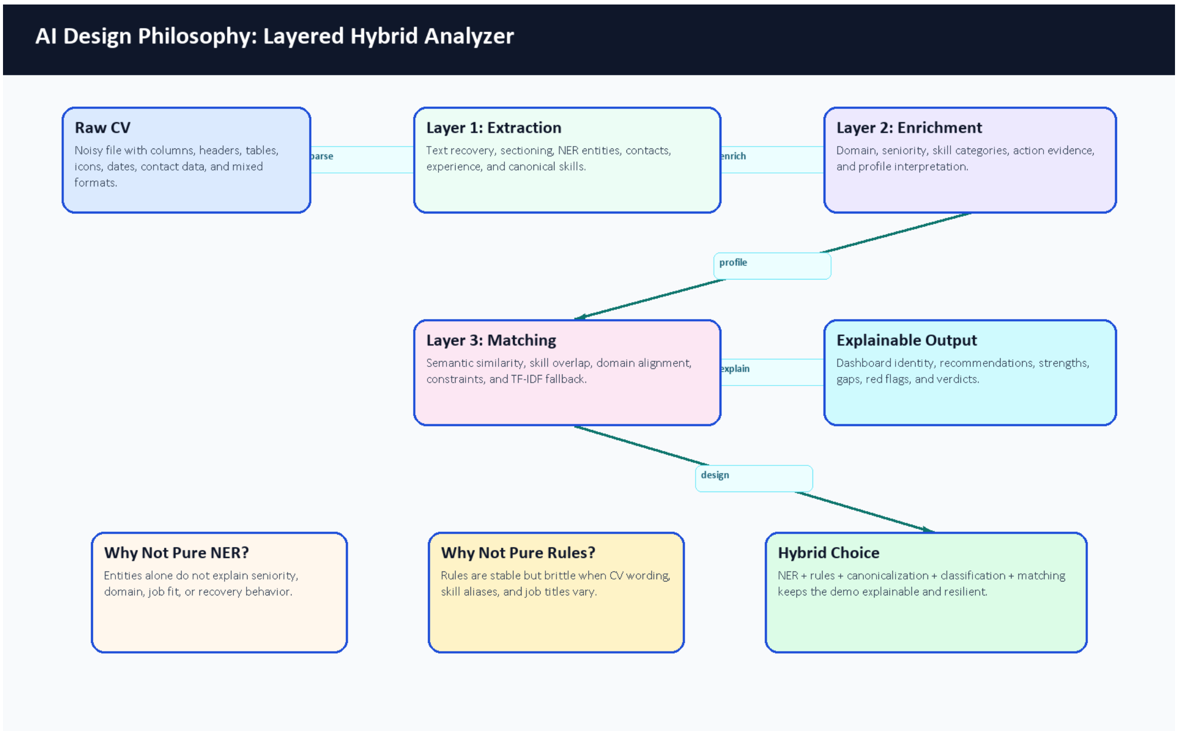


Figure 19. AI design philosophy for the layered hybrid analyzer.

Design Option	Advantage	Limitation	CareerCompass Decision
Pure NER	Learns entity patterns from data.	Does not solve file recovery, seniority, domain, matching, or explanation by itself.	Used only as one extraction component.
Pure rules	Predictable and easy to inspect.	Brittle when CVs use new wording, layouts, and aliases.	Used for safety, contacts, dates, validation, and fallback behavior.
Hybrid layered AI	Combines learned extraction, deterministic checks, semantic signals, and explanation.	More components must be tested and documented.	Chosen because it fits noisy CVs and graduation-demo transparency.

Table 17. AI design alternatives comparison.

6.3 Complete CV Processing Flow

The end-to-end flow begins when the student uploads a PDF or image CV. The frontend validates the file before sending it to Laravel. Laravel sends the file to the FastAPI analyzer, stores the private CV object, persists successful structured outputs, and records parsing status. The analyzer first recovers text, then segments sections, extracts entities, estimates experience, canonicalizes skills, classifies the profile, and supports matching for recommendations and gap analysis.

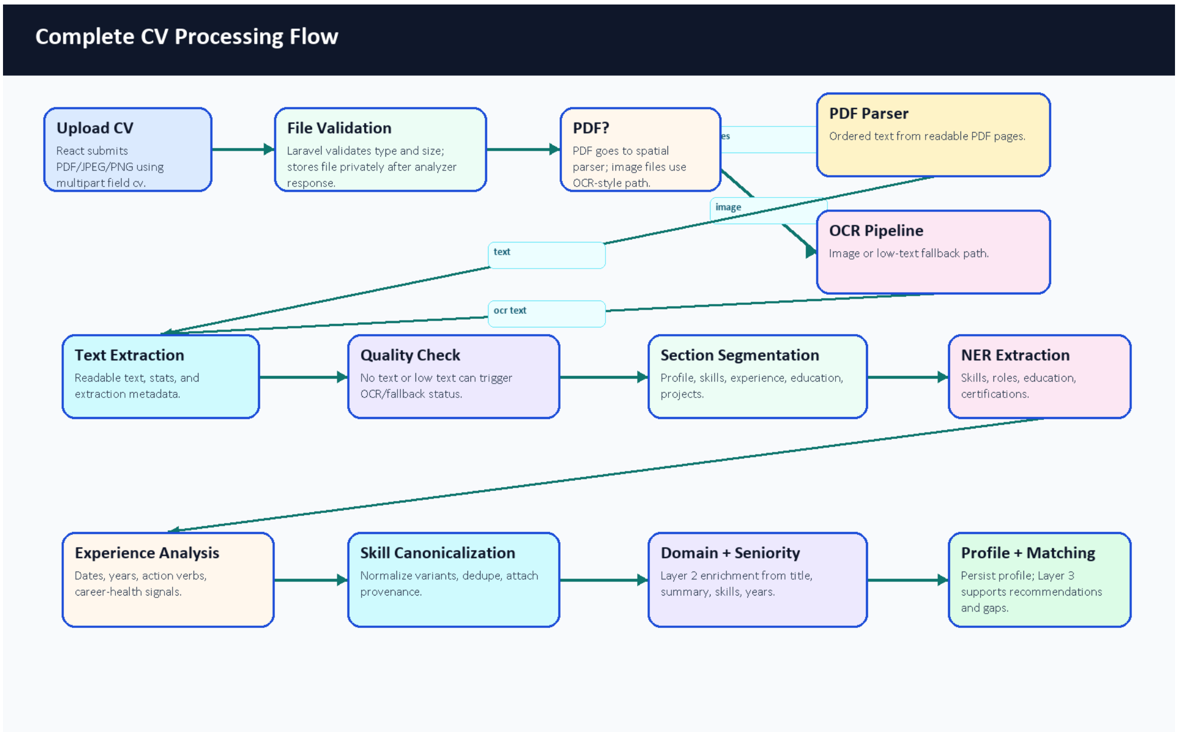


Figure 20. Complete CV processing flow.

The flowchart is intentionally more detailed than the high-level architecture diagram. It shows that the AI service performs several recoverable steps before returning data. The output is not only a list of words; it includes profile fields, skills, experience signals, domain, seniority, confidence-style values, metadata, and status.

6.4 Fault Tolerance and Recovery

CV parsing can fail for normal reasons: scanned PDFs, image-only files, weak text extraction, unsupported content, or service timeouts. The analyzer and backend are designed to report these states explicitly instead of silently overwriting good profile data with empty extraction results.

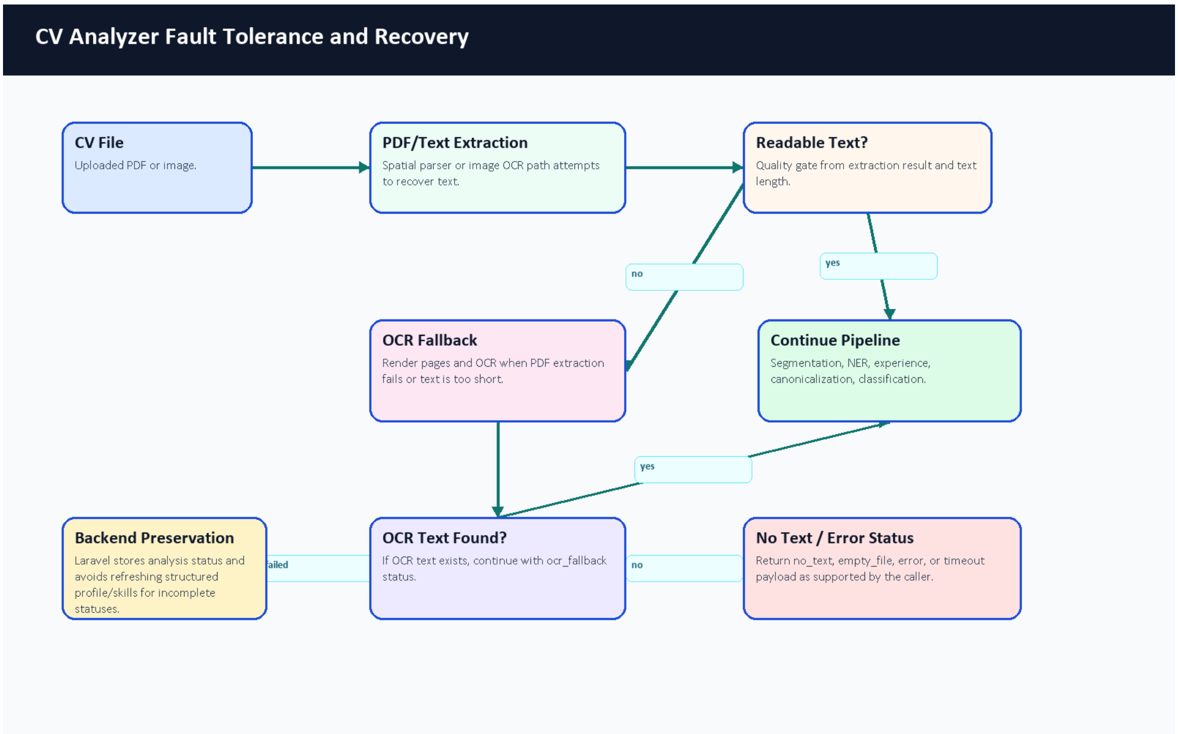


Figure 21. CV analyzer fault tolerance and recovery flow.

The FastAPI schema supports success, ocr_fallback, empty_file, no_text, and error statuses. The API-level timeout path returns a timeout payload. Laravel treats timeout, error, empty_file, and no_text as incomplete statuses and avoids refreshing profile, experience, and skills for those results. The backend still records analysis status and file metadata, then warns the frontend so the user can retry with a clearer document.

6.5 Confidence and Readiness Signals

CareerCompass uses confidence-style and readiness signals rather than a certified probability of hiring success. In Layer 1, _aggregate_confidence averages positive confidence-style values and caps the result at 1.0. Skills, profile, experience, and analysis sections can each carry confidence values. Laravel stores parser confidence_score and converts it into a completeness_score when available. The dashboard then visualizes completeness, model confidence, skill count, and experience as an estimated Career Readiness Snapshot.

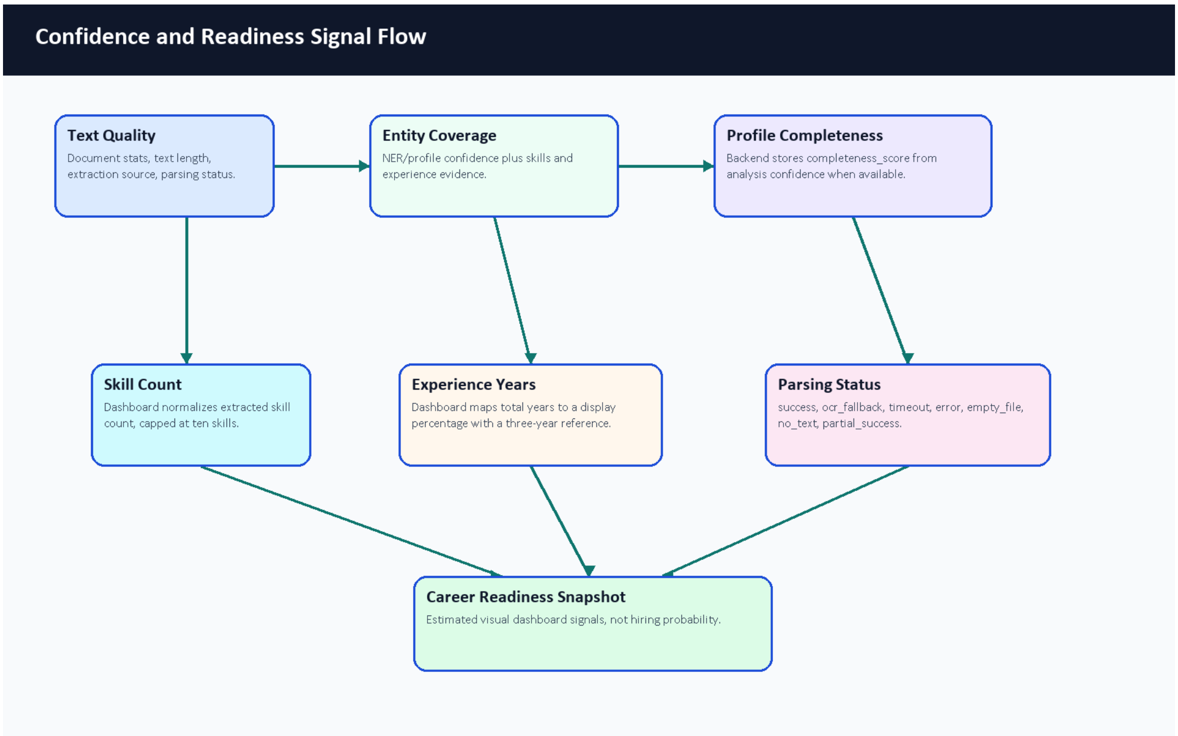


Figure 22. Confidence and readiness signal flow.

The current code-derived signal boundary is:

```
AggregateConfidence =
    min(1.0, average(positive confidence signals))

AnalysisConfidence =
    AggregateConfidence([
        skills_section.confidence_score,
        experience_section.confidence_score
    ])

BackendCompleteness =
    round(analysis.confidence_score * 100)

DashboardSkillSignal =
    min(extracted_skill_count * 10, 100)

DashboardExperienceSignal =
    min((total_experience_years / 3) * 100, 100)
```

These formulas are intentionally described as signals. They help the dashboard explain whether enough structured CV evidence was extracted, but they are not probabilities of employability, hiring success, or model correctness.

Signal	Source File or Component	Meaning	Used In	Limitation
parsing_status	main.py, schema.py, CvProcessingService.php	Whether parsing succeeded, used OCR, failed, timed out, or found no text.	Upload feedback, stored analysis status, preservation logic.	It is a status flag, not a quality score.
confidence_score	Layer 1 schema and orchestrator aggregation	Confidence-style value for extracted/derived fields.	Stored CV analysis and AI insight display.	It is not a calibrated probability of employment success.

Signal	Source File or Component	Meaning	Used In	Limitation
completeness_score	CvProcessingService.php	Backend percentage derived from analysis confidence when available.	Profile/dashboard completeness display.	It depends on parser output availability.
Skill signal	Dashboard.jsx	Extracted skill count normalized and capped at ten skills.	Career Readiness Snapshot.	More skills do not automatically mean a better candidate.
Experience signal	Dashboard.jsx	Total parsed years mapped to a display percentage using a three-year reference.	Career Readiness Snapshot.	It is a UI signal, not a universal seniority formula.
Extraction metadata	Layer 1 metadata and Laravel analysis metadata	Source, spatial status, segmentation, gaps, action-verb and experience details.	Debugging, review, and future evaluation.	Metadata quality depends on successful text recovery.

Table 18. Confidence and readiness signal summary.

6.6 Skill Canonicalization With Practical Example

Skill extraction is noisy because the same skill can appear in different forms. The canonicalizer supports exact variant mapping, exact canonical matching, RapidFuzz matching when available, normalized-key fallback, semantic embedding fallback, and pass-through behavior. The current committed config is largely industry-agnostic, so the example below is labeled illustrative of the implemented mapping stages rather than proof that every alias is already configured in source data.

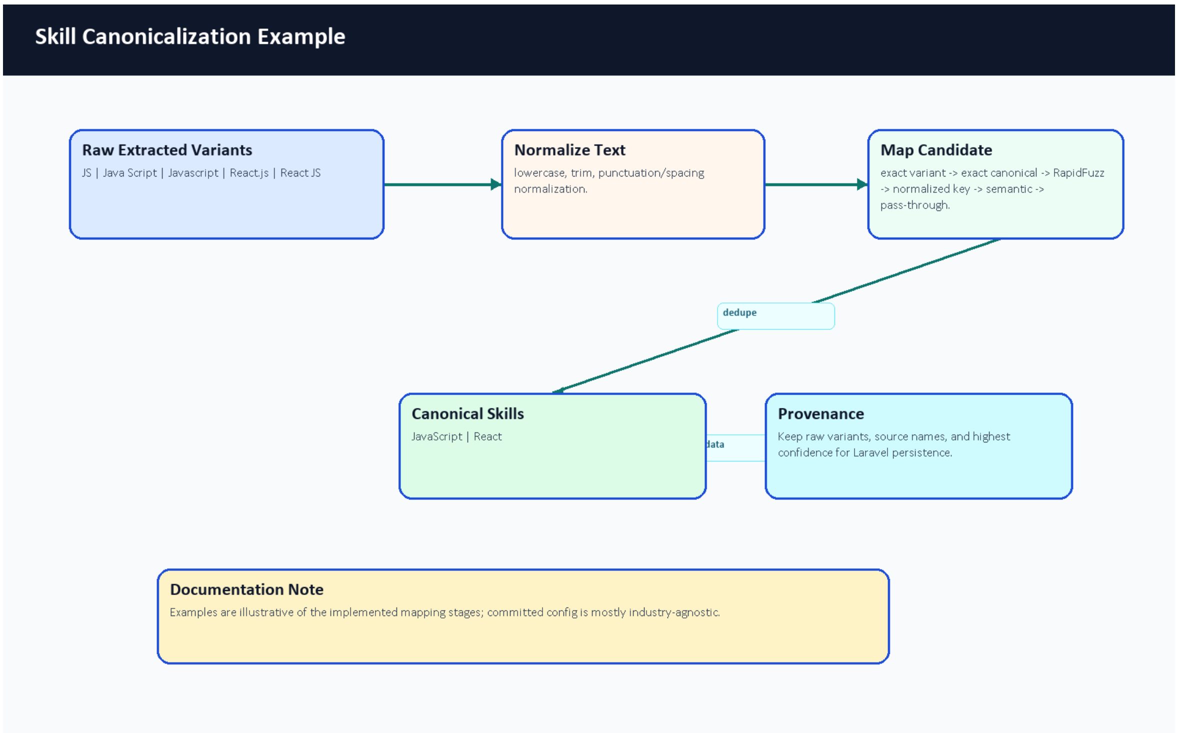


Figure 23. Skill canonicalization example.

Raw Extracted Skill	Normalized Skill	Why
JS	JavaScript	Illustrative abbreviation normalization.
Java Script	JavaScript	Illustrative spacing normalization.
Javascript	JavaScript	Illustrative casing/spelling normalization.

Raw Extracted Skill	Normalized Skill	Why
React.js	React	Illustrative framework alias normalization.
React JS	React	Illustrative punctuation and spacing normalization.

Table 19. Skill canonicalization example.

6.7 Fine-Tuned BERT NER Architecture

The NER architecture is a fine-tuning workflow, not a from-scratch language model. The training notebook uses bert-base-cased, tokenizes CV text with offsets, aligns character-span annotations to BIO token labels, trains a token-classification head, and exports career_compass_ner_final. At runtime, AdvancedNEREngine can load local ignored model weights if supplied; those weights are not committed to Git. A user-provided Colab export now provides recorded training-run metrics, but the repository alone still does not contain the final dataset, model weights, or a fully reproducible benchmark package.

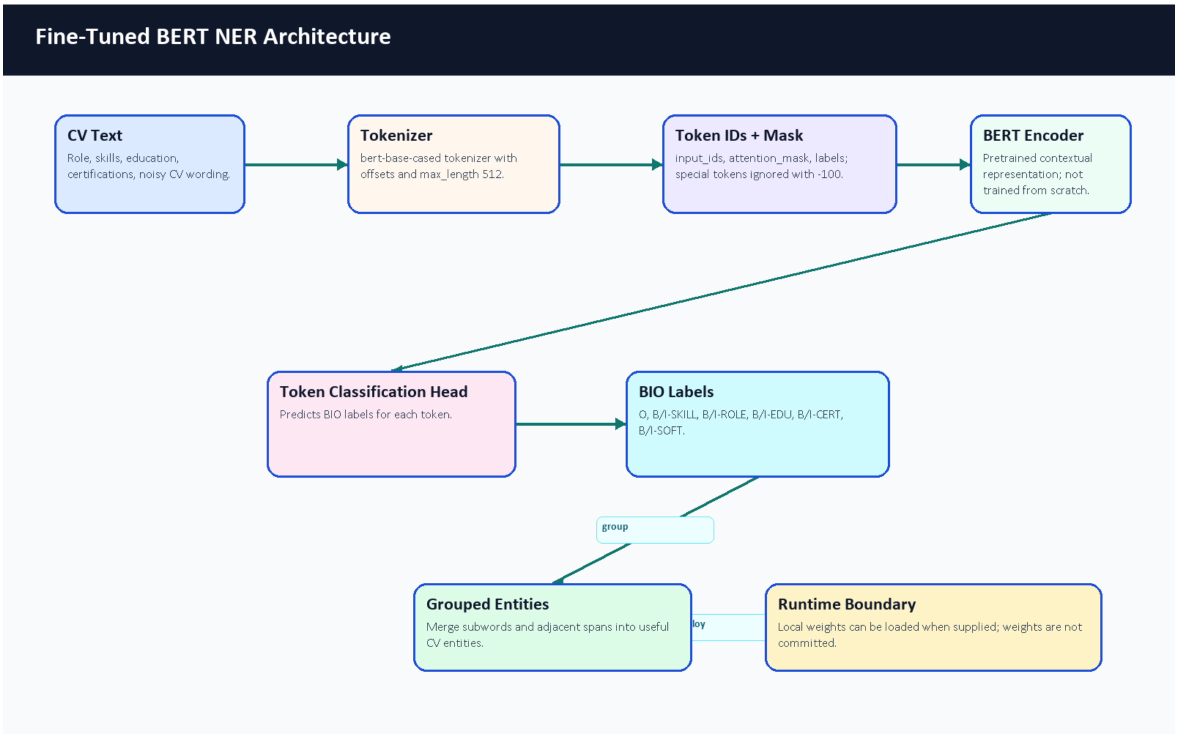


Figure 24. Fine-tuned BERT NER architecture.

The simplified BIO example in Table 14 remains valid for examiner explanation: Backend can start a ROLE entity, Developer can continue it, and Laravel or Docker can start SKILL entities. The actual model sees tokenized subwords and offsets rather than only human-readable words.

6.8 Detailed Training Pipeline

Synthetic training data is used because labeled CV NER data is not naturally available in the repository. The generator is designed to create varied technical CV snippets, including positive examples and negative decoys. Negative decoys matter because they teach the model not to tag every technical-looking phrase. The cleaner normalizes samples and validates entity spans before the notebook performs token alignment and fine-tuning.

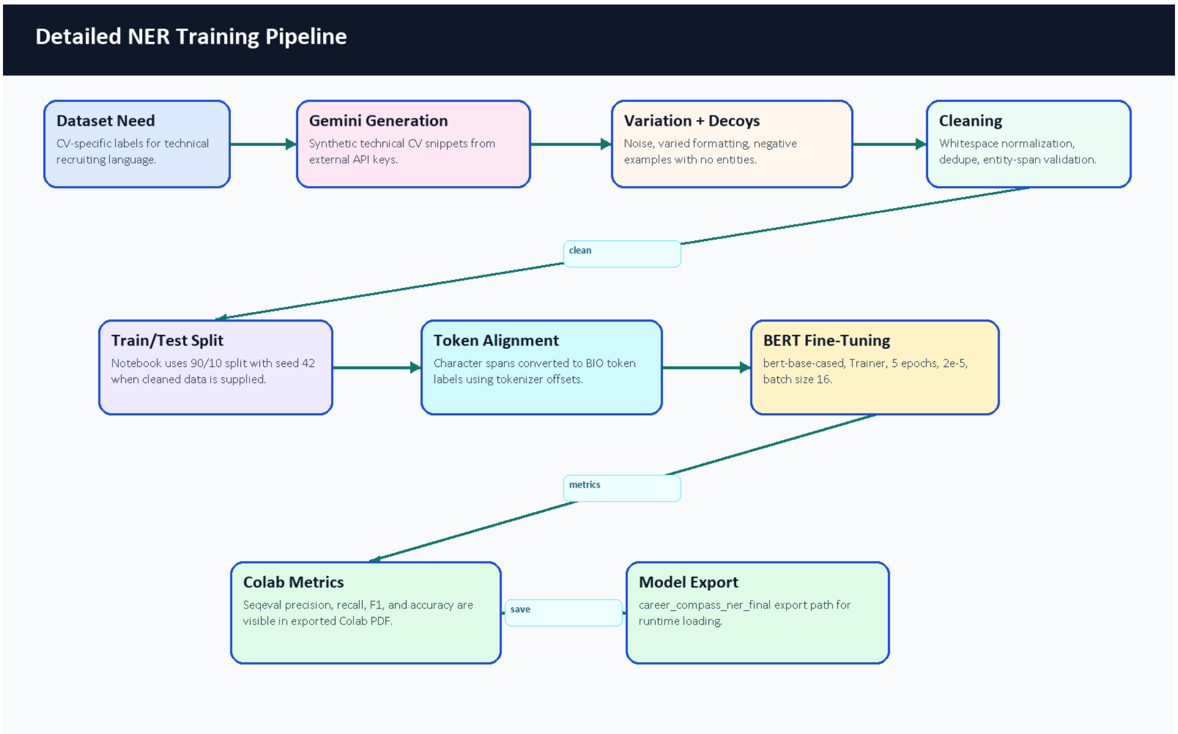


Figure 25. Detailed NER training pipeline.

The documentation pass also reviewed committed evaluation evidence and generated a dataset transparency note under docs/graduation-book/model-analysis/dataset_statistics.md. The user-provided Colab PDF gives recorded training-run output cells, including split counts and overall validation metrics. However, the cleaned dataset content and model weights are still not committed, and the PDF does not show per-label support counts. Therefore, the report includes the verified Colab metrics but still avoids a fake per-label distribution chart. Figure 29 records which evidence is available and what remains unavailable for reproducible academic review.

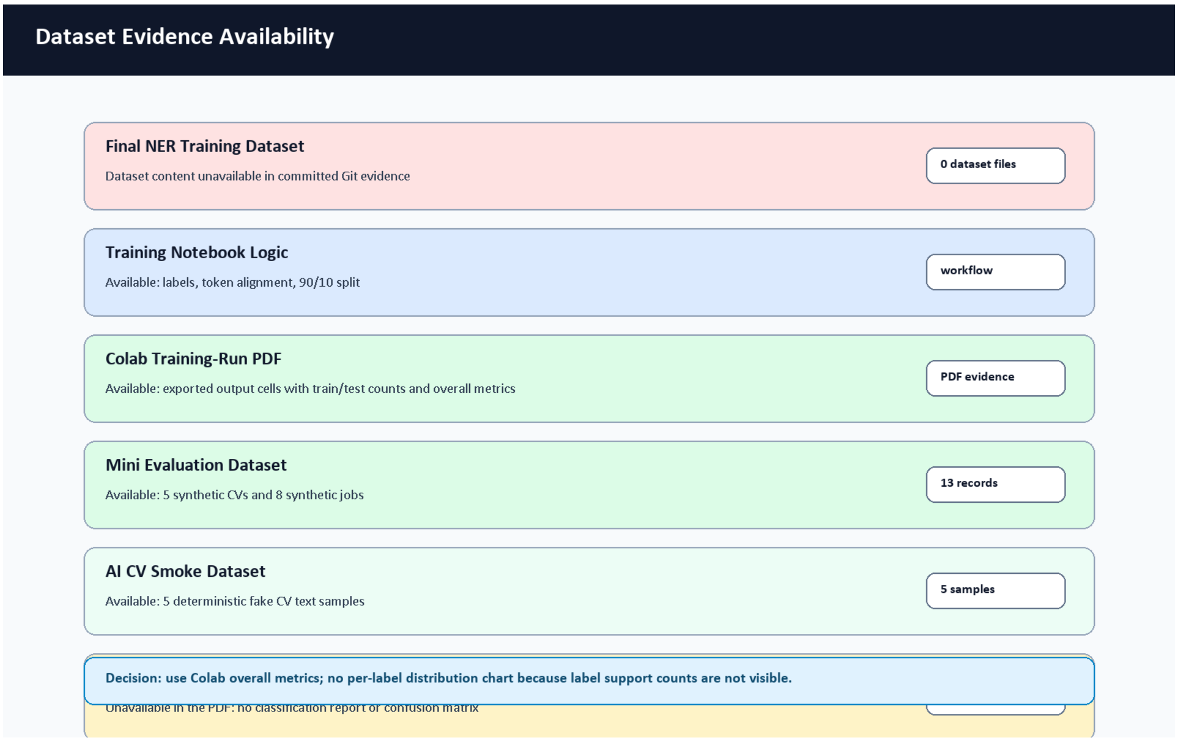


Figure 29. Dataset evidence availability summary.

Dataset Statistic	Status	Reason
Cleaned NER training samples	45,911 rows recorded in Colab PDF	The PDF shows the generated rows loaded from train_real_tech_cleaned.json; the dataset content itself is not committed.
Recorded train split	41,319 rows	Visible in the exported Colab PDF; split uses test size 0.1 and seed 42.
Recorded test split	4,592 rows	Visible in the exported Colab PDF and used for the notebook evaluation output.
Entity counts by label	Not visible in the PDF	The PDF shows labels but not per-label support counts.
Negative decoy count	Not available from committed evidence	Generator/cleaner support decoys, but final generated data is absent.
Mini evaluation CV samples	5	Generated documentation mini dataset under docs/graduation-book/evaluation/; separate from NER training.
Mini evaluation job samples	8	Generated documentation mini dataset; not a production benchmark.
AI CV Analyzer smoke samples	5	Deterministic text-only smoke set created for this evidence pass; evaluates parser-style labels and dependency availability, not transformer weights.
Final NER label distribution chart	Not generated	No per-label support counts are visible in the PDF, and no committed final labeled dataset exists to count SKILL/ROLE/EDU/CERT/SOFT/O labels honestly.

Table 20. Dataset availability and transparency.

6.8.1 Colab NER Fine-Tuning Results

The team exported the Google Colab notebook train_ner.ipynb as a PDF with visible output cells. This PDF was copied into docs/graduation-book/model-analysis/colab_train_ner_results.pdf after inspection. It is treated as supporting training evidence for the NER fine-tuning process. The PDF shows the notebook title train_ner.ipynb - Colab, timestamp 6/7/26, 3:55 AM, the heading CareerCompass AI Engine: Global Skill NER training (Autonomous), and a synthetic data augmentation strategy. It also shows the cleaned dataset path train_real_tech_cleaned.json, 11 BIO labels, train/test row counts, tokenization completion, model initialization from bert-base-cased, training arguments, and epoch-level metrics.

These numbers improve the academic evidence for the training workflow, but they should be interpreted carefully. They are Colab-run validation outputs for the generated/synthetic dataset and notebook split visible in the PDF. They are not production accuracy, not a large real-world CV benchmark, and not reproducible from the repository alone unless the same dataset, runtime, and exported model artifacts are supplied.

Parameter	Value from Colab PDF	Source
Notebook	train_ner.ipynb - Colab	PDF header
Export timestamp	6/7/26, 3:55 AM	PDF header
Dataset file	train_real_tech_cleaned.json	PDF data-loading cell
Total loaded rows	45,911	PDF dataset output

Parameter	Value from Colab PDF	Source
Train rows	41,319	PDF dataset output
Test rows	4,592	PDF dataset output
Split	test size 0.1, seed 42	PDF data-loading cell
Base checkpoint	bert-base-cased	PDF model initialization cell
Labels	O plus B/I for SKILL, ROLE, EDU, CERT, SOFT	PDF label configuration cell
Max length	512 tokens	PDF tokenization cell
Epochs	5	PDF training arguments
Learning rate	2e-5	PDF training arguments
Batch size	16 train / 16 eval	PDF training arguments
Weight decay	0.01	PDF training arguments
Best model metric	F1	PDF training arguments

Table 44. Colab NER training run configuration.

Epoch	Training Loss	Validation Loss	Precision	Recall	F1	Accuracy
1	0.077623	0.069118	0.921027	0.928206	0.924603	0.973227
2	0.061530	0.064051	0.915886	0.941504	0.928518	0.974912
3	0.053831	0.063463	0.928387	0.943469	0.935867	0.976233
4	0.044553	0.064025	0.932287	0.937967	0.935118	0.977018
5	0.037280	0.068058	0.933307	0.940521	0.936900	0.976376

Table 45. Colab NER epoch metrics.

Metric	Final Epoch Value	Source
Precision	0.933307	Colab PDF output, epoch 5
Recall	0.940521	Colab PDF output, epoch 5
F1-score	0.936900	Colab PDF output, epoch 5
Accuracy	0.976376	Colab PDF output, epoch 5
Training loss	0.037280	Colab PDF output, epoch 5
Validation loss	0.068058	Colab PDF output, epoch 5

Table 46. Colab NER final metric summary.

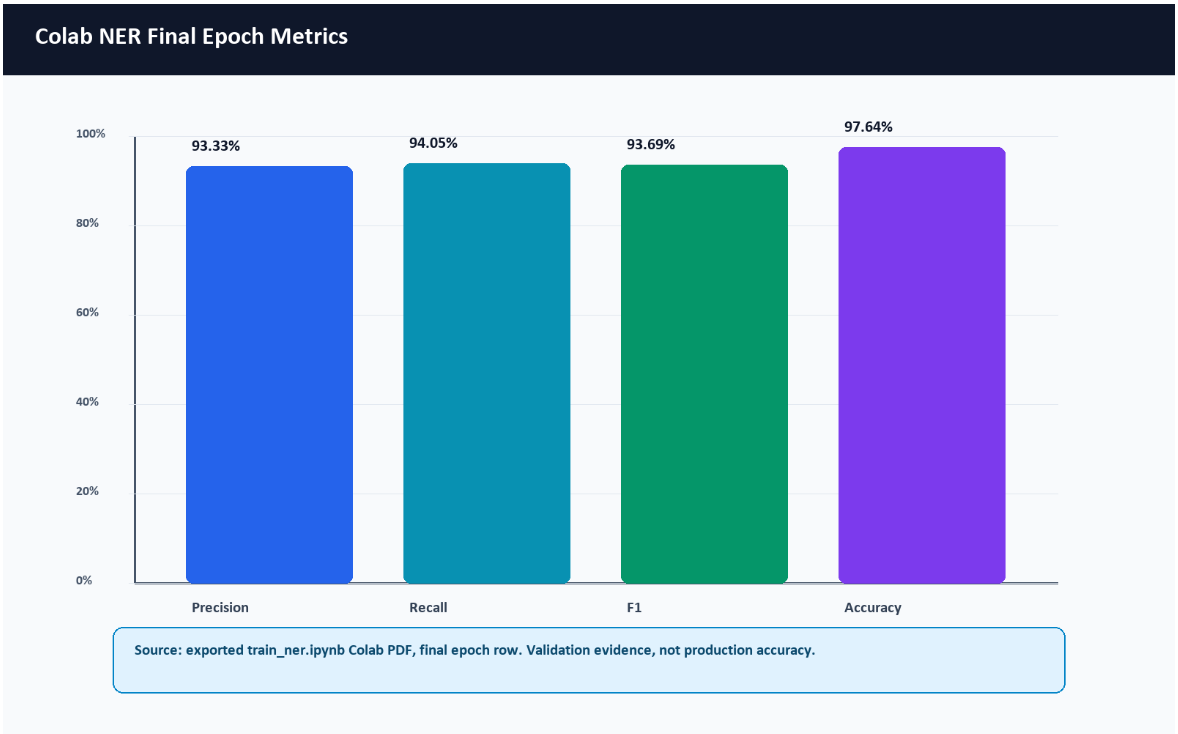


Figure 50. Colab NER final epoch metrics.

No per-label classification report or confusion matrix is visible in the PDF. Therefore, this report does not invent per-label SKILL/ROLE/EDU/CERT/SOFT support, precision, recall, F1, or a confusion-matrix chart.

6.9 Matching Score Formula and Penalty Logic

The Layer 3 matcher exposes a clear score-composition formula in similarity.py and matching_config.json. It is exact for the code path implemented by IntelligentMatcher.calculate_match; however, it still depends on upstream component scores such as semantic similarity, skill similarity, and domain similarity.

```
BaseScore =
    semantic_score * w_semantic
  + skills_score   * w_skills
  + domain_score   * w_domain

FinalScore =
    clamp(BaseScore - total_penalty + bonus_boost, 0.0, 1.0)

MatchScorePercent = round(FinalScore * 100, 2)
```

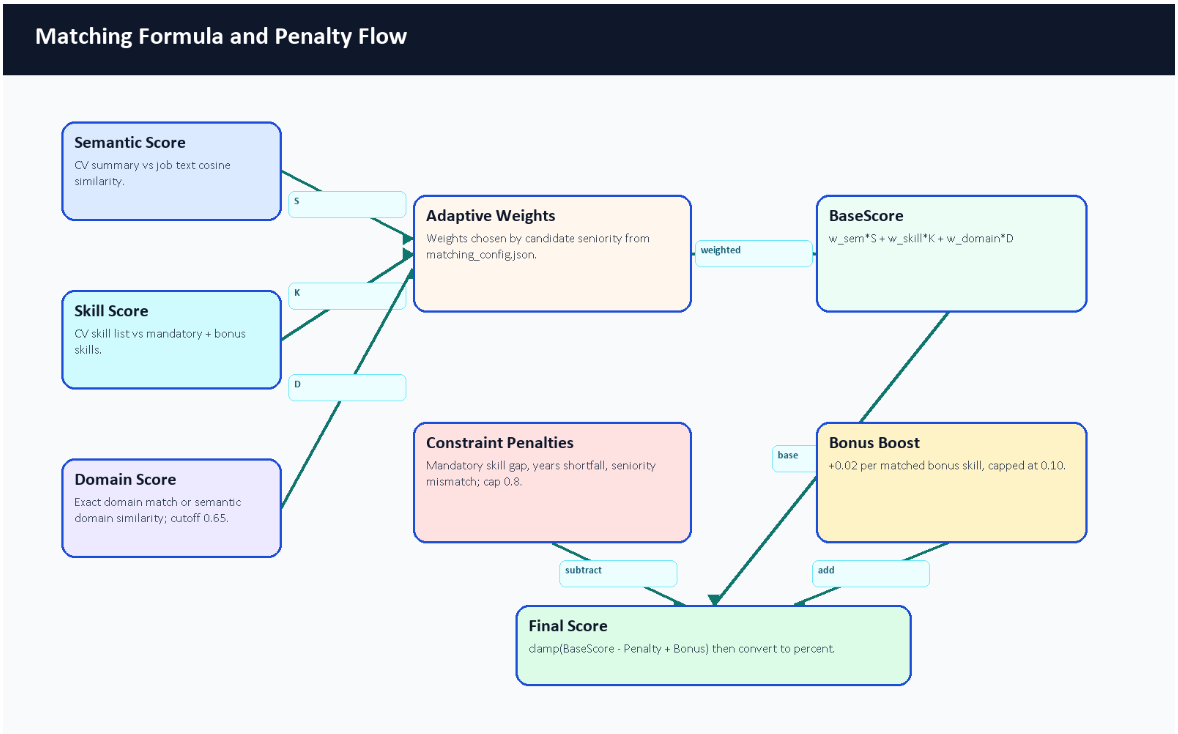


Figure 26. Matching formula and penalty flow.

Seniority	Semantic Weight	Skill Weight	Domain Weight	Notes
intern	0.30	0.60	0.10	Early roles emphasize concrete skill overlap.
junior	0.40	0.40	0.20	Balanced summary and skill evidence.
mid	0.35	0.35	0.30	Adds more domain importance.
senior	0.25	0.25	0.50	Domain alignment becomes more important.
lead	0.20	0.20	0.60	Leadership roles emphasize domain/role alignment.
default	0.35	0.35	0.30	Fallback profile.

Table 21. Seniority-aware matching weights.

Constraint penalties are also code-derived. Missing mandatory skills subtract 15 percent each, capped at 50 percent. Experience shortfall subtracts a proportional penalty capped at 30 percent. Seniority mismatch subtracts 20 percent. Total validation penalty is capped at 80 percent. Bonus skills add 2 percent each, capped at 10 percent. The /api/hybrid-match endpoint additionally blends the Layer 3 semantic/adaptive result with TF-IDF when TF-IDF is available: 60 percent semantic/adaptive and 40 percent TF-IDF.

6.10 Explainable AI Recommendation Output

The analyzer does not only return a single percentage. It also returns supporting evidence that can be shown to users and examiners: score breakdowns, missing mandatory skills, strengths, gaps, red flags, and a fit verdict. This is important academically because it makes the recommendation process inspectable rather than opaque.

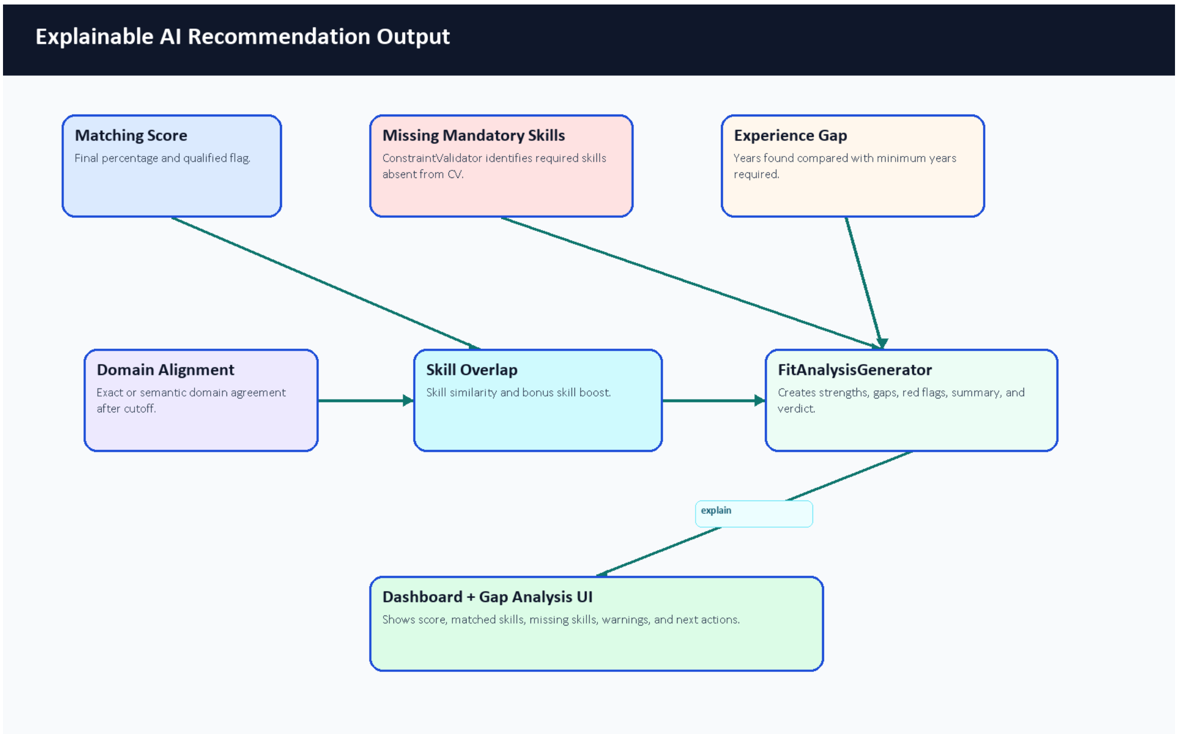


Figure 27. Explainable AI recommendation output.

Output Type	Example	Why It Helps
Score	78 percent	Gives a quick summary of estimated fit.
Matched skills	Laravel, Docker, MySQL	Shows evidence supporting the recommendation.
Missing skills	Kubernetes	Turns the gap into a learning target.
Red flags	Significant seniority mismatch	Warns that a numeric score should not be read alone.
Verdict	Strong Match or Potential Fit	Converts score ranges into readable guidance.
Gaps	Experience shortfall or missing mandatory skills	Explains why a candidate may need improvement before applying.

Table 22. Recommendation explanation output types.

6.11 AI Analyzer Sequence

The analyzer is synchronous during CV upload: Laravel calls FastAPI and receives a structured parse result before updating the returned user resource. The stored profile, skills, experiences, CV analysis, and private file metadata then support later dashboard, recommendation, and gap-analysis requests.

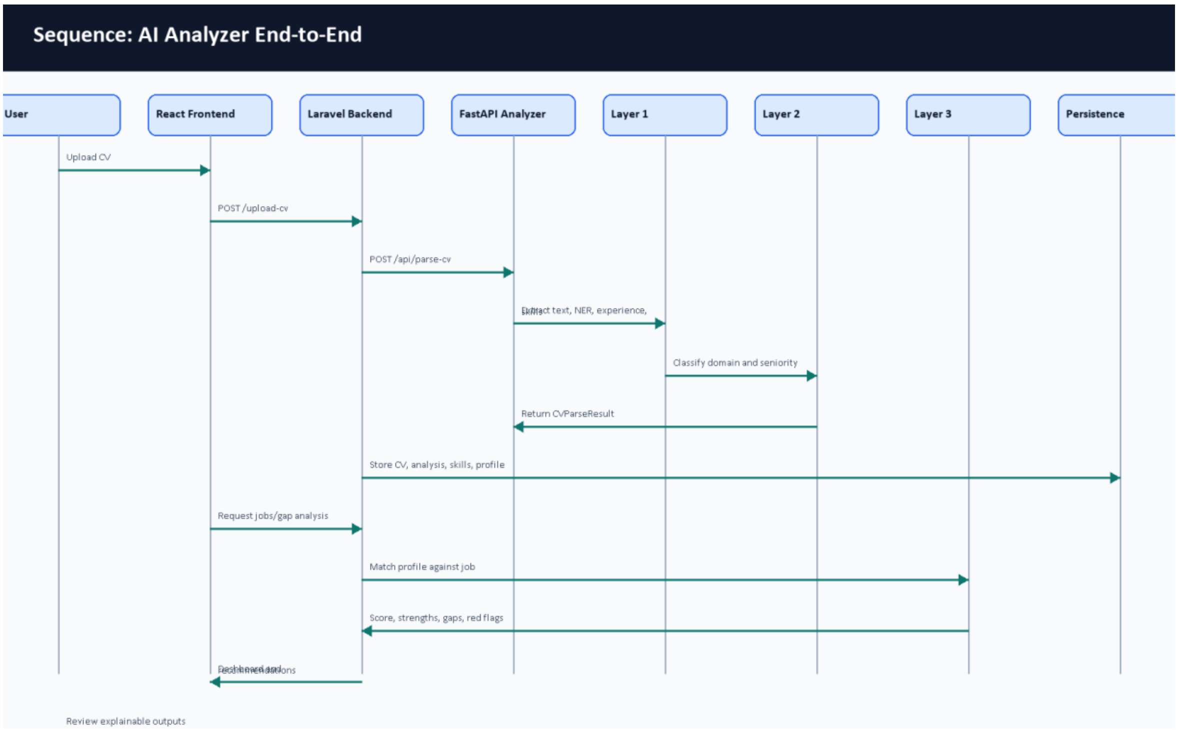


Figure 28. AI analyzer sequence diagram.

6.12 Computational Complexity Overview

The following complexity statements are approximate code-level descriptions, not formal proofs. They describe how cost grows with input size and component counts.

Component	Approximate Complexity	Explanation
Text cleanup and line processing	O(n)	Scans the recovered text and lines, where n is text length.
Spatial PDF row grouping	Approximately O(w log w)	Word positions are grouped/sorted, where w is extracted word count.
NER chunking	O(c x model_cost)	Long CV text is split into c chunks; transformer inference dominates per chunk.
Skill canonicalization	O(s x k) for fuzzy/semantic comparison	s extracted skills may be compared against k configured/canonical names.
Domain classification	O(d) embedding comparisons	CV context is compared against d taxonomy domain descriptions.
Matching one job	O(s + r) plus embedding calls	Compares candidate skills with job requirements r and computes semantic/domain signals.
Ranking many jobs	O(j x match_cost)	j jobs require repeated matching or backend fallback comparison.
TF-IDF fallback	O(t) vectorization plus sparse cosine	Depends on token count t in the two texts.

Table 23. Computational complexity overview.

6.13 Raw CV Input to Structured Output Example

The example below is an illustrative walkthrough designed for examiner readability. It is not a live model benchmark. It uses a small CV fragment and a small job description to show how the three layers cooperate. Because the full transformer/OCR runtime dependencies were not available in the documentation Python environment, the JSON block is labeled as an illustrative schema example based on the actual schema.py response structure rather than a live model run.

Raw CV fragment:

```
Ahmed Mohamed
Backend Developer
Skills: Laravel, Docker, MySQL, REST APIs
Experience: Backend Intern at TechWave, 2024-2025
Education: Computer Science
```

Raw Evidence	Extracted Output	Schema Area
Backend Developer	Predicted role/title evidence.	profile.current_title, analysis.predicted_role
Laravel, Docker, MySQL, REST APIs	Four hard technical skills.	skills.items[]
Backend Intern at TechWave, 2024-2025	One experience item with company, title, period, and technologies.	experience.items[]
Computer Science	Education evidence.	Profile/education metadata when recovered
Text CV fragment	Successful text parsing path; no OCR needed in this example.	parsing_status, stats, analysis.metadata

Table 24. Raw CV fragment extraction example.

Illustrative structured response:


```
{
  "parsing_status": "success",
  "profile": {
    "full_name": "Ahmed Mohamed",
    "current_title": "Backend Developer",
    "email": null,
    "phone": null,
    "location": null,
    "summary": null
  },
  "stats": {
    "page_count": 1,
    "char_count": 168,
    "word_count": 24,
    "language_hint": "en"
  },
  "skills": {
    "items": [
      {
        "name": "Laravel",
        "category": "hard",
        "confidence_score": 0.65,
        "evidence": "ner_or_rule"
      },
      {
        "name": "Docker",
        "category": "hard",
        "confidence_score": 0.65,
        "evidence": "ner_or_rule"
      },
      {
        "name": "MySQL",
        "category": "hard",
        "confidence_score": 0.65,
        "evidence": "ner_or_rule"
      },
      {
        "name": "REST APIs",
```

```
        "category": "hard",
        "confidence_score": 0.65,
        "evidence": "ner_or_rule"
      }
    ],
    "confidence_score": 0.65
  },
  "experience": {
    "items": [
      {
        "title": "Backend Intern",
        "company": "TechWave",
        "start_date": "2024-01-01",
        "end_date": "2025-01-01",
        "is_current": false,
        "description": [
          "Built REST APIs."
        ],
        "technologies": [
          "Laravel",
          "Docker",
          "MySQL"
        ],
        "confidence_score": 0.85
      }
    ],
    "confidence_score": 0.85
  },
  "analysis": {
    "predicted_role": "Backend Developer",
    "seniority": "junior",
    "primary_domain": "Backend Development",
    "strengths": [
      "Relevant backend API and database evidence."
    ],
    "gaps": [],
    "red_flags": [],
    "confidence_score": 0.75,
    "metadata": {
      "experience": {
        "total_experience_years": 1.0
      },
      "extraction": {
        "source": "pdf_text"
      }
    }
  }
}
```


Classification	Result	Reason
Domain	Backend Development	Backend role plus Laravel, MySQL, REST API evidence.
Seniority	Intern/Junior style estimate	Internship evidence and limited years.
Skill category	Hard technical skills	Laravel, Docker, MySQL, and APIs are technical implementation skills.

Table 25. Layer 2 interpretation example.

Example job: Junior Backend Developer requiring Laravel, MySQL, Docker, and REST APIs.

Matching Evidence	Result
Matched skills	Laravel, MySQL, Docker, REST APIs
Missing skills	None in this simplified example
Fit interpretation	Good illustrative fit for a junior backend role
Explanation	Skill overlap is strong; seniority appears compatible; final production score would require live matcher execution.

Table 26. Layer 3 matching evidence example.

6.14 Why Not Use a Direct LLM-Only Approach?

Direct LLM analysis can be powerful, especially for summarizing complex CVs and producing natural-language feedback. CareerCompass still avoids a direct LLM-only runtime because the graduation/demo system must be reproducible, inspectable, containerized, and privacy-aware. The Gemini-based code is used for synthetic training-data generation, not for sending private uploaded CVs to an external LLM during the normal runtime flow. The runtime analyzer is decomposed into layers so each part can be tested, explained, and improved independently.

Direct LLM-Only Approach	CareerCompass Hybrid Analyzer
Requires an external inference service for each private CV unless self-hosted.	Runs the analyzer locally/containerized in the demo stack.
Responses can vary across prompts, model versions, and temperatures.	Uses deterministic rules, typed schema validation, and explicit status values.
Harder to benchmark each sub-step because extraction, reasoning, and explanation are blended.	Layers can be inspected separately: text recovery, NER, rules, classification, matching, and explanation.
Privacy risk is higher if real CVs are sent to a remote service.	Runtime CV analysis can stay inside the deployed services.
Can explain textually but may hallucinate unsupported fields.	Outputs structured strengths, gaps, red flags, metadata, and confidence-style signals from code paths.
May be faster to prototype but harder to reproduce academically.	Fits Docker-based graduation evaluation and component-level evidence.

Table 27. AI approach comparison.

6.15 AI Analyzer Limitations and Future Work

The limitations below make the AI contribution more academically honest, not weaker. They identify exactly what should be improved before the system is treated as a stronger research or production artifact.

- OCR quality affects scanned or image-heavy CV extraction.
- Synthetic data may not cover all real CV styles, languages, layouts, and informal wording.
- The Colab PDF provides recorded NER training-run metrics, but the final cleaned NER dataset and exported model weights are not committed, so the run is not reproducible from repository files alone.
- More real or human-reviewed labeled CV data is needed for trustworthy per-label precision, recall, and F1.
- A larger fixed benchmark should evaluate parsing, role prediction, domain classification, seniority, matching, and gap analysis together.
- Arabic and multilingual CV support can be improved beyond the current UI localization.
- The role/domain taxonomy and skill alias catalog should expand through reviewed labor-market evidence.

- A model card and dataset card should be added for any exported model artifact.
- Human-in-the-loop review can improve reliability for important student guidance decisions.
- Privacy-preserving training and evaluation workflows should be designed before using real CV data.

6.16 AI Chapter Summary

The standalone AI chapter was added because the analyzer is a core project contribution. The system is best described as a transparent, layered hybrid analyzer for a graduation/demo environment. It does not claim production-grade AI accuracy, a certified hiring probability, or repository-alone reproducible final NER benchmarking. The new Colab PDF strengthens training-run evidence, while the academic value remains the integration of document recovery, NER, rules, canonicalization, classification, matching, explanation, and honest fallback behavior.

Chapter 7: Testing and Evaluation

7.1 Introduction

Evaluation was performed using repository-aware commands and browser evidence. The goal was to verify that each major part of the graduation/demo system runs and to document limitations honestly.

7.2 Testing Strategy

The testing strategy combined automated tests, build checks, configuration checks, service health probes, and manual functional evaluation. Automated tests provide repeatable evidence. Screenshots provide visible workflow evidence. Manual tables document behavior that is difficult to fully automate in the available environment.

7.3 Backend Testing

Backend validation was executed inside the backend container. Composer dependencies were already installed. php artisan config:clear, php artisan route:list, migrations, and tests passed. The route list confirmed 131 routes. The Laravel test suite passed with 39 tests and 297 assertions.

7.4 Frontend Testing

The frontend was validated using the existing frontend/node_modules and the bundled Node runtime. ESLint passed with 9 warnings and 0 errors. The warnings were related to React fast-refresh export conventions and hook dependency notes. The Vite production build passed and transformed 2904 modules.

7.5 Python Services Testing

The AI Job Miner pytest suite passed with 75 tests. Python syntax compilation passed for both AI services. The AI CV Analyzer pytest command could not run because pytest was not installed in that container; this is recorded as a skipped/blocked validation step rather than a failure of the application code.

7.6 Docker and Integration Testing

Docker Compose configuration validation passed for both development and production overlay configurations. A full compose build/start was attempted; the initial full build exceeded the 15-minute command timeout, but the stack continued building and was later brought up successfully with targeted frontend/Nginx rebuild/start. All main containers reached healthy or running state during final checks.


Validation Command Evidence

```
Validation summary captured for the graduation book:

docker compose config --quiet: PASSED
docker compose -f docker-compose.yml -f docker-compose.prod.yml config --quiet: PASSED
docker compose up -d --build: stack built; full initial build exceeded 15 minutes, then targeted rebuild/start completed
composer install in backend-api container: PASSED
php artisan config:clear: PASSED
php artisan route:list: PASSED (131 routes)
php artisan migrate --force --no-interaction: PASSED (nothing to migrate)
php artisan test: PASSED (39 tests, 297 assertions)
frontend ESLint: PASSED with 9 warnings and 0 errors
frontend Vite build: PASSED (2904 modules transformed)
ai-job-miner pytest: PASSED (75 passed)
ai-cv-analyzer compileall: PASSED
ai-job-miner compileall: PASSED
ai-cv-analyzer pytest: SKIPPED/blocked because pytest is not installed in that container
HTTP probes: /, /api/health, /api/ready, /status, AI CV Analyzer root, and Job Miner health returned 200
```

Figure 49. Validation command evidence.

7.7 CI/CD Validation

GitHub Actions workflow files were reviewed as part of repository inspection. A live GitHub Actions status screenshot was not captured before the draft PR because PR checks only become meaningful after the branch is pushed and GitHub schedules workflows. The manual review checklist asks the team to inspect CI status on the opened draft PR.

7.8 AI CV Analyzer Model Evidence

The AI CV Analyzer training workflow was inspected from repository files, helper documentation, and the exported Colab training-results PDF. Full model training was not rerun during this documentation update because the generator requires external Gemini API keys, the cleaned training dataset content is not committed, the runtime dependencies for transformer inference were not installed in the bundled documentation Python environment, and the notebook is designed for a Colab/T4-style GPU runtime.

The important distinction is reproducibility scope. The repository alone still does not provide the final labeled evaluation dataset, final model weights, or a one-command reproducible training run. However, the user-provided Colab PDF does provide recorded output cells from the actual notebook run, including train/test counts and overall epoch metrics. Those values are reported as Colab-run training evidence on the generated/synthetic notebook split, not as production accuracy. A local ignored artifact folder was present on this workstation and safe metadata was inspected, but model binaries remain outside Git. The mini evaluation below remains useful for regression-style demonstration, but it is separate from the Colab NER training metrics.

Evidence Item	Status	What It Proves	What It Does Not Prove
Runtime NER artifact path	Code checks ai-cv-analyzer/models/ner_weights/career_compass_ner_final; the folder is ignored by Git.	The service can load a local token-classification model when deployed.	It does not prove the artifact is committed or provide a final held-out F1 score.

Evidence Item	Status	What It Proves	What It Does Not Prove
Local ignored metadata	config.json and tokenizer metadata were inspected locally without copying weights.	The local artifact uses a BERT token-classification configuration and cased tokenizer.	It is not a portable repository artifact.
Training notebook	Present under ai-cv-analyzer/training/train_ner.ipynb	The training process, label map, token alignment, Trainer setup, metrics code, and export steps are documented.	It does not include the cleaned dataset or model weights.
Colab training-results PDF	Exported PDF under docs/graduation-book/model-analysis/colab_train_ner_results.pdf	Shows recorded NER fine-tuning/evaluation outputs: 41,319 train rows, 4,592 test rows, and final epoch precision 0.933307, recall 0.940521, F1 0.936900, accuracy 0.976376.	It does not prove readiness for real deployment and may not be reproducible without the same dataset, runtime, and model artifacts.
Dataset generator	Present under ai-cv-analyzer/training/generate_tech_dataset.py	Synthetic labeled data can be generated from Gemini with key rotation and negative decoys.	It was not run here because it requires API keys and would generate a large dataset.
Dataset cleaner	Present under ai-cv-analyzer/training/clean_dataset.py	Dataset normalization, deduplication, and span validation are part of the workflow.	The cleaned dataset file itself is not committed.
API tests	Present under ai-cv-analyzer/tests/test_service_api.py	FastAPI status handling and hybrid-match formula behavior are covered with fakes.	These tests do not measure real NER model accuracy.
Dependency probe	Local bundled Python lacked transformers, torch, sentence_transformers, OCR/PDF packages, and Gemini client libraries.	Explains why live model inference and training were not rerun during documentation generation.	It does not reflect what the Docker image may install at runtime.
Mini evaluation	Generated under docs/graduation-book/evaluation/	Synthetic skill/recommendation/gap logic can be checked repeatably.	It is not a statistical live-model benchmark.

Table 28. Model evaluation evidence.

7.9 NER Extraction Examples

The table below documents expected extraction behavior from the inspected NER labels and runtime post-processing. It is intentionally marked as example evidence rather than measured per-label accuracy because transformer dependencies were not available in the local documentation runtime and the Colab PDF reports overall metrics rather than a per-label classification report.

Example CV Text	Expected NER Entities	Downstream Use	Evidence Type
Experienced Backend Developer with Laravel, Docker, and MySQL.	ROLE: Backend Developer; SKILL: Laravel, Docker, MySQL	Predicted role, extracted skills, backend/domain matching.	Illustrative example from label schema and code path.

Example CV Text	Expected NER Entities	Downstream Use	Evidence Type
Graduated from Faculty of Computers and Information, Kafr El-Sheikh University.	EDU: Faculty of Computers and Information, Kafr El-Sheikh University	Education/profile evidence.	Illustrative example from EDU label behavior.
AWS Cloud Practitioner certified with Kubernetes deployment experience.	CERT: AWS Cloud Practitioner; SKILL: Kubernetes	Certification and cloud/DevOps skill evidence.	Illustrative example from CERT/SKILL labels.
Leadership, communication, and teamwork across agile projects.	SOFT labels exist in training setup; runtime grouping mainly returns SKILL/ROLE/EDU/CERT.	Soft-skill interpretation is handled mostly by taxonomy and rule layers.	Limitation observed from runtime grouping code.

Table 29. NER extraction examples.

7.10 Semantic Matching vs TF-IDF Fallback Examples

The semantic matching path could not be executed locally during this documentation update because sentence-transformer dependencies were unavailable in the bundled Python environment. The pure Python TF-IDF matcher was executed directly as a small deterministic fallback check. It gave a positive score for overlapping backend skills and zero for an unrelated mobile-role comparison.

Pair	Semantic Path Status	TF-IDF Fallback Result	Interpretation
CV: Laravel Docker MySQL REST APIs; Job: Backend developer with Laravel Docker MySQL	Not executed locally; dependencies unavailable.	0.4316	Keyword overlap confirms a backend-oriented match signal.
CV: Flutter Dart mobile UI; Job: Backend developer with Laravel Docker MySQL	Not executed locally; dependencies unavailable.	0.0000	No meaningful keyword overlap, so fallback does not inflate score.
Expected runtime behavior	Sentence embeddings plus TF-IDF in /api/hybrid-match when both paths are available.	60 percent semantic/adaptive plus 40 percent TF-IDF in that endpoint.	The fallback helps explainable matching but is not a substitute for full semantic evaluation.

Table 30. Semantic matching and TF-IDF example results.

7.11 Model Evaluation Limitations

- The Colab PDF provides recorded overall training-run metrics, but the repository alone still does not include the dataset/model artifacts needed to reproduce the run.
- The cleaned labeled dataset used for final training is not committed.
- The model-weight folder is ignored by Git; safe local metadata was inspected, but binary weights were not copied or benchmarked.
- Local documentation Python did not include transformer, sentence-transformer, OCR, PDF, or Gemini packages, so live model inference and training were not rerun here.

- The Colab PDF does not show a per-label classification report, per-label support counts, or a confusion matrix.
- The examples in Table 29 are expected-behavior examples, while the TF-IDF values in Table 30 are actual small local fallback checks.
- A stronger final defense package should add a fixed labeled CV test set, saved per-label NER metrics, and CI-friendly inference smoke tests.

7.12 CV Analyzer Mini Dataset Evaluation

A sample PDF CV was generated for the screenshot workflow and uploaded through the running system. The upload succeeded, and the dashboard showed parsed CV data, backend role inference, extracted skills, and profile completeness. To strengthen the evaluation beyond that smoke test, this revision adds a mini synthetic dataset under docs/graduation-book/evaluation/.

The mini CV evaluation is explicitly offline and deterministic. It uses fake CV text, expected skill labels, and a keyword/role inference evaluator. It does not claim live model accuracy. The live AI CV Analyzer endpoint can be added to this mini-evaluation later, but the current document records only metrics that were actually computed from the synthetic dataset.

7.13 Recommendation Mini Dataset Evaluation

The recommendation mini evaluation ranks synthetic jobs for each synthetic CV using skill overlap plus domain and seniority bonuses. This validates the recommendation concept and provides a repeatable regression check for report evidence. It is not a production recommender benchmark, and the report does not claim complete job-market coverage.

7.14 Gap Analysis Mini Dataset Evaluation

The gap-analysis mini evaluation compares expected matched and missing skills with computed matched and missing skills for selected CV/job pairs. This directly validates the explanation structure used by the gap-analysis workflow: matched skills should be shown separately from missing skills.

Mini Dataset Files

The mini evaluation uses fake synthetic CVs and fake synthetic job records stored under docs/graduation-book/evaluation/. It is intentionally small and preliminary. It is useful for graduation validation and regression checks, but it is not statistically representative and should not be used as a production benchmark.

Sample ID	Expected Role	Seniority	Domain	Expected Skills
cv_backend_laravel	Backend Laravel Developer	junior	backend_web	PHP, Laravel, MySQL, REST API, Docker, Git
cv_frontend_react	Frontend React Developer	intern	frontend_web	React, JavaScript, Vite, HTML, CSS, API integration
cv_data_ml	Data and ML Student	student	data_ml	Python, pandas, scikit-learn, NLP, data analysis
cv_full_stack	Full Stack Developer	junior	full_stack_web	Laravel, React, MySQL, Docker, REST API, Git
cv_qa_testing	QA Testing Engineer	intern	quality_assurance	testing, test cases, pytest, API testing, bug reporting

Table 31. Mini CV dataset.

Job ID	Title	Domain	Required Skills
job_laravel_backend	Junior Laravel Backend Developer	backend_web	PHP, Laravel, MySQL, REST API, Docker, Git
job_react_frontend	React Frontend Intern	frontend_web	React, JavaScript, Vite, HTML, CSS, API integration
job_full_stack_web	Full Stack Web Developer	full_stack_web	Laravel, React, MySQL, Docker, REST API, Git
job_data_analyst	Junior Data Analyst	data_ml	Python, pandas, scikit-learn, NLP, data analysis
job_qa_intern	QA Automation Intern	quality_assurance	testing, test cases, pytest, API testing, bug reporting
job_devops_docker	DevOps Docker Intern	devops	Docker, Git, CI, Linux, monitoring
job_php_api	PHP API Developer	backend_web	PHP, Laravel, REST API, MySQL, Git, Docker
job_nlp_assistant	NLP Assistant Intern	data_ml	Python, NLP, scikit-learn, data analysis, testing

Table 32. Mini job dataset.

Metric Definitions

Skill precision measures how many extracted skills are expected labels. Skill recall measures how many expected skills were extracted. Skill F1 is the harmonic mean of precision and recall [30]. Recommendation top-1 and top-3 relevance compare ranked jobs against manual relevance labels. Gap agreement compares computed matched/missing skills against expected matched/missing skills.

Area	Metric	Value	Notes
CV offline	Macro skill precision	1.000	Keyword extraction over synthetic CV text
CV offline	Macro skill recall	1.000	Compared with expected skill labels
CV offline	Macro skill F1	1.000	F1 computed from precision/recall [30]
CV offline	Role match rate	1.000	Rule-based role inference on synthetic data
CV offline	Seniority match rate	1.000	Rule-based seniority inference
CV offline	Domain match rate	1.000	Rule-based domain inference
Recommendation offline	Top-1 relevance	1.000	Top recommendation belongs to manual relevant set
Recommendation offline	Top-3 relevance	1.000	Any top-3 job belongs to manual relevant set
Recommendation offline	Mean precision@3	0.800	Relevant jobs among top three
Gap offline	Matched skill agreement F1	1.000	Computed matched skills vs. expected matched skills
Gap offline	Missing skill agreement F1	1.000	Computed missing skills vs. expected missing skills

Table 33. Mini evaluation metrics.

Recommendation Ranking Details

CV Sample	Expected Relevant Jobs	Top 3 Offline Recommendations	P@3

CV Sample	Expected Relevant Jobs	Top 3 Offline Recommendations	P@3
cv_backend_laravel	job_laravel_backend, job_php_api, job_full_stack_web, job_devops_docker	job_laravel_backend, job_php_api, job_full_stack_web	1.000
cv_frontend_react	job_react_frontend, job_full_stack_web	job_react_frontend, job_full_stack_web, job_qa_intern	0.667
cv_data_ml	job_data_analyst, job_nlp_assistant	job_data_analyst, job_nlp_assistant, job_laravel_backend	0.667
cv_full_stack	job_full_stack_web, job_laravel_backend, job_php_api, job_react_frontend, job_devops_docker	job_full_stack_web, job_laravel_backend, job_php_api	1.000
cv_qa_testing	job_qa_intern, job_nlp_assistant	job_qa_intern, job_nlp_assistant, job_react_frontend	0.667

Table 34. Recommendation ranking details.

Gap Analysis Pair Details

CV / Job Pair	Matched Skills	Missing Skills	Agreement
cv_backend_laravel -> job_laravel_backend	Docker, Git, Laravel, MySQL, PHP, REST API	None	matched F1=1.000; missing F1=1.000
cv_frontend_react -> job_full_stack_web	React	Docker, Git, Laravel, MySQL, REST API	matched F1=1.000; missing F1=1.000
cv_data_ml -> job_nlp_assistant	NLP, Python, data analysis, scikit-learn	testing	matched F1=1.000; missing F1=1.000
cv_qa_testing -> job_qa_intern	API testing, bug reporting, pytest, test cases, testing	None	matched F1=1.000; missing F1=1.000
cv_full_stack -> job_react_frontend	React	API integration, CSS, HTML, JavaScript, Vite	matched F1=1.000; missing F1=1.000

Table 35. Gap analysis pair details.

7.15 AI CV Analyzer Smoke Evaluation

The smoke evaluation under docs/graduation-book/evaluation/ uses five short, fake CV text samples: backend, data analyst, frontend, DevOps/cloud, and low-information/noisy input. It is deterministic and useful as a small reproducibility check. It does not run the full transformer NER model and must not be reported as final model accuracy.

The script also probes the local documentation runtime. In this run, full analyzer import was unavailable because ModuleNotFoundError: No module named 'pdfplumber'. The pure Python TF-IDF matcher was available, with a backend-overlap probe score of 0.5101.

Package	Status

Package	Status
easyocr	Unavailable
pdfplumber	Unavailable
pydantic	Available
sentence_transformers	Unavailable
torch	Unavailable
transformers	Unavailable

Dependency probe for the AI CV Analyzer smoke evaluation.

Area	Metric	Value	Notes
AI analyzer smoke	Macro skill precision	0.971	Five manually labeled text samples
AI analyzer smoke	Macro skill recall	1.000	Expected skill labels defined in smoke sample JSON
AI analyzer smoke	Macro skill F1	0.985	Deterministic text smoke metric, not NER benchmark
AI analyzer smoke	Role match rate	1.000	Simple role inference over sample text
AI analyzer smoke	Domain match rate	1.000	Simple domain evidence markers
AI analyzer smoke	Seniority match rate	0.800	One mismatch preserved as evidence of limitation
AI analyzer smoke	Parsing status match rate	1.000	Includes low-information abstention sample

AI CV Analyzer smoke evaluation metrics.

Sample	Skill F1	Role	Domain	Seniority	Status
smoke_backend_laravel	1.000	Pass	Pass	Check	Pass
smoke_data_analyst	1.000	Pass	Pass	Pass	Pass
smoke_frontend_react	0.923	Pass	Pass	Pass	Pass
smoke_devops_cloud	1.000	Pass	Pass	Pass	Pass
smoke_low_information	1.000	Pass	Pass	Pass	Pass

Per-sample AI CV Analyzer smoke evaluation results.

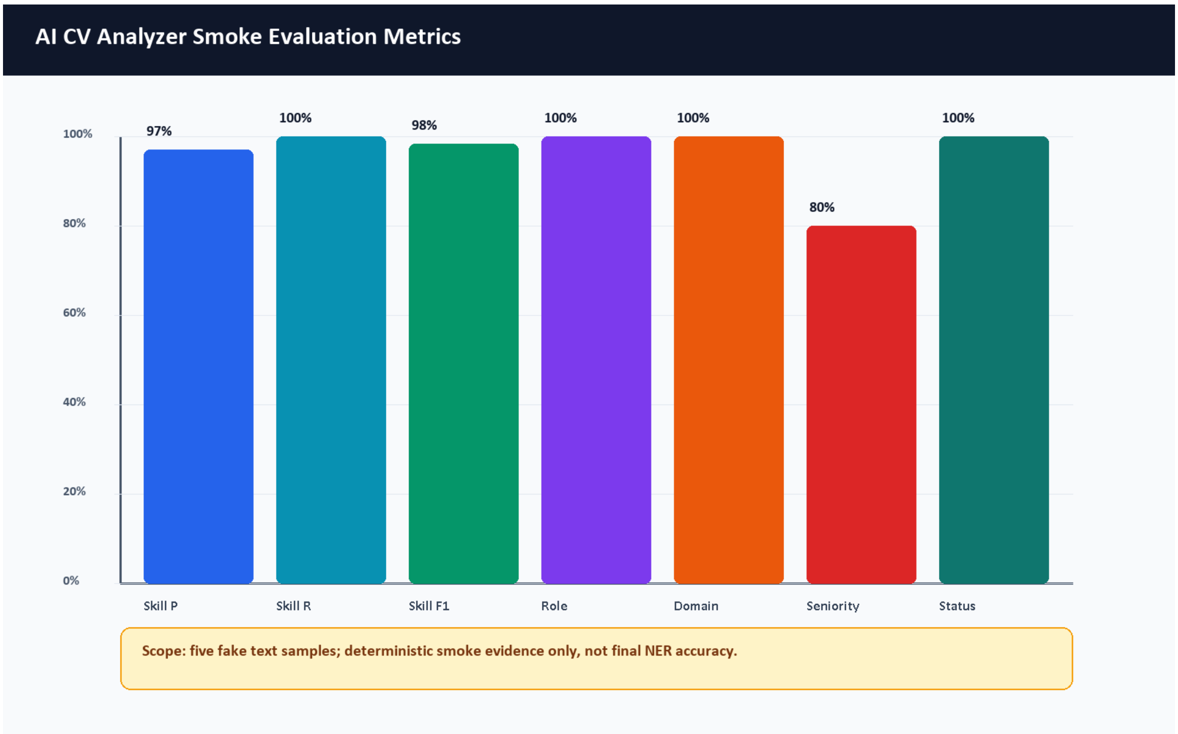


Figure 30. AI CV Analyzer smoke evaluation metrics.

7.16 Job Miner Evaluation

The AI Job Miner test suite passed with 75 tests. Admin source diagnostics displayed active sources and source state. The job database contained imported/demo jobs visible in the admin dashboard. Because external job sources can change or throttle scraping, long-term data quality evaluation should be repeated near the final defense.

7.17 Application Tracker Evaluation

The application tracker was evaluated by saving a selected job to the tracker and loading the Applications page. The screenshot shows saved opportunity state. Backend tests also include application tracker behavior.

7.18 Admin Dashboard Evaluation

Admin login and admin dashboard access were tested with the demo admin account. The admin dashboard, admin jobs, admin sources, and admin target roles pages were captured. The dashboard displayed 22 users, 209 imported jobs, 9 active sources, and 13 target roles at capture time.

7.19 Performance Observations

The local Docker stack is heavy because it runs frontend, backend, multiple Laravel workers, MySQL, MinIO, two Python AI services, Prometheus, and Grafana. Initial full build can exceed a short command timeout on a Windows laptop. Once images are built, targeted service startup and HTTP checks are practical for a graduation demo.

7.20 Evaluation Limitations

- The AI CV Analyzer pytest suite was not executed because pytest was absent in that container.
- The browser CV upload remains a smoke test, and the mini dataset is synthetic rather than statistically representative.

- The AI CV Analyzer training notebook and exported Colab PDF were inspected, but full model training was not rerun because the cleaned training dataset and external generation keys are not committed and the workflow is designed for Colab GPU execution.
- The Colab metrics are recorded training-run evidence on the notebook split, not a production benchmark on a large real-world CV dataset.
- The AI CV Analyzer smoke evaluation uses five deterministic text samples and does not measure the transformer NER model weights.
- The recommendation score shown in screenshots is an estimated local demo output.
- External scraping reliability depends on source availability and changing website/API behavior.
- Production security, privacy, and performance audits remain future work.

7.21 Summary of Results

Area	Command or Scenario	Result	Evidence
Docker config	docker compose config --quiet	Passed	Terminal evidence
Docker prod config	docker compose -f docker-compose.yml -f docker-compose.prod.yml config --quiet	Passed	Terminal evidence
Backend dependencies	composer install --no-interaction --prefer-dist	Passed	Command output
Backend routes	php artisan route:list	Passed, 131 routes	Command output
Backend tests	php artisan test	Passed, 39 tests, 297 assertions	Command output
Frontend lint	ESLint	Passed, 9 warnings, 0 errors	Command output
Frontend build	Vite build	Passed, 2904 modules transformed	Command output
AI Job Miner tests	python -m pytest	Passed, 75 tests	Command output
AI CV Analyzer syntax	python -m compileall .	Passed	Command output
AI CV Analyzer pytest	python -m pytest	Skipped, pytest missing	Command output
HTTP probes	/, /api/health, /api/ready, /status, AI services	200 responses	Command output

Table 36. Automated validation results.

Test ID	Module	Scenario	Status	Evidence
M-01	Authentication	Register demo user	Passed	Register screenshot/API output
M-02	Authentication	Login student	Passed	Figure 34
M-03	CV upload	Upload valid PDF	Passed	Figures 35-37
M-04	CV upload	Invalid file handling	Not Run Manual	Backend validation tests
M-05	Recommendations	Open jobs page after CV	Passed	Figure 38
M-06	Gap analysis	Analyze selected job	Passed	Figure 40
M-07	Tracker	Save job	Passed	Figure 41
M-08	Admin	Login admin and open dashboard	Passed	Figure 44

Test ID	Module	Scenario	Status	Evidence
M-09	Admin sources	Open diagnostics	Passed	Figure 46
M-10	Status	Open system status page	Passed	Figure 43

Table 37. Manual functional evaluation matrix.

Test ID	Expected vs Actual Observation
M-01	Expected user creation and token return; actual user was created with an accepted Gmail-style address.
M-02	Expected dashboard access after login; actual dashboard loaded.
M-03	Expected CV acceptance and parsing; actual upload parsed successfully.
M-04	Expected validation error for invalid file; actual manual browser repetition was not run because backend validation/tests already cover the rule.
M-05	Expected recommendations with estimated matches; actual jobs page loaded.
M-06	Expected match and skill breakdown; actual gap page loaded.
M-07	Expected saved job in tracker; actual application page loaded with saved item.
M-08	Expected admin-only dashboard; actual dashboard visible after admin login.
M-09	Expected source diagnostics; actual diagnostics page visible.
M-10	Expected health UI; actual system status page visible.

Table 38. Manual functional observations.

Chapter 8: Security and Privacy

8.1 Introduction

Security in CareerCompass is implemented for a graduation/demo context. The system includes meaningful controls, but it should not be presented as production-grade without future hardening, legal review, and operational security work.

8.2 Authentication and Authorization

User authentication uses token-based API access through Laravel. Laravel Sanctum supports API token and SPA authentication use cases [2]. CareerCompass stores an auth token in the browser and attaches it to API requests. Authorization is role-aware: student routes require authentication, while admin routes require the admin role. Authentication security should follow established password and session guidance in production [28].

8.3 Admin Access Control

Admin access is enforced server-side by admin middleware. Frontend route protection improves user experience, but the backend check is the important control. The demo admin account is generated by a seeder and should be changed or disabled in any non-demo environment.

8.4 CV File Privacy

CV files contain personal data. CareerCompass validates file type and size, stores files privately, and avoids public direct file exposure. OWASP's file upload guidance emphasizes validation, restricted storage, and safe handling [27].

8.5 Private Storage and Signed Downloads

Uploaded CV files are stored through private storage and accessed through signed or temporary URLs. MinIO/S3-compatible storage supports object-based file storage and access control patterns [9]. The graduation demo should still avoid uploading real sensitive CVs unless the environment is controlled.

The model-training workflow is intentionally documented separately from runtime CV processing. Synthetic training snippets may be generated through Google AI developer tooling [34], [35], but real student CV uploads should not be sent to external AI APIs without explicit consent, a privacy notice, retention rules, and supervisory approval. In the demonstrated runtime, Laravel sends the uploaded file to the local FastAPI analyzer service and stores the file privately; the Gemini-based generator is a training-support script, not the normal CV-upload path.

8.6 Internal Service Tokens

Scraper import routes use a service token middleware. This reduces accidental public ingestion, but service tokens must be rotated, stored securely, and monitored in production.

8.7 Payload and File Validation

Laravel form requests validate registration, login, CV upload, applications, and profile updates. File validation includes MIME type, extension, and size checks. Validation reduces risk but does not replace malware scanning, deep file inspection, or content disarm in production.

8.8 Logging and Request IDs

The API client attaches request IDs, and backend logging records important events such as CV processing status and AI gateway errors. For production, logs should avoid sensitive CV content and should be retained according to a privacy policy.

8.9 Demo Security Limitations

Area	Current Demo Control	Production Hardening Needed
Admin account	Demo seeder account	Secret rotation, SSO/MFA, audit logs
CV files	Private storage and signed URLs	Malware scanning, retention policy, consent model
Tokens	Bearer tokens	Token rotation, secure cookie strategy, revocation review
Scraper service	Internal token	Secret manager, network isolation, rate limits
Monitoring	Local Prometheus/Grafana	Auth, TLS, dashboard access control
Privacy	Local demo posture	Legal review, privacy notice, data minimization

Table 39. Security and privacy controls.

8.10 Future Production Hardening

Future work should include HTTPS-only deployment, secure cookie/session strategy, administrator MFA, centralized secrets management, object scanning, retention policies, audit logging, rate-limit review, CSRF/CORS review, dependency vulnerability scanning, and a privacy impact assessment.

Chapter 9: Conclusion and Future Work

9.1 Conclusion

CareerCompass demonstrates a practical AI-assisted career guidance workflow for a graduation project. It integrates CV parsing, profile normalization, skill extraction, imported job records, estimated recommendation, gap analysis, application tracking, admin diagnostics, and containerized deployment.

9.2 Project Achievements

- Built a working multi-service web application with frontend, backend, AI services, database, object storage, proxy, and monitoring.
- Implemented student authentication, dashboard, CV upload, profile view, jobs, gap analysis, and application tracking.
- Implemented admin dashboard, job management, source diagnostics, and target role management.
- Documented the AI CV Analyzer runtime architecture, optional local NER artifact path, synthetic data-generation workflow, and Colab training notebook.
- Added tests and validation commands across backend, frontend, Python services, Docker, and HTTP probes.
- Captured browser screenshots from the running local system.
- Generated a formal report with diagrams, references, and evaluation notes.

9.3 Educational Value

The project demonstrates practical learning in software architecture, service decomposition, secure file handling, API design, database schema design, frontend state management, AI service integration, testing, Docker operations, monitoring, and academic documentation.

9.4 Current Limitations

- The system is a graduation/demo platform and not a production product.
- Recommendation and gap analysis outputs are estimates.
- AI evaluation needs larger labeled datasets, committed training artifacts, per-label reports, and repeatable model scoring.
- The exported NER model artifact path is supported by the runtime, but model weights are ignored by Git and the repository does not include a reproducible final metric run from the training notebook.
- The Colab PDF records overall NER training metrics, but the final labeled NER dataset was not available in committed evidence, so label distribution and final per-label NER precision/recall/F1 were not claimed.
- OCR and PDF text-recovery quality can affect scanned CVs, image-heavy layouts, multi-column documents, and unusual fonts.
- Synthetic training examples may not represent all real student CV styles, Arabic/multilingual CVs, or informal job-market wording.
- External scraping sources can be unstable.
- The AI CV Analyzer container needs pytest installed to run its test suite.
- Security and privacy controls need production hardening before real deployment.

9.5 Future Work

- Add a larger CV/job evaluation dataset with manual labels.

- Add a reproducible NER evaluation pipeline that runs on a fixed labeled test set and records per-label precision, recall, and F1.
- Store model cards, dataset cards, and training-run summaries for each exported model artifact.
- Add a privacy-preserving workflow for any future human-reviewed real CV dataset.
- Improve Arabic/multilingual parsing, OCR, and field extraction.
- Expand the role taxonomy, skill aliases, and labor-market evidence with periodic review.
- Add human-in-the-loop review for high-impact guidance and ambiguous CV parsing results.
- Improve skill normalization through reviewed aliases and false-positive checks.
- Add production-grade authentication and administrator controls.
- Add malware scanning and retention policies for uploaded CV files.
- Improve recommendation explanations and calibration.
- Add automated browser end-to-end tests.
- Extend CI to run all containerized test suites consistently.
- Improve observability dashboards and alerting.
- Add deployment documentation for a secure cloud environment.

9.6 Final Remarks

CareerCompass is an original graduation project that connects academic software engineering requirements with a practical career guidance problem. Its strongest value is the integration of many realistic system parts into one demonstrable workflow, while preserving honest language about limitations and future production work.

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Appendices

Appendix A: API Request and Response Examples

This appendix expands the endpoint summary with JSON-oriented examples. The examples are intentionally small and use placeholders such as Authorization: Bearer <token>. They document the implemented API shape for examiners and future maintainers; they do not expose real tokens, private CV contents, or production secrets. The format follows an OpenAPI-style request/response documentation pattern [39].

Group	Example Endpoints	Purpose
Health	/api/health, /api/ready, /api/metrics	Liveness, readiness, and Prometheus metrics.
Auth	/api/v1/register, /api/v1/login, /api/v1/logout, /api/v1/user	User identity and token lifecycle.
CV	/api/v1/upload-cv, /api/v1/user/skills, /api/v1/user/cv-analysis	CV upload, parsed analysis, and extracted skills.
Jobs	/api/v1/jobs, /api/v1/jobs/recommended, /api/v1/jobs/{id}	Job listing, details, and recommendations.
Gap Analysis	/api/v1/gap-analysis/job/{jobId}, /api/v1/gap-analysis/role/{roleId}	Skill comparison and recommendations.
Applications	/api/v1/applications	Save and update tracked opportunities.
Admin	/api/v1/admin/dashboard/stats, /api/v1/admin/jobs, /api/v1/admin/scraping-sources, /api/v1/admin/target-roles	Admin dashboards and diagnostics.
Internal Scraper	/api/jobs/import, /api/jobs/import/check, /api/proxies/active	Service-token protected import routes.

Table 40. API endpoint summary.

A.1 Student CV Upload Endpoint

Method and URL: POST /api/v1/upload-cv

Purpose: Upload a PDF/image CV to Laravel, send it to the local FastAPI AI CV Analyzer, persist successful structured outputs, and return warnings/status for the dashboard.

Request example:

```
Authorization: Bearer <token>
Content-Type: multipart/form-data

cv=@sample_cv.pdf
```

Successful response example:

```
{
  "success": true,
  "message": "CV parsed successfully.",
  "parsing_status": "success",
```

```
  "analysis_id": 12,
  "skills_count": 4,
  "predicted_role": "Backend Developer",
  "profile_updated": true,
  "retry_available": false,
  "download_url": "https://example.local/api/cv-files/12?signature=...",
  "data": {
    "analysis_id": 12,
    "parsing_status": "success",
    "skills_count": 4,
    "predicted_role": "Backend Developer",
    "warnings": []
  }
}
```

Error response example:

```
{
  "success": false,
  "message": "The AI engine is currently unavailable. Please try again in a moment.",
  "parsing_status": "error",
  "retry_available": true,
  "warnings": [
    {
      "code": "ai_unavailable",
      "message": "The AI engine could not be reached. No profile data was changed."
    }
  ]
}
```

A.2 AI Analyzer Parse-CV Endpoint

Method and URL: POST /api/parse-cv

Purpose: FastAPI service endpoint that accepts an uploaded CV file and returns the typed parse schema used by Laravel.

Request example:

```
Content-Type: multipart/form-data

file=@sample_cv.pdf
```

Response example:

```
{
  "parsing_status": "success",
  "profile": {
    "full_name": "Ahmed Mohamed",
    "current_title": "Backend Developer",
    "email": "ahmed@example.com"
  },
  "stats": {
    "page_count": 1,
    "word_count": 340,
    "language_hint": "en"
  },
  "skills": {
    "items": [
      {"name": "Laravel", "category": "hard", "confidence_score": 0.65},
      {"name": "Docker", "category": "hard", "confidence_score": 0.65}
    ],
    "confidence_score": 0.65
  },
  "experience": {
    "items": [],
    "confidence_score": 0.0
  },
```

```
  "analysis": {
    "predicted_role": "Backend Developer",
    "seniority": "junior",
    "primary_domain": "Backend Development",
    "strengths": [],
    "gaps": [],
    "red_flags": [],
    "confidence_score": 0.65
  },
  "request_id": "example-request-id"
}
```

Error response example:

```
{
  "detail": "Empty file uploaded."
}
```

A.3 AI Hybrid Match Endpoint

Method and URL: POST /api/hybrid-match

Purpose: Compare CV text/skills with a job description using Layer 3 adaptive matching plus TF-IDF when available.

Request example:

```
{
  "cv_skills": ["Laravel", "Docker", "MySQL"],
  "cv_text": "Backend developer with Laravel Docker MySQL REST APIs.",
  "job_description": "Junior backend developer required with Laravel, Docker, MySQL, and REST API
experience.",
  "job_skills": ["Laravel", "Docker", "MySQL", "REST APIs"]
}
```

Response example:

```
{
  "hybrid_match_score": 78.4,
  "semantic_match_pct": 82.0,
  "tfidf_score_pct": 73.0,
  "missing_skills": [],
  "formula": "Final = (Adaptive Layer 3 x 60%) + (TF-IDF x 40%)",
  "matching_mode": "hybrid",
  "request_id": "example-request-id"
}
```

Validation error example:

```
{
  "detail": "cv_text must not be empty."
}
```

A.4 Recommended Jobs Endpoint

Method and URL: GET /api/v1/jobs/recommended

Purpose: Return personalized job recommendations for an authenticated student when CV/profile context exists, otherwise return recent usable jobs.

Request example:

```
Authorization: Bearer <token>
Accept: application/json
```

Response example:


```
{
  "success": true,
  "job_title": "Backend Developer",
  "data": [
    {
      "id": 101,
      "title": "Junior Backend Developer",
      "company": "DemoTech",
      "location": "Cairo",
      "source": "demo",
      "match_percentage": 84.5,
      "skills_count": 4
    }
  ],
  "meta": {
    "total": 1,
    "based_on": "Your CV title: Backend Developer"
  }
}
```


A.5 Gap Analysis Endpoint

Method and URL: GET /api/v1/gap-analysis/job/{jobId}

Purpose: Compare the authenticated student's extracted skills/profile with one job and return matched skills, missing skills, recommendations, and CV-analysis context.

Request example:

```
Authorization: Bearer <token>
Accept: application/json
```

Response example:


```
{
  "success": true,
  "data": {
    "job": {
      "id": 101,
      "title": "Junior Backend Developer",
      "company": "DemoTech"
    },
    "analysis": {
      "match_percentage": 80.0,
      "match_level": "Good Match",
      "matched_skills": ["Laravel", "Docker", "MySQL"],
      "missing_skills": ["Kubernetes"]
    },
    "recommendations": [
      "Practice Kubernetes deployment basics before applying."
    ],
    "cv_analysis": {
      "parsing_status": "success",
      "completeness_score": 75,
      "strengths": [],
      "gaps": [],
      "red_flags": []
    }
  }
}
```


Error response example:

```
{
  "success": false,
  "message": "Upload a CV first so the system can extract skills and profile data."
}
```

A.6 Health and Readiness Endpoints

Methods and URLs: GET /api/health, GET /api/ready

Purpose: Confirm whether Laravel is alive and whether dependent services are ready.

Health response example:

```
{
  "success": true,
  "status": "ok",
  "service": "CareerCompass API",
  "request_id": "example-request-id"
}
```

Readiness response example:

```
{
  "success": true,
  "status": "ready",
  "checks": {
    "database": {"ok": true},
    "cache": {"ok": true},
    "ai": {"ok": true, "status": 200},
    "scraper": {"ok": true, "status": 200}
  },
  "request_id": "example-request-id"
}
```

Appendix B: Quick Start Guide for Examiners

This guide is intended for a local graduation/demo run. It assumes Docker Desktop is installed and running. It should not be read as a production deployment guide.

B.1 Start the Stack

```
git clone https://github.com/YousefAlTohamy/CareerCompass.git
cd CareerCompass
cp .env.example .env
docker compose -f docker-compose.yml -f docker-compose.prod.yml up -d --build
docker compose exec backend-api php artisan migrate --seed
```

B.2 Useful URLs

- Frontend: http://localhost
- Backend health: http://localhost/api/health
- Backend readiness: http://localhost/api/ready
- System status page: http://localhost/status
- Admin dashboard: http://localhost/admin/dashboard
- Grafana, if exposed by the local compose stack: http://localhost:3000

B.3 Demo Accounts and Flow

The demo admin account is seeded from environment variables. The current example values are:

Variable	Demo Value
DEMO_ADMIN_EMAIL	careercompassadmin@gmail.com
DEMO_ADMIN_PASSWORD	CareerCompassAdmin2026

Demo admin environment values.

Student demo flow:

1. Open http://localhost.
2. Register a new student account or log in with an existing demo student.
3. Upload a small PDF CV from the dashboard.
4. Review dashboard readiness signals, extracted role, extracted skills, and profile page.
5. Open Jobs and review recommended jobs.
6. Open Gap Analysis for a selected job and review matched/missing skills.
7. Save a job to Applications.
8. Log in as the demo admin and review dashboard, jobs, sources, and target roles.

B.4 Validation Commands

```
docker compose config --quiet
docker compose -f docker-compose.yml -f docker-compose.prod.yml config --quiet
docker compose exec backend-api php artisan test
cd frontend
npm run lint
npm run build
```

Documentation/evaluation checks:

```
python docs/graduation-book/evaluation/run_mini_evaluation.py
python docs/graduation-book/evaluation/run_ai_cv_analyzer_smoke_eval.py
python docs/graduation-book/scripts/generate_graduation_book.py
python -m compileall ai-cv-analyzer
```

B.5 Troubleshooting Notes

- If the frontend appears stale after rebuild, hard-refresh the browser or rebuild the frontend/nginx containers.
- If Docker startup is slow, wait for MySQL, MinIO, Laravel workers, Python AI services, Prometheus, and Grafana to settle before judging readiness.
- AI model weights are not committed to Git; the runtime supports ai-cv-analyzer/models/ner_weights/career_compass_ner_final when supplied locally.
- NER training requires a cleaned labeled dataset and external API keys for synthetic-data generation; those are not included in the repository.
- Demo admin credentials should be changed for any non-demo environment.

Appendix C: Database Tables Summary

Table	Purpose
users	Authentication identity, role, and banned status.
user_profiles	Student profile details and contact metadata.
skills	Canonical skill catalog.

Table	Purpose
user_skills	User-to-skill relationship and proficiency metadata.
user_experiences	Extracted or entered experience records.
cv_analyses	CV parsing status, predicted role, confidence, file metadata, strengths, gaps, and warnings.
job_postings	Imported/demo job data.
job_skills	Job-to-skill relationship and requirement metadata.
applications	Saved opportunities and status tracking.
scraping_sources	Admin-configured job source definitions.
target_job_roles	Target role list for scraping and market exploration.
scraping_jobs	Scraping batch execution state.

Table 41. Database tables summary.

Appendix D: Docker Services Summary

Service	Role
nginx	Public gateway and reverse proxy.
frontend	React/Vite web UI container.
backend-api	Laravel API runtime.
backend-worker variants	Queue processing for default, high, AI, emails, and scraping work.
backend-scheduler	Scheduled Laravel commands.
db	MySQL database.
minio	S3-compatible CV object storage.
ai-cv-analyzer	FastAPI CV parsing service.
ai-job-miner	FastAPI job mining service.
prometheus	Metrics collection.
grafana	Metrics visualization.

Table 42. Docker services summary.

Appendix E: Screenshots

The screenshot set was reviewed during this pass. Some dashboard states are similar, but they document different examiner-visible states: before CV upload, upload UI, and after analysis. No screenshot merge was performed because combining them would reduce traceability and could disturb existing figure references in the generated List of Figures.

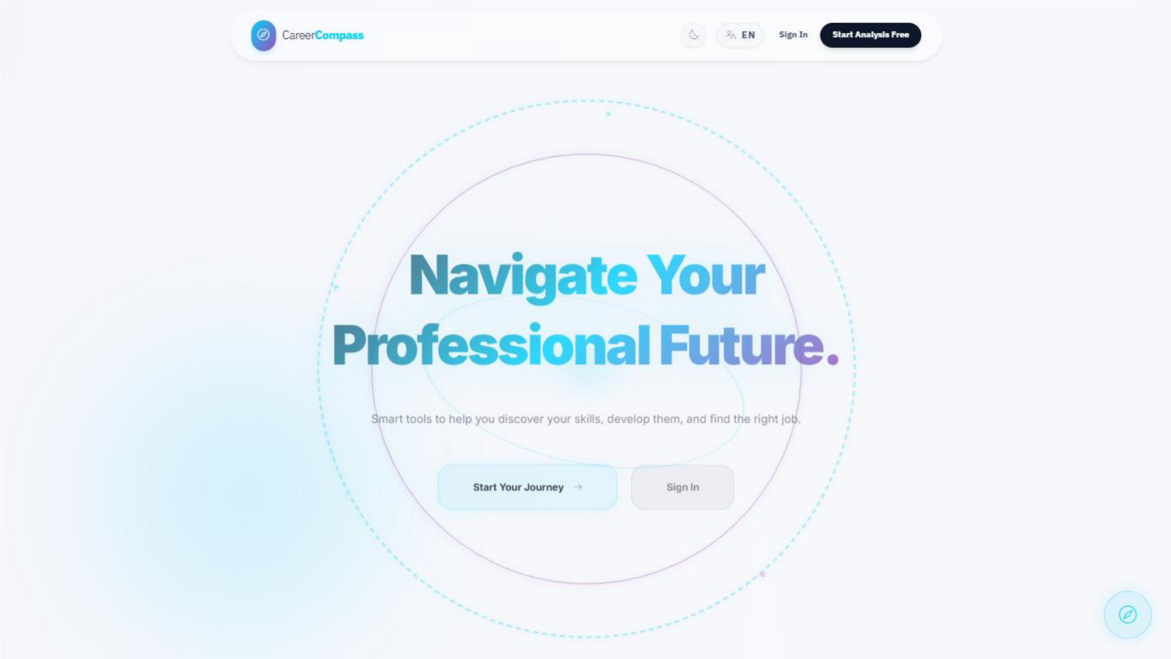

Figure 31. Home page.

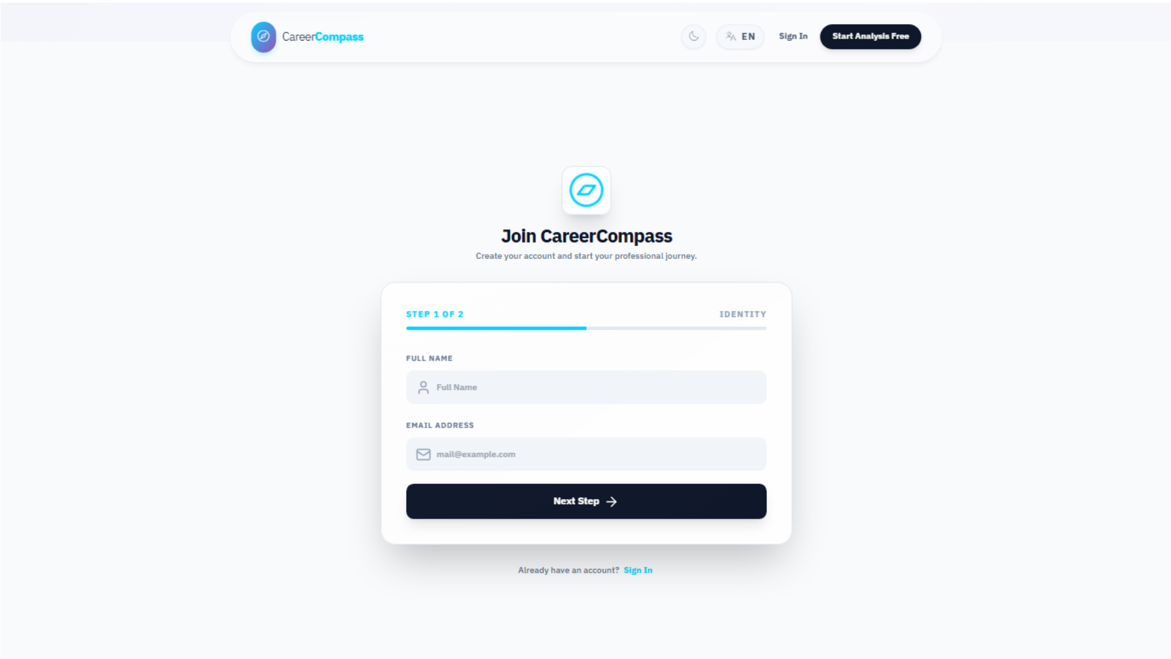

Figure 32. Register page.

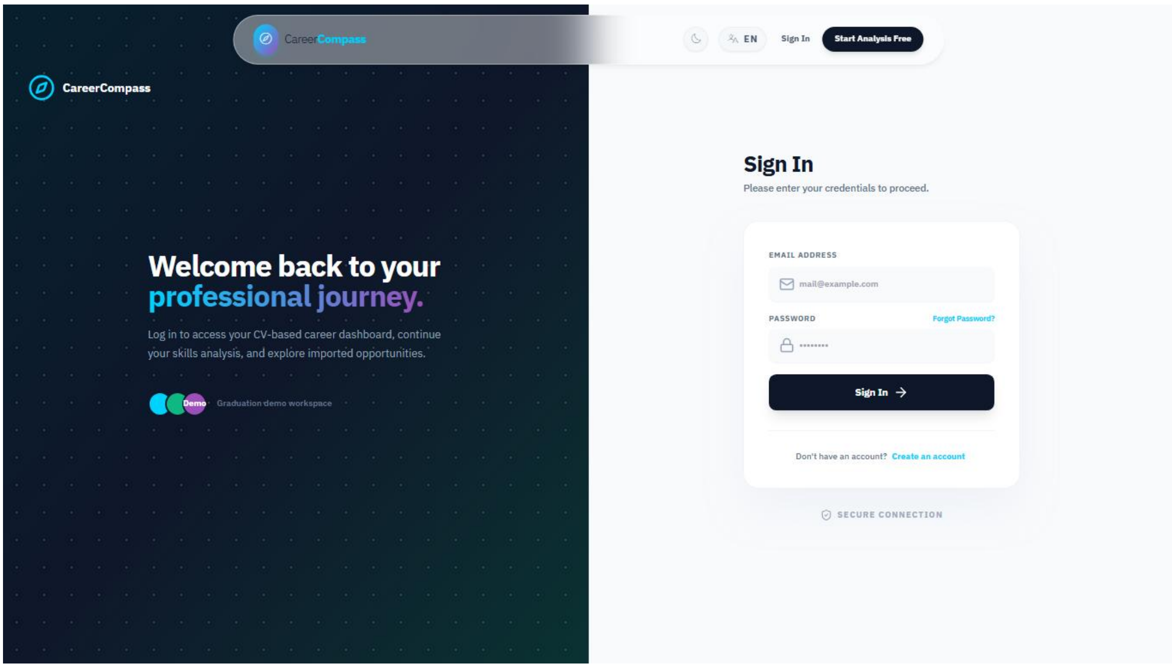

Figure 33. Login page.

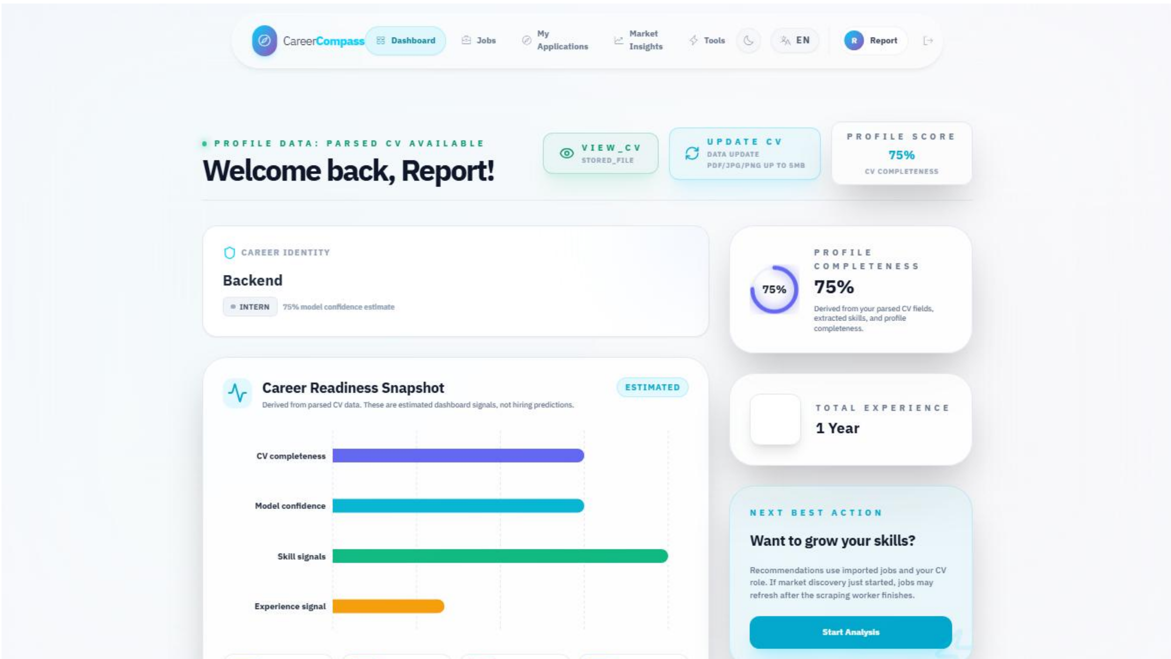

Figure 34. Student dashboard before CV upload.

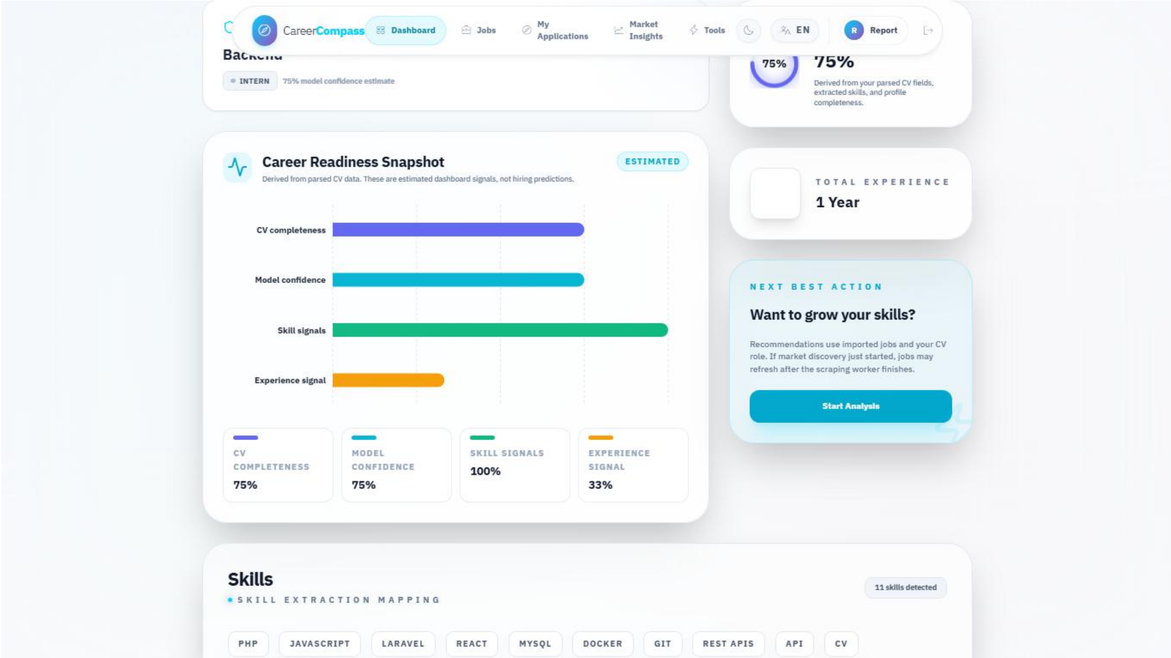

Figure 35. CV upload user interface.

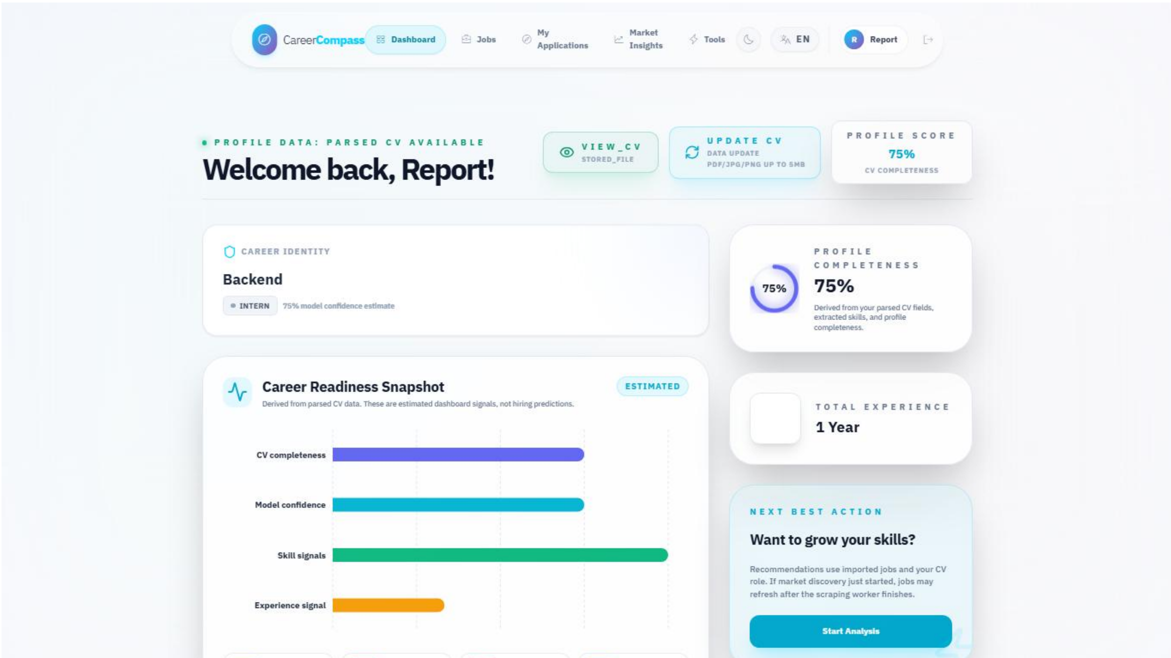

Figure 36. Dashboard after successful CV parsing.

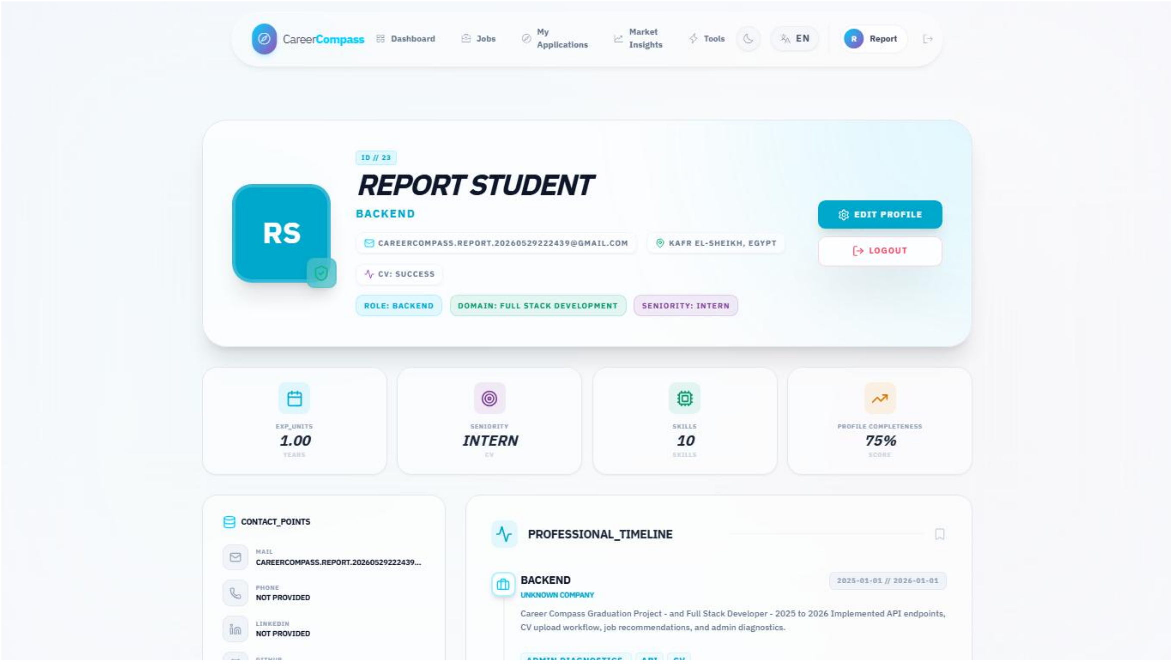

Figure 37. Extracted profile and skills page.

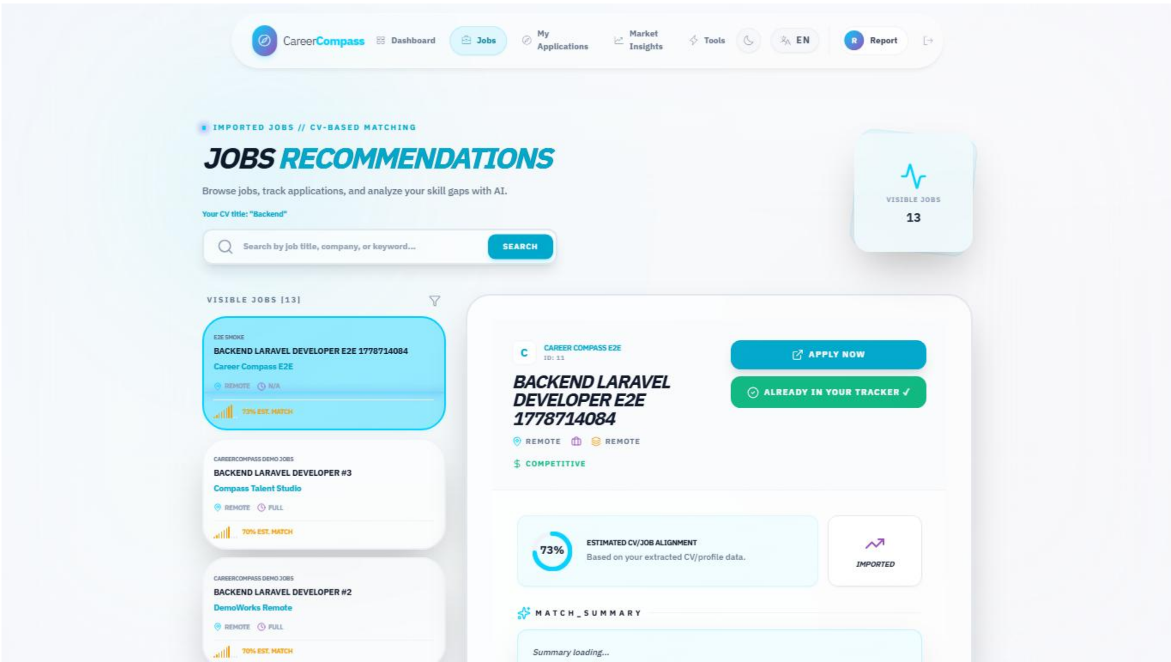

Figure 38. Jobs recommendations page.

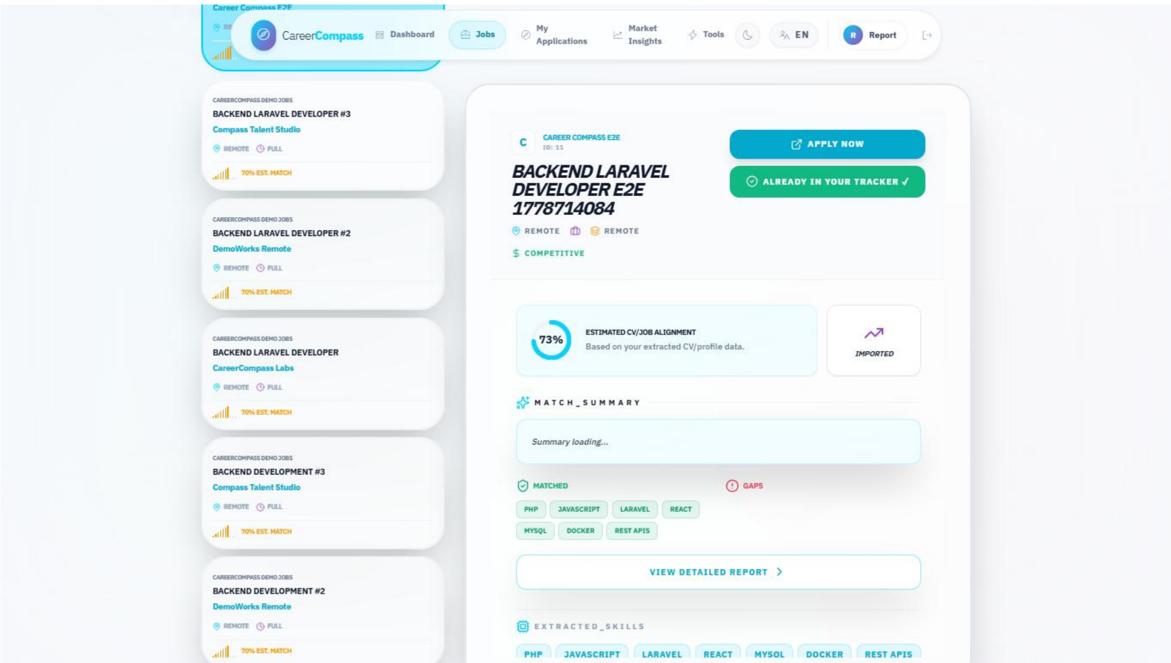

Figure 39. Job detail and inline gap panel.

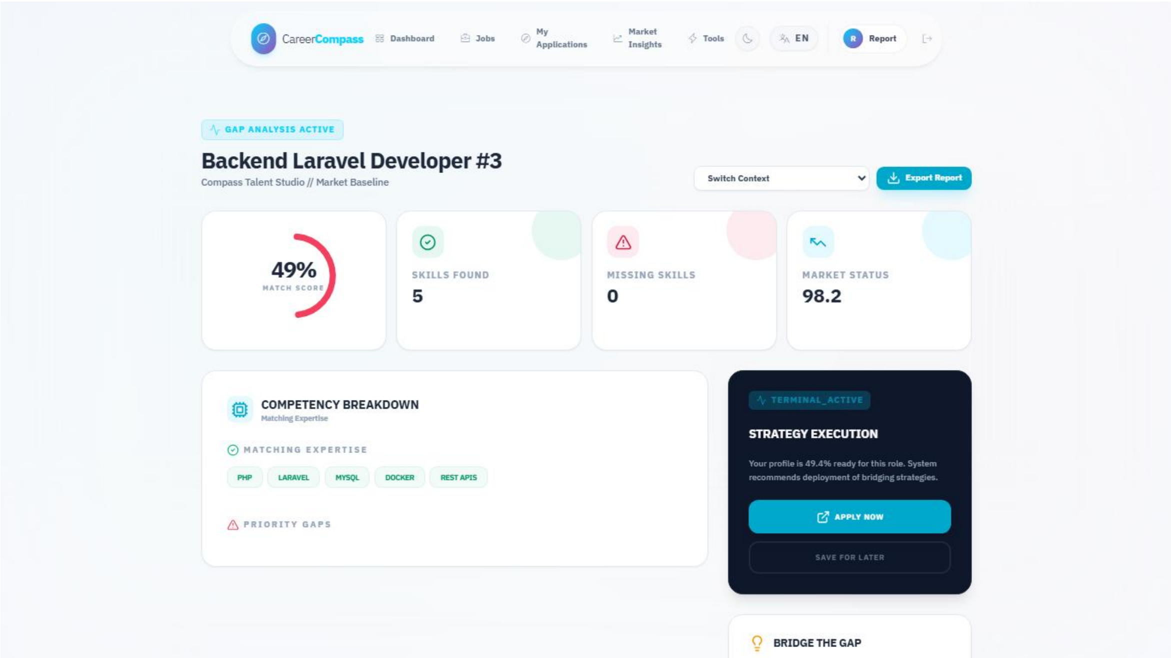

Figure 40. Gap analysis page.

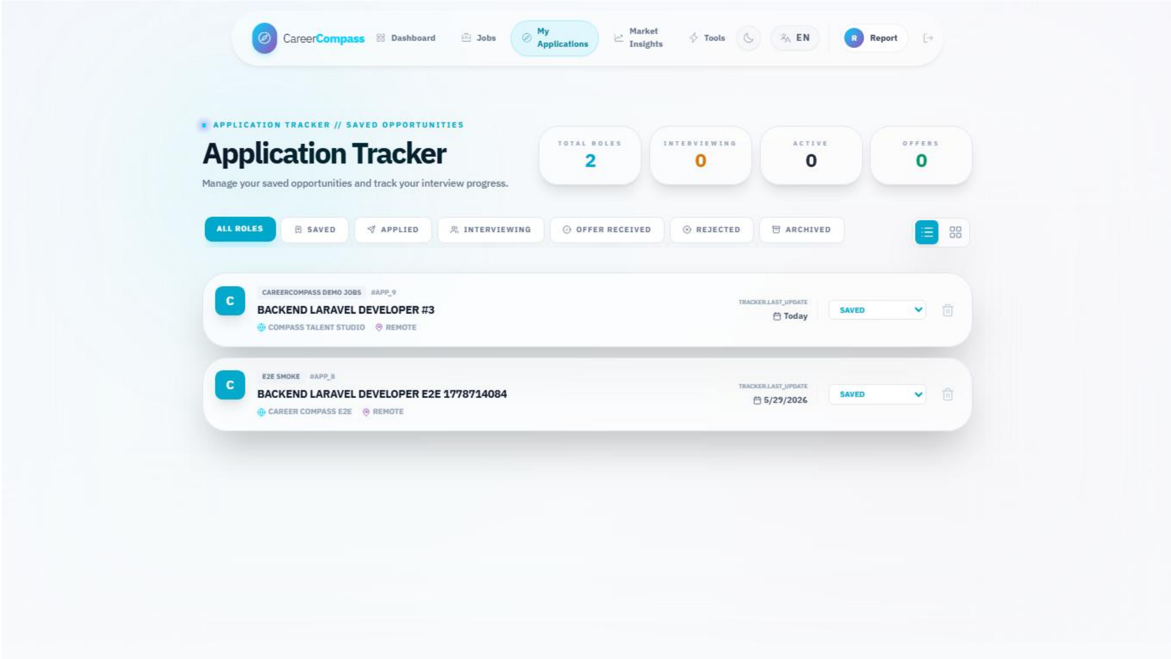

Figure 41. Applications tracker page.

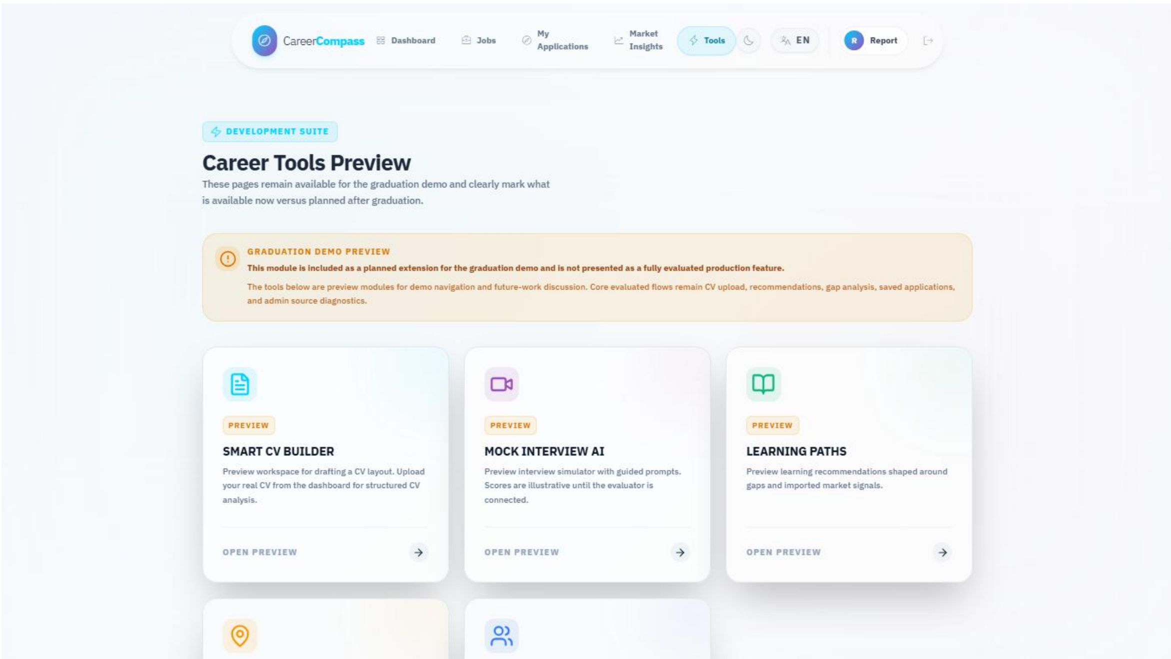

Figure 42. Tools Hub preview page.

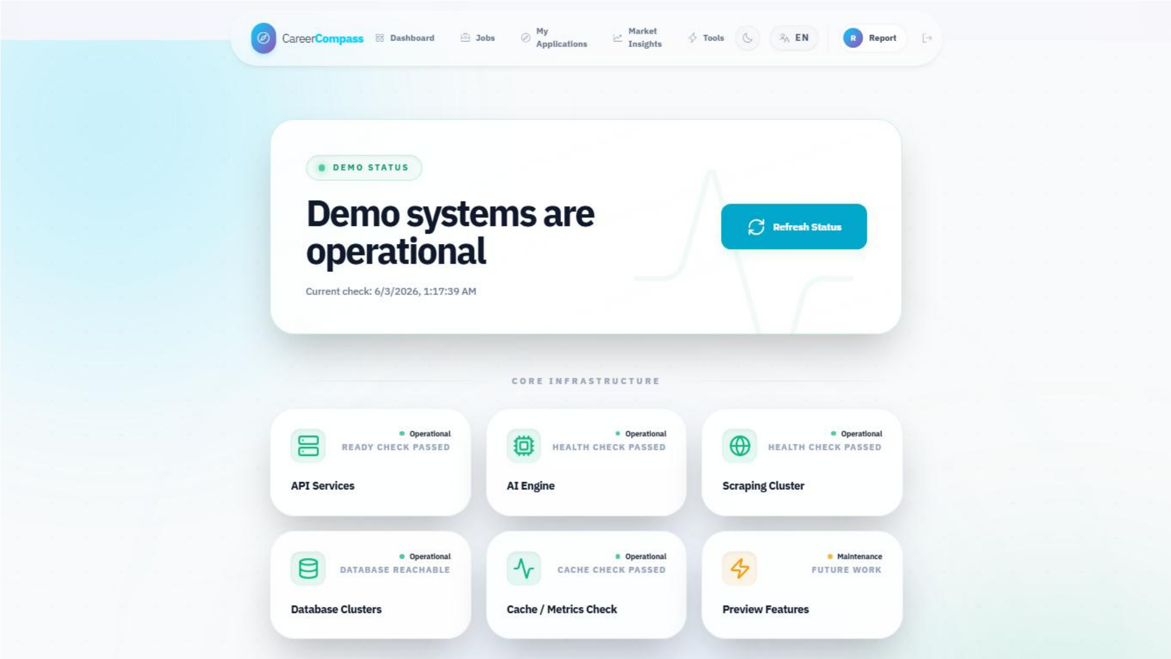

Figure 43. System status page.

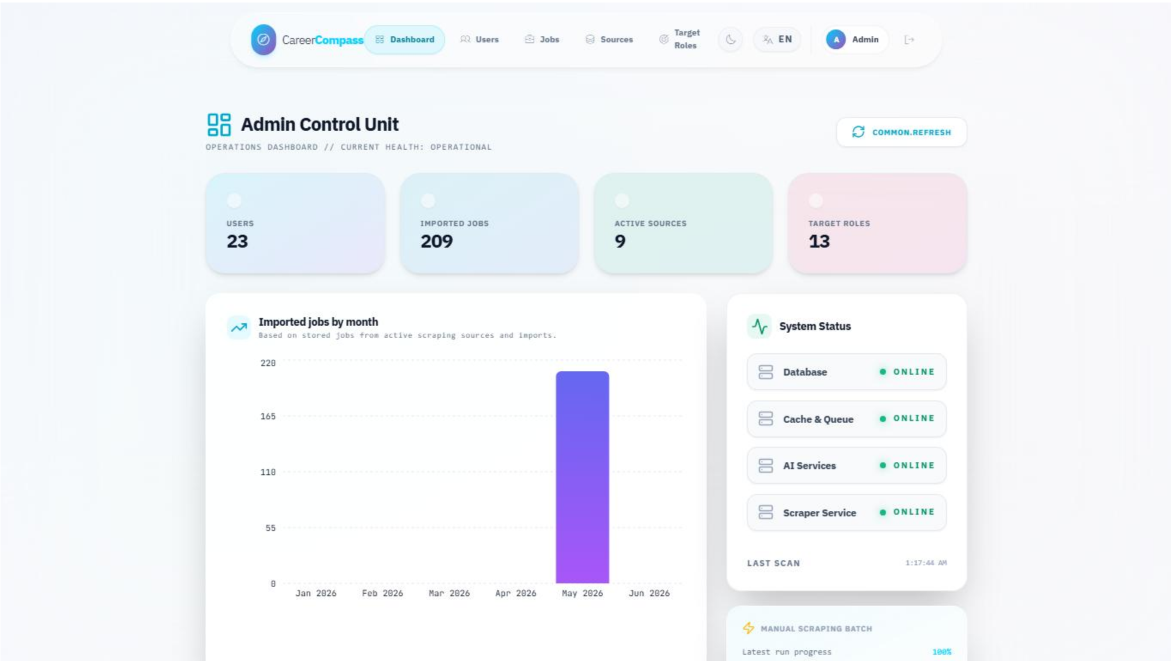

Figure 44. Admin dashboard.

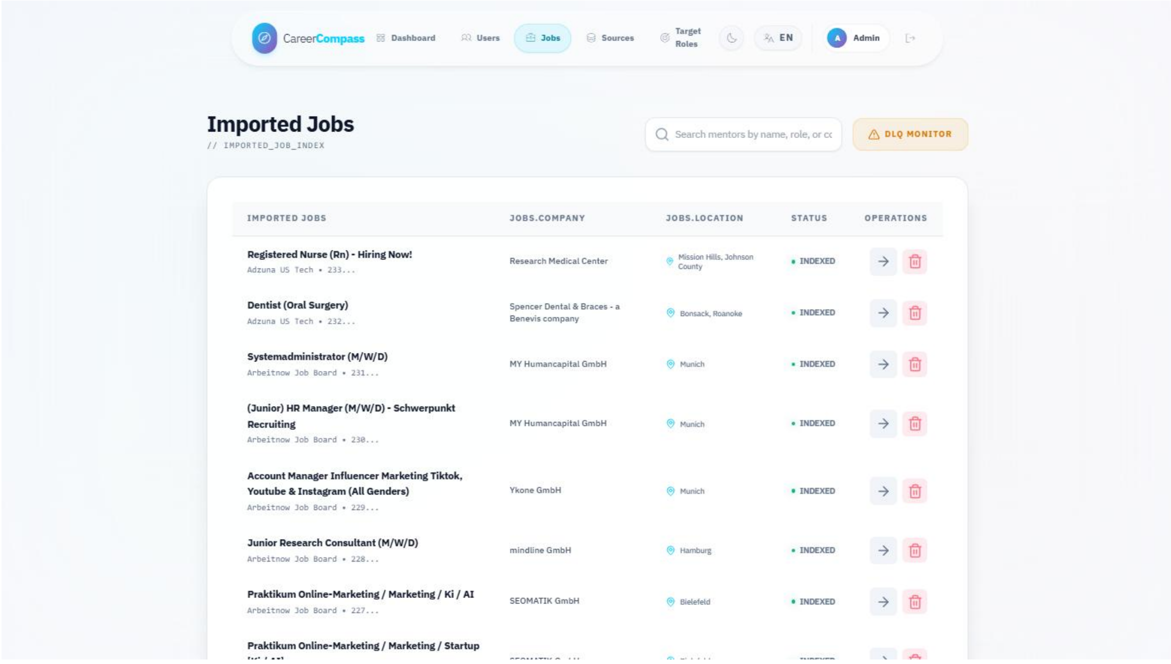

Figure 45. Admin jobs page.

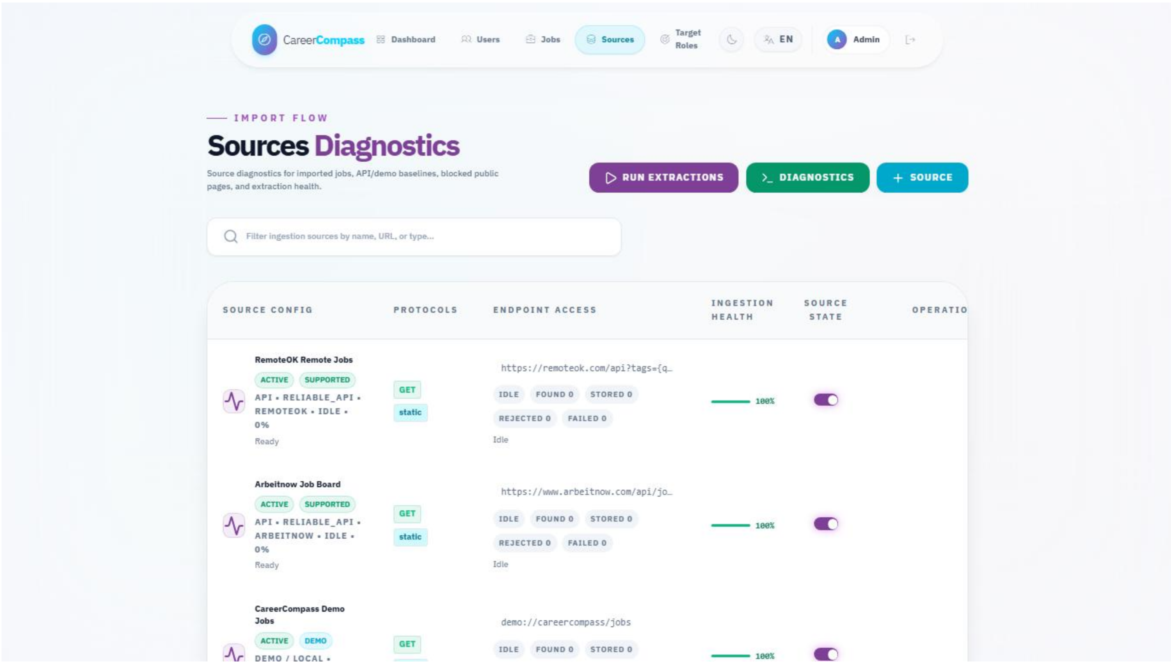

Figure 46. Admin sources diagnostics page.

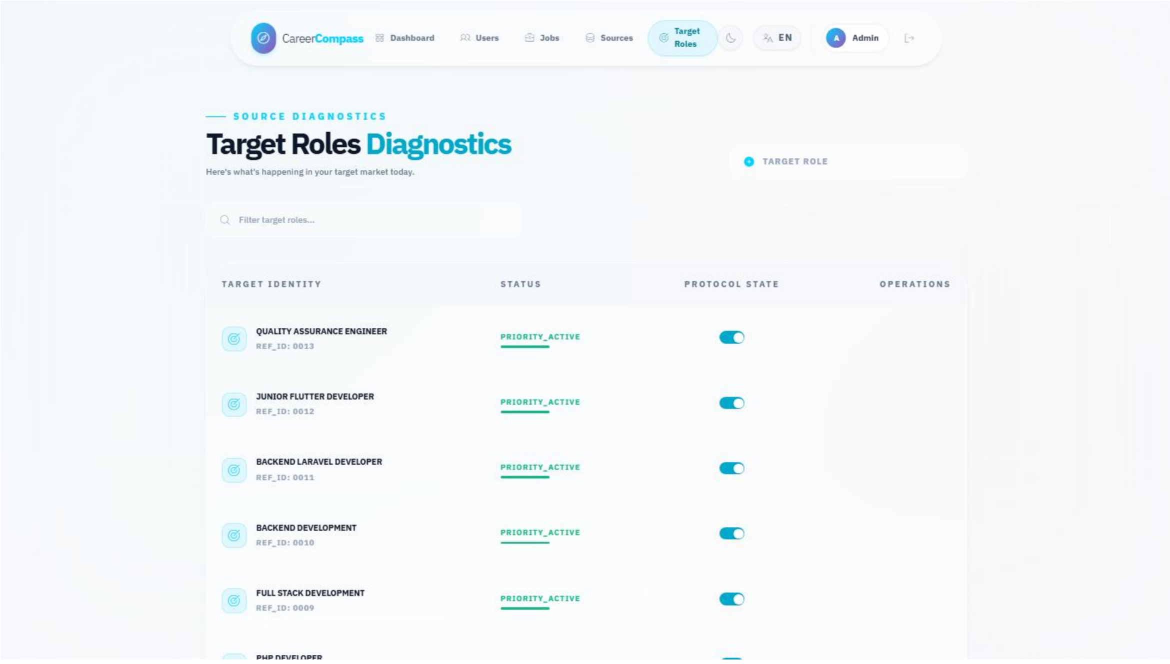

Figure 47. Admin target roles page.

Docker Compose Services Evidence

NAME	IMAGE	COMMAND	SERVICE	CREATED
cc-backend	graduation-project-backend-api	"/usr/local/bin/cc-eâ€¦"	backend-api	2 days ago
cc-backend-scheduler	graduation-project-backend-scheduler	"/usr/local/bin/cc-eâ€¦"	backend-scheduler	4 days ago
cc-backend-worker	graduation-project-backend-worker	"/usr/local/bin/cc-eâ€¦"	backend-worker	4 days ago
cc-backend-worker-ai	graduation-project-backend-worker-ai	"/usr/local/bin/cc-eâ€¦"	backend-worker-ai	4 days ago
cc-backend-worker-emails	graduation-project-backend-worker-emails	"/usr/local/bin/cc-eâ€¦"	backend-worker-emails	4 days ago
cc-backend-worker-high	graduation-project-backend-worker-high	"/usr/local/bin/cc-eâ€¦"	backend-worker-high	4 days ago
cc-backend-worker-scraping	graduation-project-backend-worker-scraping	"/usr/local/bin/cc-eâ€¦"	backend-worker-scraping	4 days ago
cc-cv-analyzer	graduation-project-ai-cv-analyzer	"uvicorn main:app --â€¦"	ai-cv-analyzer	4 days ago
cc-db	mysql:8.0	"docker-entrypoint.sâ€¦"	db	2 weeks ago
cc-frontend	graduation-project-frontend	"docker-entrypoint.sâ€¦"	frontend	4 minutes ag
cc-job-miner	graduation-project-ai-job-miner	"uvicorn service_apiâ€¦"	ai-job-miner	4 days ago
cc-nginx	nginx:stable-alpine	"/docker-entrypoint.â€¦"	nginx	2 days ago
graduation-project-grafana-1	grafana/grafana-oss:11.4.0	"/run.sh"	grafana	2 weeks ago
graduation-project-minio-1	minio/minio:latest	"/usr/bin/docker-entâ€¦"	minio	2 weeks ago
graduation-project-prometheus-1	prom/prometheus:v2.55.1	"/bin/sh -c 'sed \"s/â€¦"	prometheus	2 weeks ago

Figure 48. Docker services evidence.

Figure 49. Validation command evidence.

Appendix F: Test Cases

The manual test matrix in Chapter 7 should be repeated before final submission. Additional recommended tests include invalid CV uploads, banned user login, expired signed download URLs, failed AI service behavior, scraper token rejection, admin route rejection for normal users, and browser checks on a clean database.

The mini dataset evaluation files are:

- evaluation/mini_cv_dataset.json
- evaluation/mini_jobs_dataset.json
- evaluation/expected_labels.json
- evaluation/run_mini_evaluation.py
- evaluation/mini_evaluation_results.json
- evaluation/mini_evaluation_summary.md
- evaluation/ai_cv_analyzer_smoke_samples.json
- evaluation/run_ai_cv_analyzer_smoke_eval.py
- evaluation/ai_cv_analyzer_smoke_results.json
- evaluation/ai_cv_analyzer_smoke_summary.md

Appendix G: GitHub Actions / CI Summary

The repository contains GitHub Actions workflow definitions. After the draft PR is opened, reviewers should inspect PR checks, rerun failed jobs if needed, and confirm that branch protection expectations are satisfied before merging. A CI screenshot was not embedded in this generated report because live PR checks were not available until after branch push.

Appendix H: Demo Script

9. Start Docker Desktop.

10. Run docker compose -f docker-compose.yml -f docker-compose.prod.yml up -d --build.
11. Open http://localhost.
12. Register or login as a student.
13. Upload a sample PDF CV from the dashboard.
14. Review parsed profile and skills.
15. Open Jobs and select a recommendation.
16. Open Gap Analysis and explain matched/missing skills.
17. Save the job to Applications.
18. Login as the demo admin.
19. Open Admin Dashboard, Jobs, Sources, and Target Roles.
20. Open System Status and explain health checks.
21. Discuss limitations and future work honestly.

Appendix I: AI CV Analyzer Deep Inventory

This appendix summarizes the code audit that supports Sections 5.5, Chapter 6, and Sections 7.8-7.11. The detailed companion notes are stored in docs/graduation-book/model-analysis/ and should be kept with the generated book artifacts.

I.1 Runtime Call Path

22. Laravel receives the CV upload and stores the file privately.
23. Laravel sends the uploaded file to the FastAPI analyzer /api/parse-cv endpoint.
24. main.py chooses image or PDF handling and wraps processing in timeout/error fallbacks.
25. CVOrchestrator extracts ordered text, triggers OCR fallback when needed, segments sections, runs NER, extracts contacts and dates, canonicalizes skills, and validates the strict output schema.
26. Layer 2 enriches the result with domain, seniority, and skill-category classification.
27. Laravel persists normalized profile, skills, experiences, and analysis metadata.
28. Recommendations and gap analysis reuse the stored profile together with Layer 3 matching and backend services.

I.2 Function Inventory Summary

Area	Classes and Functions Audited	Main Purpose
main.py	_get_orchestrator, health_check, metrics, hybrid_match, analyze_cv, _process_with_timeout, process_file, _timeout_result, _error_result	API endpoints, service lifecycle, timeout/fallback behavior, and hybrid match composition.
Layer 1 NER/contact/sections	AdvancedNEREngine, extract_contacts, SemanticSegmenter, DataCanonicalizer, ExperienceEngine, spatial/OCR helpers	Converts raw CV text into profile, skills, experience, and confidence-style structured data.
Layer 2 classification	ClassificationOrchestrator, DomainEngine, SeniorityEngine, SkillEngine, load_taxonomy	Adds domain, seniority, and skill-category interpretation.

Area	Classes and Functions Audited	Main Purpose
Layer 3 matching	SemanticEmbedder, IntelligentMatcher, ConstraintValidator, FitAnalysisGenerator, JobDescriptionEngine, RankingOrchestrator, tfidf.match_score	Produces explainable job-fit scores, penalties, verdicts, and ranking behavior.
Training and diagnostics	generate_tech_dataset.py, clean_dataset.py, train_ner.ipynb, verify_phase*.py, test_service_api.py, trace_cv.py	Supports synthetic dataset generation, cleaning, notebook training, phase verification, service tests, and trace output.

Table 43. AI CV Analyzer function inventory summary.

I.3 Training Summary

The notebook trains a BERT-family token-classification model from bert-base-cased, maps character spans to BIO token labels, uses a 90/10 train/evaluation split with seed 42, trains for five epochs with learning rate 2e-5 and batch size 16, computes seqeval precision/recall/F1/accuracy, and exports career_compass_ner_final. The exported Colab PDF records overall final-epoch metrics: precision 0.933307, recall 0.940521, F1 0.936900, and accuracy 0.976376. The final cleaned dataset content and model weights are not committed.

I.4 Generated Dataset Summary

The synthetic dataset script asks Google Gemini tooling to generate varied CV snippets across technical domains, rotates API keys/models from environment variables, includes negative decoys, and writes labeled examples for cleaning. The cleaner normalizes text, deduplicates exact samples, validates spans, and filters malformed or overly broad entities. This supports the training workflow, but it is separate from normal private CV upload processing.

I.5 Local Artifact and Secrets Boundary

.env and ai-cv-analyzer/models/ are ignored by Git. This protects secrets and avoids committing large model weights. Documentation may mention the runtime path and safe local metadata, but should not imply that the committed repository contains the model binary or private API keys.